Numerical Methods

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Contents

1 Floating-Point Numbers **1**
1.1 Numeral Systems . . . 1
1.2 Floating-Point Numbers . . . 2
1.3 IEEE 754 Standard . . . 2
1.4 Rounding and Precision . . . 5
1.5 Numerical Stability . . . 6

2 Error **8**
2.1 Rounding and Truncation Error . . . 8
2.2 Precision of Approximation . . . 9
2.3 Absolute and Relative Error . . . 9
2.4 Big-O Notation . . . 10
2.5 Properties of big-O Notation . . . 12

3 Building Blocks of Numerical Methods **15**
3.1 Taylor Polynomials . . . 15
3.2 Linearisation . . . 17
3.3 Array programming . . . 19

4 Numerical Integration **21**
4.1 Midpoint Rule . . . 21
4.2 Trapezoidal Rule . . . 26
4.3 Simpson's Rule . . . 28

5 Solutions to Systems of Equations **32**
5.1 Newton's Method . . . 33
5.2 Newton's Method for Systems of Equations . . . 36
5.3 Fixed Point Method . . . 43
5.4 Jacobi Method for Linear Systems . . . 50

6 Differential Equations **56**
6.1 Euler's Method . . . 56
6.2 Midpoint Method . . . 62
6.3 Runge–Kutta Methods . . . 66
6.4 Numerical Solutions to Systems of Differential Equations . . . 69
6.5 Implicit Methods . . . 69

7 Optimisation **71**
7.1 Motivation: Training Neural Networks 71
7.2 Univariate Gradient Descent 73
7.3 Multivariate Gradient Descent 77
7.4 Function Composition and the Chain Rule 83
7.5 Automatic Differentiation 85

1 Floating-Point Numbers

This section discusses how numbers are represented in a computer. Design choices have to be made: there are infinitely many real numbers yet computational resources are finite.

1.1 Numeral Systems

The familiar *decimal system* system represents numbers in powers of 10. For example, the number 1453.529 expands as

$$1453.529 = 1 \cdot 10^3 + 4 \cdot 10^2 + 5 \cdot 10^1 + 3 \cdot 10^0 + 5 \cdot 10^{-1} + 2 \cdot 10^{-2} + 9 \cdot 10^{-3}.$$

In this *positional notation* the contribution of a digit to the value of a number is determined by its value multiplied by a factor that equals a chosen *base* (10 in the decimal system) raised to a power that depends on the position of the digit. The position is counted from the *decimal separator* with non-negative powers on the left and negative on the right. For example, because 5 is the second digit to the left from the separator, it receives power 1 (because counting starts from zero on the left) and thus contributes $5 \cdot 10^1 = 50$. Similarly, because 9 is the third digit to the right, it receives power -3 and contributes $9 \cdot 10^{-3} = 0.009$.

Definition 1.1 (Positional numeral system)**.** Let $b \geq 2$ be an integer called *base* (or *radix*). A number in base b is written as

$$(a_n a_{n-1} \cdots a_0 . c_1 c_2 \cdots)_b = a_n \cdot b^n + a_{n-1} \cdot b^{n-1} + \cdots + a_0 \cdot b^0 + c_1 \cdot b^{-1} + c_2 \cdot b^{-2} + \cdots,$$

where the *digits* $a_0, \ldots, a_n$ and $c_1, c_2, \ldots$ are integers between 0 and $b-1$.

The subscript b is typically not used for decimal numbers (i.e., when $b = 10$) except for emphasis.

Example 1.2. To convert the octal (i.e., base $b = 8$) number $(35.07)_8$ to decimal we expand

$$(35.07)_8 = 3 \cdot 8^1 + 5 \cdot 8^0 + 0 \cdot 8^{-1} + 7 \cdot 8^{-2} = 3 \cdot 8 + 5 + \frac{0}{8} + \frac{7}{8^2} = 29.109375.$$

Example 1.3. In the hexadecimal (i.e., base $b = 16$) the letters from A to F are used to represent the digits from 10 to 15. To convert the hexadecimal number $(\mathrm{AE.1B})_{16}$ to decimal we expand

$$\begin{aligned}(\mathrm{AE.1B})_{16} &= 10 \cdot 16^1 + 14 \cdot 16^0 + 1 \cdot 16^{-1} + 11 \cdot 16^{-2} \\ &= 10 \cdot 16 + 14 + \frac{1}{16} + \frac{11}{256} \\ &= 174.10546875.\end{aligned}$$

Example 1.4. To convert $(220.1)_3$ to decimal we expand

$$(220.1)_3 = 2 \cdot 3^2 + 2 \cdot 3^1 + 0 \cdot 3^0 + 1 \cdot 3^{-1} = 2 \cdot 9 + 2 \cdot 3 + \frac{1}{3} = 24.333\ldots$$

Observe that a finite representation in one base, such as $(220.1)_3$, need not to finite in a different base.

Example 1.5. To convert the decimal number 33.25 to binary (i.e., base $b = 2$) we expand

$$\begin{aligned}(33.25)_{10} = 32 + 1 + \frac{1}{4} &= 1 \cdot 2^5 + 0 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 + 0 \cdot 2^{-1} + 1 \cdot 2^{-2} \\ &= (100001.01)_2.\end{aligned}$$

Binary system. In the binary system the digits (i.e., 0 and 1) are called binary digits, more commonly referred to as *bits*. For example, the binary number $(110.1)_2 = 6.5$ has four bits. The largest number that can be represented with n bits is

$$(1\underbrace{0\cdots00}_{n \text{ times}})_2 - (1)_2 = (\underbrace{11\cdots11}_{n \text{ times}})_2 = 1 \cdot 2^{n-1} + 1 \cdot 2^{n-2} + \cdots + 1 \cdot 2^1 + 1 \cdot 2^0 = 2^n - 1. \tag{1.1}$$

1.2 Floating-Point Numbers

Although any base can be used to represent any number (possibly in a non-terminating manner), this is not possible in practice due to limited memory. When implementing numbers on a computer one has to choose which numbers to represent. This is done via floating-point numbers.

Definition 1.6 (Floating-point number)**.** A *floating-point number* $a \in \mathbb{R}$ in base b has the form

$$a = s \cdot c \cdot b^e,$$

where $s \in \{-1, 1\}$ is the *sign*, $c \in \mathbb{R}_{\geq 0}$ the *mantissa* (or *significand*), and $e \in \mathbb{Z}$ the *exponent*.

The floating-point representation is not unique. For example, 123.45 can be written as

$$12345 \cdot 10^{-2} \quad \text{or} \quad 1234.5 \cdot 10^{-1},$$

or in infinitely many other ways as a decimal (i.e., base $b = 10$) floating-point number. In order to represent floating-point numbers on a computer one must select ranges for the mantissa and exponent. These choices determine the precision and memory cost of the resulting floating-point format.

1.3 IEEE 754 Standard

The *IEEE Standard for Floating-Point Arithmetic* (IEEE 754) defines a collection of floating-point formats used in virtually all computing. The binary IEEE 754 formats represent floating-point numbers in the form

$$a = s \cdot (1 + f) \cdot 2^{e - e_{\text{bias}}}, \tag{1.2}$$

where e_{bias}, the *exponent bias*, is a positive integer and $f \in (0, 1)$ the *fraction*. Both $f \in (0, 1)$ and $e \in \mathbb{Z}_{\geq 0}$ are expressed in binary with chosen numbers of bits. That is,

$$f = (0.\bar{b}_1 \bar{b}_2 \cdots \bar{b}_p)_2 \quad \text{and} \quad e = (\hat{b}_1 \hat{b}_2 \cdots \hat{b}_q)_2$$

for some positive integeres p and q and bits $\bar{b}_i, \hat{b}_j, \in \{0, 1\}$. The sign is represented by a sign b_s in the form $s = (-1)^{b_s}$, so that

$$b_s = 0 \implies s = 1 \qquad \text{and} \qquad b_s = 1 \implies s = -1.$$

To represent a floating-point number in this format the $n = p + q + 1$ bits (p bits for the fraction, q for the exponent and one for the sign) are ordered to give an IEEE 754 binary representation of a floating-point number:

$$(b_s \hat{b}_q \hat{b}_{q-1} \cdots \hat{b}_1 \bar{b}_p \bar{b}_{p-1} \cdots \bar{b}_1)_2 = (b_{p+q} b_{p+q-1} \cdots b_0)_2. \tag{1.3}$$

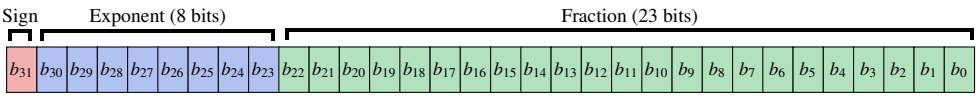


Figure 1.1: Bit layout for 32-bit IEEE 754 floats. See Table 1.1.

Figure 1.1 illustrates the bit scheme. In IEEE 754 the integer e_{bias} in (1.2) that determines the *actual exponent* of two is set as $e_{\text{bias}} = 2^{q-1} - 1$. This means that the actual exponent ranges [recall (1.1)]

$$\text{from} \quad e_{\min} = 0 - e_{\text{bias}} = -2^{q-1} + 1 \quad \text{to} \quad e_{\max} = 2^q - 1 - e_{\text{bias}} = 2^{q-1}(2-1) = 2^{q-1}. \tag{1.4}$$

Example 1.7. Consider IEEE 754 format floating-point numbers with $p = 3$ bits for the fraction and $q = 2$ bits for the exponent. The total number of bits is $n = p + q + 1 = 6$. The binary number $(101110)_2$ represents a floating-point number of the form (1.2) with $s = (-1)^1 = -1$, $e = (01)_2 = 1$, and $f = (0.110)_2 = 0.75$. Because $e_{\text{bias}} = 2^{q-1} - 1 = 1$, $(101110)_2$ corresponds to the decimal number

$$s \cdot (1+f) \cdot 2^{e - e_{\text{bias}}} = (-1) \cdot (1 + 0.75) \cdot 2^{1-1} = -1.75.$$

by (1.2).

Special quantities in IEEE 754. There are exceptions related to the exponents $e = (0 \cdots 0)_2$ and $e = (1 \cdots 1)_2$:

- If the exponent is zero, $e = (0 \cdots 0)_2$:
  - If the *fraction is zero*, $f = (0.0 \cdots 0)_2$, the number is interpreted as zero. The sign bit determines the whether the zero is "positive" or "negative", which has no effect on most computations. That is,

    $$\text{“positive” zero} = (10 \cdots 0)_2 \quad \text{and} \quad \text{“negative” zero} = (00 \cdots 0)_2.$$

  - If the *fraction is not zero*, the floating-point number is called *subnormal* and represents a decimal number of the form

    $$a_{\text{subnormal}} = s \cdot f \cdot 2^{-e_{\text{bias}}+1}, \tag{1.5}$$

    which differs from a *normal* number of the form (1.2) by $1 + f \in (1,2)$ being replaced by $f \in (0,1)$. Smaller numbers than what would be possible with normal numbers alone can be represented with subnormal numbers.
- If the exponent is maximal, $e = (1 \cdots 1)_2$:
  - If the *fraction is zero*, $f = (0.0 \cdots 0)_2$, the number is interpreted as positive (∞) or negative infinite ($-\infty$) depending on the sign bit.
  - If the *fraction is not zero*, the number is interpreted as *not a number* (NaN), meaning that the number is undefined. This can occur when trying to compute $0/0$.

IEEE 754 binary formats. Table 1.1 contains the specifications and key properties of the three most common IEEE 754 binary floating-point formats. These are typically referred to as *half-precision* numbers (16 bit), *floats* or *single-precision numbers* (32 bits), and *doubles* or *double-precision numbers* (64 bits).

Table 1.1: Key properties of IEEE 754 half, single-precision float, and double-precision formats. Note that each format includes one bit for the sign in addition to the exponent and fraction bits. *Machine epsilon* quantifies the precision of a floating-point system. It refers to the difference between 1 and the next larger floating-point number. One can work out that $\varepsilon = 2^{-p}$, where p is the number of bits in the fraction. *The number bits in the fraction determines precision and the number of bits in the exponent determines how small and large numbers can be represented.*

Property	**Half (16-bit)**	**Float (32-bit)**	**Double (64-bit)**
Total bits	16	32	64
Exponent bits (q)	5	8	11
Fraction bits (p)	10	23	52
Exponent bias (e_{bias})	15	127	1023
Smallest positive normal	2^{-14}	2^{-126}	2^{-1022}
Smallest positive subnormal	2^{-24}	2^{-149}	2^{-1074}
Largest finite value ($\approx$)	6.55×10^4	3.40×10^{38}	1.80×10^{308}
Machine epsilon (ε)	2^{-10}	2^{-23}	2^{-52}

Example 1.8. Let us find the IEEE 754 float (32 bits) representation of the decimal number $a = -3.75$. Since the sign is $s = (-1)^{b_s}$, the sign bit is one, $b_s = 1$. By (1.4), the exponent bias is $e_{\text{bias}} = 2^{q-1} - 1 = 2^7 - 1 = 127$. Because $1 + f \in (1,2)$ and $(1+f) \cdot 2^1 \in (2,4)$, the actual exponent must be 1, meaning that $1 = e - e_{\text{bias}} = e - 127$. Therefore $e = 128 = 2^7 = (10\cdots0)_2$, where there are seven zeros. From $3.75 = (1+f) \cdot 2$ we solve $f = \frac{1}{2} \cdot 3.75 - 1 = 0.875 = (0.111)_2$. Therefore the float representation of -3.75 is

$$11000000011100000000000000000000,$$

where we use the same colours as in Figure 1.1 to represent the sign, exponent, and fraction, and have appended zeros to the fraction to ensure that the representation has 32 bits.

Example 1.9. Let us proceed to the opposite direction and find out which decimal number a the float represention

$$00001010001100000000000000000000$$

corresponds to. The sign bit is zero ($b_s = 0$), so that the sign is positive because $s = (-1)^{b_s} = (-1)^0 = 1$. The exponent is $e = (00010100)_2 = 2^4 + 2^2 = 20$. The fraction is $f = (0.01100000000000000000000)_2 = 2^{-2} + 2^{-3} = 0.375$. Therefore

$$a = s \cdot (1+f) \cdot 2^{e - e_{\text{bias}}} = 1 \cdot (1 + 0.375) \cdot 2^{20-127} = 1.375 \cdot 2^{-107} \approx 8.47 \times 10^{-33}.$$

Example 1.10. Let us verify some properties of floats in Table 1.1. By (1.4), the exponent bias is $e_{\text{bias}} = 2^{q-1} - 1 = 2^7 - 1 = 127$. The smallest possible positive normal number is the smallest positive floating-point number of the form (1.2). This means that we want (a) a fraction that is as small as possible and (b) a *positive* exponent (if the exponent were zero

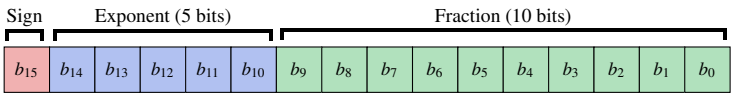


Figure 1.2: Bit layout for 16-bit IEEE 754 half-precision floating point numbers. See Table 1.1.

the number would be subnormal) that is as small as possible. This means that

$$f = (0.\underbrace{0\cdots0}_{23\text{ times}})_2 = 0 \quad \text{and} \quad e = (\underbrace{0\cdots0}_{7\text{ times}}1)_2 = 1.$$

Therefore the smallest positive *normal* float is

$$a = 1\cdot(1+f)\cdot 2^{e-e_{\text{bias}}} = 1\cdot(1+0)\cdot 2^{1-127} = 2^{-126} \approx 1.18\times 10^{-38}.$$

To get the smallest positive *subnormal* float we set the exponent to zero, select the smallest positive fraction, $f = (0.0\cdots01)_2 = 2^{-23}$ and use the subnormal form (1.5), which gives

$$a = 1\cdot f\cdot 2^{-127+1} = 2^{-23}\cdot 2^{-126} = 2^{-149} \approx 1.40\times 10^{-45}.$$

The largest finite float has largest possible fraction and an exponent that is one minus the largest possible exponent (because the largest exponent is interpreted as infinity or NaN):

$$f = (0.\underbrace{1\cdots1}_{23\text{ times}})_2 = \sum_{k=1}^{23} 2^{-k} = 1-2^{-23} \quad \text{and} \quad e = (\underbrace{1\cdots1}_{7\text{ times}}0)_2 = \sum_{k=1}^{7} 2^{k} = 254.$$

These give the largest finite float

$$a = 1\cdot(1+f)\cdot 2^{e-e_{\text{bias}}} = (2-2^{-23})\cdot 2^{127} \approx 3.40\times 10^{38}.$$

Floating-point numbers in Python. In Python, one way to recover the 32-bit binary representation of a floating-point number is as follows:

```
import numpy as np
a = -3.75
res = np.binary_repr(np.float32(a).view(np.int32), width=32)
print(res)
```

Running these lines of code should print `11000000011100000000000000000000`, which matches with the representation we obtained in Example 1.8. However, note that in Python the term "float" refers to double-precision (64-bit) floating point numbers.

1.4 Rounding and Precision

Floating-point numbers provide a *finite-precision arithmetic*. That is, not all real numbers can be represented exactly and the results of arithmetic operations can differ from *exact arithmetic*. For example, the familiar decimal number 0.1 cannot be represented exactly as a non-terminating binary number:

$$(0.1)_{10} = (0.00011001100110011\ldots)_2.$$

Consequently, such numbers cannot be represented exactly as binary floating-point numbers and are rounded to a representable floating-point number according the rounding rules specified in the IEEE 754 standard. This can be readily observed in computations. For example, in Python

a simple multiplication whose value in exact arithmetic is 653082380.34 becomes something different:

```
>>> print(27869.01 * 23434)
653082380.3399999
```

We can observe the same phenomenon when printing, for example, 50 decimal digits of $a = 0.1$:

```
>>> a = 0.1
>>> print(f"{a:.{50}f}")
0.10000000000000000555111512312578270211815834045410
```

Precision. Floating-point numbers are *not uniform in precision*. This is both an advantage (a huge range of numbers can be represented) and a disadvantage (rounding errors are inevitable when operating with large numbers). For example, with doubles (64 bits) one can represent numbers between 1 and 2 with precision 2^{-52}, meaning that all numbers on the interval $[1,2]$ round to a multiple of 2^{-52}. Precision decreases as numbers become larger, being 2^{-51} for numbers between 2 and 4 and eventually 2^{971} for numbers between 2^{1023} and 2^{1024}. We can easily see this in action in Python (which uses doubles by default). The following computations return positive values as they should:

```
>>> b = 1.0
>>> a = 1.0 + 2.0**(-52)
>>> print(a - b)
2.220446049250313e-16
>>> b = 2.0**1023
>>> a = 2.0**1023 + 2.0**971
>>> print(a - b)
1.99584030953472e+292
```

Note that in the second computation it is essential to write "`2.0`" rather than "2" to ensure that Python interprets the numbers as floating-point numbers rather than integers, which are handled differently. However, if we make a slightly smaller it is rounded to b and the difference is incorrectly (in comparison to exact arithmetic) evaluated as zero:

```
>>> b = 1.0
>>> a = 1.0 + 0.5*2.0**(-52)
>>> print(a - b)
0.0
>>> b = 2.0**1023
>>> a = 2.0**1023 + 0.5*2.0**971
>>> print(a - b)
0.0
```

Underflow and overflow. Similar phenomena occur when one tries to do arithmetic with numbers smaller than the *underflow level* (the row with smallest positive subnormals in Table 1.1) rounding to zero and numbers larger than the *overflow level* (the row with largest finite values in Table 1.1) rounding to infinity or throwing an error. For 64-bit doubles these levels are 2^{-1074} and approximately 1.80×10^{308}.

1.5 Numerical Stability

The inherent inadequacies of floating-point numbers need to be taken into account when designing numerical algorithms. *Numerical stability* of an algorithm refers to its ability to tolerate numerical

errors. One should be particularly careful when subtracting nearby floating-point numbers and avoid doing this if possible.

Example 1.11. Computing $\sqrt{1+x}-\sqrt{x}>0$ is prone to numerical cancellations when x is large because in that case both $\sqrt{1+x}$ and $\sqrt{x}$ are large and nearby but far away from nearest representable floating-point numbers due to their sparsity (recall Section 1.4). For example, for $x=2^{60}$ this expression evaluates to zero in the 64-bit arithmetic used in Python. In contrast, the equivalent form

$$\sqrt{1+x}-\sqrt{x}=\frac{(\sqrt{1+x}-\sqrt{x})(\sqrt{1+x}+\sqrt{x})}{\sqrt{1+x}+\sqrt{x}}=\frac{1}{\sqrt{1+x}+\sqrt{x}},$$

which used the identity $(a-b)(a+b)=a^2-b^2$, evaluates to a positive value even for $x=2^{1023}$.

Example 1.12. Similar phenomenon can occur when computing a^2-b^2 for large a and b. This computation can be made more stable by using the identity $(a-b)(a+b)=a^2-b^2$. While this identity contains a subtraction $a-b$, the numbers being subtracted are smaller than in a^2-b^2 whenever $a,b>1$. Therefore floating-point rounding is less extreme.

The following lines of Python illustrate Example 1.12:

```
>>> a = 2.0**50
>>> b = a - 1.0
>>> print(a**2 - b**2)
2251799813685248.0
>>> print((a - b)*(a + b))
2251799813685247.0
```

The correct value of a^2-b^2 for $a=2^{50}$ and $b=2^{50}-1$ is 2251799813685247, which the computation based on the more numerically stable form $(a-b)(a+b)$ returned.

2 Error

This section discusses the nature of numerical errors in computation and how errors are measured.

2.1 Rounding and Truncation Error

Numerical errors arise from both the inherent inaccuracy of finite-precision arithmetic and numerical approximations of mathematical processes.

Definition 2.1 (Rounding error)**.** *Rounding error* is the error caused by the use of finite-precision arithmetic (e.g., floating-point numbers) as opposed to exact arithmetic.

Example 2.2. The rounding and precision problems that we observed in Sections 1.4 and 1.5 are examples of rounding errors.

Definition 2.3 (Truncation error)**.** *Truncation error* is the error caused by a numerical approximation of a mathematical process.

Example 2.4. From basic analysis we recall that the exponential constant $e \approx 2.718$ can be expressed as the infinite series

$$e = \sum_{k=0}^{\infty} \frac{1}{k!},$$

where $n! = 1 \cdot 2 \cdots (n-1) \cdot n$ is the factorial and $0! = 1$. To approximate e one can truncate this series after n terms (here $n \in \{0,1,2,3\}$):

$$\begin{aligned}
e \approx e_0 &= \sum_{k=0}^{0} \frac{1}{k!} = \frac{1}{0!} = \frac{1}{1} = 1, \\
e \approx e_1 &= \sum_{k=0}^{1} \frac{1}{k!} = \frac{1}{0!} + \frac{1}{1!} = \frac{1}{1} + \frac{1}{1} = 2, \\
e \approx e_2 &= \sum_{k=0}^{2} \frac{1}{k!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} = \frac{1}{1} + \frac{1}{1} + \frac{1}{2} = 2.5, \\
e \approx e_3 &= \sum_{k=0}^{3} \frac{1}{k!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} = \frac{1}{1} + \frac{1}{1} + \frac{1}{2} + \frac{1}{2 \cdot 3} = 2.666\ldots
\end{aligned}$$

Any of the differences $e - e_n$ for $n \in \{0,1,2,3\}$ is a truncation error.

However, truncation error does not refer only to explicit truncation as in Example 2.4 but to any type of numerical approximation.

Example 2.5. Recall again from basic analysis that the derivative of a function $f \colon \mathbb{R} \to \mathbb{R}$ at a point $x_0 \in \mathbb{R}$ is defined as the limit

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

of the difference quotient $[f(x_0 + h) - f(x_0)]/h$. One can therefore approximate the derivative by computing the difference quotient for a small h. For example, the derivative of

$f(x) = x^2$ at $x_0 = 1$ can be approximated as (with $h = 1/10$)

$$f'(x_0) \approx \frac{f(x_0+h) - f(x_0)}{h} = \frac{(1+1/10)^2 - 1^2}{1/10} = \frac{21}{10}.$$

Since $f'(x) = 2x$, the true derivative is $f'(x_0) = f'(1) = 2$. The error

$$f'(x_0) - \frac{21}{10} = 2 - \frac{21}{10} = -\frac{1}{10}$$

is a truncation error even though no summation is being truncated. However, even here one can think in terms of a truncation: one is "truncating the limiting process $h \to 0$" by picking a small h and using that rather than going all the way to zero.

2.2 Precision of Approximation

Definition 2.6 (Precision of approximation)**.** An approximation $\tilde{x}$ to x is said to *approximate x to d decimal places* if

$$|x - \tilde{x}| < \frac{1}{2}10^{-d}.$$

For example, if we want $\tilde{x}$ to approximate x to the nearest integer (i.e., to zero decimal places), then it must hold that $|x - \tilde{x}| < 0.5$. This means that the distance between x and $\tilde{x}$ is always rounded to zero when approximated to the nearest integer. Note that this does not mean that both numbers are rounded to the same value when approximated to the nearest integer. For example, if $x = 0.51$ and $\tilde{x} = 0.49$, then $|0.51 - 0.49| = 0.02 < 0.5$ but $x \approx 1$ and $\tilde{x} \approx 0$. Note also that if $\tilde{x}$ approximates x to d decimal places, it also approximates it to $d-1$ decimal places, and so on. What matters is, of course, the highest decimal precision.

Example 2.7. The number 3.14 approximates π to two decimal places since

$$|\pi - 3.14| = 0.00159\ldots = 0.159\ldots \cdot 10^{-2} < 0.5 \cdot 10^{-2}.$$

The number 3.14 approximates π also to one decimal place since

$$|\pi - 3.14| = 0.00159\ldots = 0.0159\ldots \cdot 10^{-1} < 0.5 \cdot 10^{-1}.$$

The numbe 3.14 does *not* approximate π to three decimal places since

$$|\pi - 3.14| = 0.00159\ldots = 1.59\ldots \cdot 10^{-3} > 0.5 \cdot 10^{-3}.$$

2.3 Absolute and Relative Error

Error can be measured in either absolute or relative terms.

Definition 2.8 (Absolute and relative error)**.** Let $\tilde{x}$ be an approximation of x. Then

$$\textit{absolute error} = |x - \tilde{x}| \quad \text{and} \quad \textit{relative error} = \left|\frac{x - \tilde{x}}{x}\right|.$$

The absolute error is simply the magnitude of the difference between the true quantity of interest x and its approximation $\tilde{x}$. The relative error measures the error *in proportion to the magnitude of the true value*. Note that when computing the relative error one divides by the magnitude of the true value, not of the approximation. The relative error is often expressed in percent.

Example 2.9. If the number $x = \pi$ is approximated to two decimal places as in Example 2.7 (i.e., $\tilde{x} = 3.14$), then the absolute error of the approximation is

$$|\pi - 3.14| = 0.00159\ldots$$

and the relative error is

$$\left|\frac{\pi - 3.14}{\pi}\right| = 0.000507\ldots = 0.0507\ldots\,\%.$$

Relative error is often the more useful of the two errors.

Example 2.10. In Section 1.4 we observed that in Python's 64-bit double-precision arithmetic the number

$$x = 2^{1023} + 0.5 \cdot 2^{971} \quad \text{is rounded to} \quad \tilde{x} = 2^{1023}.$$

Therefore the absolute rounding error is

$$|x - \tilde{x}| = |2^{1023} + 0.5 \cdot 2^{971} - 2^{1023}| = 0.5 \cdot 2^{971} \approx 9.9792 \cdot 10^{291}.$$

This somewhat large number is very small in proportion to the size of x as the relative rounding error is minuscule:

$$\left|\frac{x - \tilde{x}}{x}\right| = \left|\frac{2^{1023} + 0.5 \cdot 2^{971} - 2^{1023}}{2^{1023} + 0.5 \cdot 2^{971}}\right| \approx 1.11 \cdot 10^{-16} = 1.11 \cdot 10^{-14}\,\%.$$

2.4 Big-O Notation

Asymptotic big-O notation provides a convenient and concise way to describe the asymptotical, or limiting, behaviour of an approximation.

Definition 2.11 (Big-O notation for functions)**.** Let $x_0 \in (a,b)$ and let $f, g\colon (a,b) \to \mathbb{R}$ be functions. We write

$$f(x) = O(g(x)) \quad \text{as} \quad x \to x_0$$

if there are $C \geq 0$ and $\delta > 0$ such that $|f(x)| \leq Cg(x)$ for all $x \in (x_0 - \delta, x_0 + \delta)$.

Recall that a sequence $(a_n)_{n=1}^{\infty} = (a_1, a_2, \ldots)$ is an ordered collection of real numbers.

Definition 2.12 (Big-O notation for sequences)**.** Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two sequences. We write

$$a_n = O(b_n) \quad \text{as} \quad n \to \infty$$

if there is $C \geq 0$ such that $|a_n| \leq Cb_n$ for all $n \geq 1$.

The notation $O(g(x))$ is read as "of order $g(x)$", "big-oh of $g(x)$", or "oh of $g(x)$". If clear from the context, the limit ($x \to x_0$ or $n \to \infty$) is often omitted. On this course we use big-O notation to describe the behaviour of numerical approximations. However, this notation is very versatile, being also used to describe the time complexity and space complexity of algorithms.[1] If $g(x) \to 0$ as $x \to x_0$ (or $b_n \to 0$ as $n \to \infty$), the statement $f(x) = O(g(x))$ gives information on how fast $f(x)$ tends to zero as $x \to x_0$. Namely, this means that $f(x)$ tends to zero *at least* as fast as $g(x)$. In such a situation one wants to find g that tends to zero as fast as possible such that $f(x) = O(g(x))$ remains true.

[1] *Data Structures and Algorithms* (BM40A1500).

Example 2.13. Suppose that we want to determine the order of $f(x) = x^2 + x^3$ as $x \to 0$ (i.e., $x_0 = 0$). By Definition 2.11 we need to find a function g and constants $C \geq 0$ and $\delta > 0$ such that $|f(x)| \leq Cg(x)$ for all $x \in (-\delta, \delta)$. Select $\delta = 1$ (any other value less than one would do as well). Then, for all $x \in (-1, 1)$,

$$|f(x)| = x^2 + |x|^3 \leq x^2 + |x|^2 = 2x^2,$$

where we used the fact that $|a|^n \leq |a|^m$ if $|a| \leq 1$ and $n \geq m$. Therefore we can select $C = 2$ and $g(x) = x^2$, meaning that $f(x)$ is of order x^2 [$f(x) = O(x^2)$] as $x \to 0$. However, note that we also have

$$|f(x)| = x^2 + |x|^3 \leq x^2 + |x|^2 \leq |x| + |x| = 2|x|,$$

so that we could have chosen $g(x) = |x|$.

In Example 2.13 the order $g(x) = x^2$ is better than $g(x) = |x|$ because it describes the behaviour of f close to $x_0 = 0$ more accurately. Namely, as $x \to 0$,

$$f(x) = O(x^2) \quad \text{and} \quad f(x) = O(|x|)$$

but

$$x^2 = O(|x|) \quad \text{and} \quad |x| \neq O(x^2),$$

where $|x| \neq O(x^2)$ means that $|x|$ is *not* of order x^2 as $x \to 0$. When written down these four big-O statements are

$$f(x) \text{ tends to zero at least as fast as } x^2 \quad \text{and} \quad f(x) \text{ tends to zero at least as fast } |x|$$

but

$$x^2 \text{ tends to zero at least as fast as } |x| \quad \text{and} \quad |x| \text{ tends to zero } \textit{slower} \text{ than } x^2.$$

Figure 2.1 illustrates big-O notation for common polynomials around $x_0 = 0$.

Remark 2.14. To write $f(x) = O(g(x))$ as $x \to x_0$ is an abuse of notation. Because "=" commutes elsewhere in mathematics (i.e., $a = b$ is the same as $b = a$), $f(x) = O(g(x))$ suggests that one could write $O(g(x)) = f(x)$, which is difficult to make sense of. What $f(x) = O(g(x))$ as $x \to x_0$ means is that f *is an element of the collection of functions that*

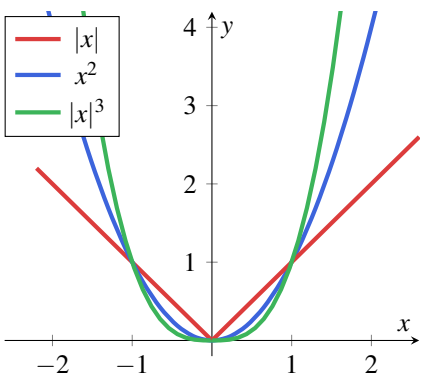


Figure 2.1: The functions $|x|$, x^2 and $|x|^3$ around $x_0 = 0$. Because $|x|$ is above x^2 when $x \in (-1, 1)$ and x^2 is above $|x|^3$, the figure illustrates that $x^2 = O(|x|)$ and $|x|^3 = O(x^2)$ as $x \to 0$. *Observe that the ordering of the graphs changes when* $|x| > 1$. If we considered a point x_0 such that $|x_0| > 1$ (say $x_0 = 2$), the big-O relations would need to be reversed: $|x| = O(x^2)$ and $x^2 = O(|x|^3)$ as $x \to 2$. Note that here we must use $|x|$ and $|x|^3$ rather than x and x^3 because x and x^3 are negative when $x < 0$ but $f(x) = O(g(x))$ enforces an upper bound on $|f(x)| \geq 0$ rather than $f(x)$.

are bounded from above by some constant times $g(x)$ around x_0. The notation must therefore be interpreted as convenient shorthand only.

Sometimes you encounter the notation

$$f(x) = g(x) + O(r(x)) \quad \text{as} \quad x \to x_0.$$

This notation means that f is the sum of g and a term that is of order $r(x)$ as $x \to x_0$. For example, if $f(x) = 3 + x^2 + x^3$, we could write

$$f(x) = 3 + O(x^2) \quad \text{as} \quad x \to 0$$

because in Example 2.13 we saw that $x^2 + x^3 = O(x^2)$ as $x \to 0$.

2.5 Properties of big-O Notation

The following theorem shows that big-O notation is *transitive*. The theorem is nothing but a big-O version of the elementary fact that $a \le b$ and $b \le c$ implies $a \le c$.

Theorem 2.15. *If $f(x) = O(g_1(x))$ and $g_1(x) = O(g_2(x))$ as $x \to x_0$, then*

$$f(x) = O(g_2(x)) \quad \text{as} \quad x \to x_0.$$

Proof. By Definition 2.11, there are $C_1, C_2 \ge 0$ and $\delta_1, \delta_2 > 0$ such that

$$|f(x)| \le C_1 g_1(x) \quad \text{for all} \quad x \in (x_0 - \delta_1, x_0 + \delta_1)$$

and

$$|g_1(x)| \le C_2 g_2(x) \quad \text{for all} \quad x \in (x_0 - \delta_2, x_0 + \delta_2).$$

These two inequalities are valid simultaneously when $x \in (x_0 - \delta, x_0 + \delta)$ for $\delta = \min\{\delta_1, \delta_2\}$. Therefore, for $x \in (x_0 - \delta, x_0 + \delta)$,

$$|f(x)| \le C_1 g_1(x) \le C_1 |g_1(x)| \le C_1 \cdot C_2 g_2(x),$$

so that we can set $g = g_2$, $C = C_1 \cdot C_2$ and $\delta = \min\{\delta_1, \delta_2\}$ in Definition 2.11. □

The next theorem shows that the constant C in Definition 2.11 does not affect the order.

Theorem 2.16. *If $f(x) = O(g_1(x))$ as $x \to x_0$ and $g_1(x) = C_{12} g_2(x)$ for $C_{12} \ge 0$, then $f(x) = O(g_2(x))$ as $x \to x_0$.*

Proof. By Definition 2.11 there are $C_1 \ge 0$ and $\delta > 0$ such that $|f(x)| \le C_1 g_1(x)$ for all $x \in (x_0 - \delta, x_0 + \delta)$. From the assumption $g_1(x) = C_{12} g_2(x)$ it follows that, for all $x \in (x_0 - \delta, x_0 + \delta)$,

$$|f(x)| \le C_1 g_1(x) = C_1 \cdot C_{12} g_2(x),$$

so that we can select $C = C_1 \cdot C_{12}$ and $g = g_2$ in Definition 2.11. □

Theorems 2.15 and 2.16 have natural variants for sequences (Definition 2.12). The following example shows a rather typical application of big-O notation in numerical approximation where one is usually interested in the behaviour of an error as some *discretisation parameter* h tends to zero.

Example 2.17. In Example 2.5 we approximated the derivative of $f(x) = x^2$ at $x_0 = 1$ with the difference quotient

$$q(h) = \frac{f(x_0 + h) - f(x_0)}{h}.$$

We now compute the order of the absolute error of this approximation as $h \to 0$. Because

$f'(x) = 2x$ and $x_0 = 1$, the absolute error $\varepsilon(h)$ is

$$\begin{aligned}\varepsilon(h) = |f'(x_0) - q(h)| = \left| f'(x_0) - \frac{f(x_0+h) - f(x_0)}{h} \right| &= \left| 2 - \frac{(1+h)^2 - 1}{h} \right| \\ &= \left| 2 - \frac{2h + h^2}{h} \right| \\ &= |2 - (2+h)| \\ &= |h|,\end{aligned}$$

where we used the identity $(a+b)^2 = a^2 + 2ab + b^2$. This shows that $\varepsilon(h) = O(|h|)$, which means that the absolute error of the derivative approximation by the difference quotient $f'(x_0) \approx q(h)$ tends to zero with *linear order* as $h \to 0$. Note that the relative error $\rho(h)$ is of the same order because

$$\rho(h) = \left| \frac{f'(x_0) - q(h)}{f'(x_0)} \right| = \frac{1}{|f'(x_0)|} \varepsilon(h),$$

where $1/|f'(x_0)| = 1/2 > 0$. This shows that $\rho(h) = O(\varepsilon(h))$. Since $\varepsilon(h) = |h|$, we can now use Theorem 2.16 to deduce that $\rho(h) = O(|h|)$.

We shall mostly work with errors $\varepsilon(h)$ that have *polynomial orders* as $h \to 0$. If a function is a polynomial, the term with the smallest degree determines the order.

Theorem 2.18. *Suppose that* $\varepsilon(h) = a_n h^n + a_{n+1} h^{n+1} + \cdots + a_{n+m} h^{n+m}$ *for* $n, m \in \mathbb{N} \cup \{0\}$ *and* $a_n, \ldots, a_{n+m} \in \mathbb{R}$. *Then*

$$\varepsilon(h) = O(|h|^n) \quad \textit{as} \quad h \to 0.$$

Proof. Select $\delta = 1$ and note that $x_0 = 0$. For $h \in (x_0 - \delta, x_0 + \delta) = (-1, 1)$,

$$|\varepsilon(h)| = |a_n h^n + a_{n+1} h^{n+1} + \cdots + a_{n+m} h^{n+m}| \leq |a_n||h|^n + \cdots + |a_{n+m}||h|^{n+m},$$

where we used the triangle inequality $|a+b| \leq |a| + |b|$. Because $|h|^{n+m} \leq |h|^n$ for $h \in (-1, 1)$, we obtain

$$|\varepsilon(h)| \leq (|a_n| + \cdots + |a_{n+m}|)|h|^n,$$

where the constant $C = |a_n| + \cdots + |a_{n+m}|$ is non-negative (i.e., ≥ 0). It follows from Definition 2.11 that $\varepsilon(h) = O(|h|^n)$. □

We are yet to discuss Definition 2.12, which defines big-O notation for sequences. On this course we use this notation to describe the error of numerical methods as n, the number of *discretisation points* or *time steps*, grows.

Example 2.19. Recall Example 2.4 where we approximated the exponential constant $e \approx 2.718$ by truncating the infinite series expansion $e = \sum_{k=0}^{\infty} 1/k!$ after n terms:

$$e \approx e_n = \sum_{k=0}^{n} \frac{1}{k!}.$$

Let us determine the order of this approximation (i.e., the order of the error $\varepsilon_n = |e - e_n|$) as $n \to \infty$. We have

$$\varepsilon_n = |e - e_n| = \left| \sum_{k=0}^{\infty} \frac{1}{k!} - \sum_{k=0}^{n} \frac{1}{k!} \right| = \left| \sum_{k=n+1}^{\infty} \frac{1}{k!} \right| = \sum_{k=n+1}^{\infty} \frac{1}{k!}. \tag{2.1}$$

For any $m \geq 1$ we can write

$$(n+m)! = 1 \cdot 2 \cdots (n+1) \cdot (n+2) \cdots (n+m) = (n+1)! \cdot (n+2) \cdots (n+m), \tag{2.2}$$

where the product $(n+2) \cdots (n+m)$ is considered equal to one if $m = 1$. By plugging (2.2) in (2.1) we obtain

$$\varepsilon_n = \sum_{k=n+1}^{\infty} \frac{1}{k!} = \sum_{m=1}^{\infty} \frac{1}{(n+m)!} = \frac{1}{(n+1)!} \left(1 + \sum_{m=2}^{\infty} \frac{1}{(n+2) \cdots (n+m)} \right).$$

The final series converges (we skip the proof), which means that we can set

$$a_n = \varepsilon_n, \quad b_n = \frac{1}{(n+1)!} \quad \text{and} \quad C = 1 + \sum_{m=2}^{\infty} \frac{1}{(n+2) \cdots (n+m)}$$

in Definition 2.12. Therefore

$$\varepsilon_n = |e - e_n| = O\left(\frac{1}{(n+1)!} \right).$$

3 Building Blocks of Numerical Methods

This section reviews basic notions and tools that recur when numerical methods are constructed and implemented.

3.1 Taylor Polynomials

The Taylor expansion of a smooth (i.e., sufficiently many times differentiable) function forms the basis of many numerical methods.

Definition 3.1 (Taylor polynomials)**.** Let $f\colon (a,b) \to \mathbb{R}$ be an n times differentiable function at a point $x_0 \in (a,b)$. The nth *Taylor polynomial* of f at the expansion point x_0 is

$$P_n(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n. \tag{3.1}$$

The nth Taylor polynomial is also called the Taylor polynomial *of degree* n.

Recall that $f^{(n)}(x)$ stands for the nth derivative of f at a point x. That is,

$$f^{(n)}(x) = \frac{\mathrm{d}^n}{\mathrm{d}x^n} f(x).$$

Moreover, the zeroth derivative is simply the function itself: $f^{(0)}(x) = f(x)$. Therefore the summation notation can be used to write the Taylor polynomial as

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k.$$

Example 3.2. Let us form the Taylor polynomials of degrees $n \in \{1,3,5\}$ for the function $f(x) = \sin(x)$ at the point $x_0 = 0$. To form these Taylor polynomials we need the values of derivatives of f up to order 5 at $x_0 = 0$. By recalling that the derivative of $\sin(x)$ is $\cos(x)$

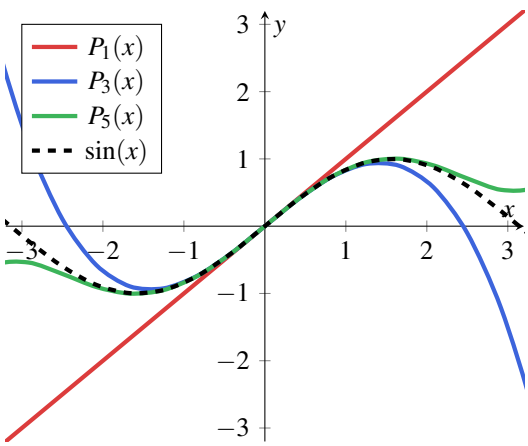


Figure 3.1: The function $f(x) = \sin(x)$ and its Taylor polynomials P_1, P_3 and P_5 of degrees $n \in \{1,3,5\}$. Taylor polynomials are defined in (3.1) and computed for the sinusoid in Example 3.2. Observe that increasing the degree of the Taylor polynomial improves the accuracy of the approximation $f(x) \approx P_n(x)$. The polynomial P_1 is a decent approximation when $|x| < 1$ whereas P_5 works well for $|x| < 2.5$.

and the derivative of $\cos(x)$ is $-\sin(x)$ we get

$$\begin{aligned}
f^{(0)}(x) &= f(x) = \sin(x) &&\implies && f^{(0)}(0) = 0,\\
f'(x) &= \cos(x) &&\implies && f'(0) = 1,\\
f''(x) &= -\sin(x) &&\implies && f''(0) = 0,\\
f^{(3)}(x) &= -\cos(x) &&\implies && f^{(3)}(0) = -1,\\
f^{(4)}(x) &= \sin(x) &&\implies && f^{(4)}(0) = 0,\\
f^{(5)}(x) &= \cos(x) &&\implies && f^{(5)}(0) = 1.
\end{aligned}$$

By plugging these values in (3.1) we obtain the Taylor polynomials

$$\begin{aligned}
P_1(x) &= f(x_0) + f'(x_0)(x - x_0) = f(0) + f'(0)x = x,\\
P_3(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 = x - \frac{1}{6}x^3,\\
P_5(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5\\
&= x - \frac{1}{6}x^3 + \frac{1}{120}x^5.
\end{aligned}$$

These polynomials are depicted in Figure 3.1.

From Example 3.2 and Figure 3.1 we observe that increasing the degree of the Taylor polynomial appears to make it a better approximation to f. Taylor's theorem familiar from basic analysis puts this observation on rigorous footing.

Theorem 3.3 (Taylor's theorem)**.** *Suppose that $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and $n+1$ times differentiable on (a,b). Let $x_0 \in (a,b)$. Then for each $x \in [a,b]$ we have*

$$f(x) - P_n(x) = R_{n+1}(x),$$

where the remainder term R_{n+1} *is given by*

$$R_{n+1}(x) = \frac{f^{(n+1)}(c_x)}{(n+1)!}(x - x_0)^{n+1}$$

for some c_x between x_0 and x.

The number c_x in Taylor's theorem depends on x and n. We can use big-O notation from Section 2.4 to rewrite Taylor's theorem if the $(n+1)$th derivative of f is uniformly bounded.

Corollary 3.4. *Suppose that $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and $n+1$ times differentiable on (a,b). Let $x_0 \in (a,b)$. If there is a constant $L \geq 0$ such that $|f^{(n+1)}(x)| \leq L$ for all $x \in (a,b)$, then*

$$R_{n+1}(x) \leq \frac{L}{(n+1)!}|x - x_0|^{n+1} = O(|x - x_0|^{n+1}) \quad \text{as} \quad x \to x_0. \tag{3.2}$$

Proof. Taylor's theorem and the assumption $|f^{(n+1)}(x)| \leq L$ for all $x \in (a,b)$ yield

$$|R_{n+1}(x)| = \left|\frac{f^{(n+1)}(c_x)}{(n+1)!}(x - x_0)^{n+1}\right| = \frac{|f^{(n+1)}(c_x)|}{(n+1)!}|x - x_0|^{n+1} \leq \frac{L}{(n+1)!}|x - x_0|^{n+1}$$

for all $x \in (a,b)$. Because $x_0 \in (a,b)$, there is $\delta > 0$ such that $(x_0 - \delta, x + \delta) \subset (a,b)$, so that the above inequality holds for all $x \in (x_0 - \delta, x + \delta)$. Therefore we can select $f(x) = R_{n+1}(x)$, $g(x) = |x - x_0|^{n+1}$ and $C = L/(n+1)!$ in Definition 2.11. This proves (3.2). □

By selecting $x = x_0 + h$ (so that $x - x_0 = h$) we can write (3.2) in the alternative form

$$R_{n+1}(x_0 + h) = O(|h|^{n+1}) \quad \text{as} \quad h \to 0$$

that shows more clearly how the remainder term tends to zero as one considers points closer and closer (i.e., $h \to 0$) to the expansion point x_0.

Example 3.5. It is important to understand that remainder term R_{n+1} can be expected to be small only in the vicinity of x_0. For example, consider the Taylor polynomial of degree $n = 2$ at the point $x_0 = 1$ for the function $f(x) = x^4 + 2x^2$. Because f is defined on the whole of $\mathbb{R}$, we can pick any interval that contains x_0 and use that in Corollary 3.4. Let us pick $[a,b] = [0,2]$. Then (as $n + 1 = 3$)

$$|f^{(3)}(x)| = \left| \frac{\mathrm{d}^2}{\mathrm{d}x^2}(4x^3 + 4x) \right| = \left| \frac{\mathrm{d}}{\mathrm{d}x}(12x^2 + 4) \right| = |24x| \leq 48 \quad \text{for all} \quad x \in [0,2].$$

Therefore we can apply Corollary 3.4 with $n = 2$, $x_0 = 1$ and $L = 48$ to conclude that

$$f(x) - P_2(x) = R_3(x) = O(|x - 1|^3) \quad \text{as} \quad x \to 1,$$

which means that the error of the degree two Taylor polynomial P_2 is *cubic* as $x \to x_0 = 1$. However, something completely different happens as one moves away from $x_0 = 1$. Because $n = 2$, we see from the definition of the Taylor polynomial in (3.1) that P_2 is a polynomial of degree two, meaning that $P_2(x) = \alpha x^2 + \beta x + \gamma$ for certain $\alpha, \beta, \gamma \in \mathbb{R}$. Since $f(x) = x^4 + 2x^2$, this means that

$$f(x) - P_2(x) = x^4 + (2 - \alpha)x^2 - \beta x - \gamma,$$

which is a polynomial of degree 4 with a positive leading coefficient (i.e., the term of degree 4 has coefficient 1). This means that

$$f(x) - P_2(x) \to \infty \quad \text{as} \quad x \to \infty \text{ or } x \to -\infty.$$

Consequently, the error can be made arbitrarily large by looking at x that is sufficiently far from $x_0 = 1$.

3.2 Linearisation

The case $n = 1$ in Taylor expansion (Definition 3.1) is one of the main tools to construct numerical methods.

Definition 3.6 (Linear approximation)**.** Let $f\colon (a,b) \to \mathbb{R}$ be a once differentiable function at a point $x_0 \in (a,b)$. The *linear approximation* to f at x_0 is the linear polynomial

$$L(x) = P_1(x) = f(x_0) + f'(x_0)(x - x_0). \tag{3.3}$$

The process of computing L and using it to approximate f is called *linearisation*.

Linear approximation is preferred over higher degree Taylor polynomials because it is so easy to compute: only $f(x_0)$ and $f'(x_0)$ are required. In applications it is often cumbersome or even impossible to obtain derivatives of orders higher than this. Moreover, linearisation

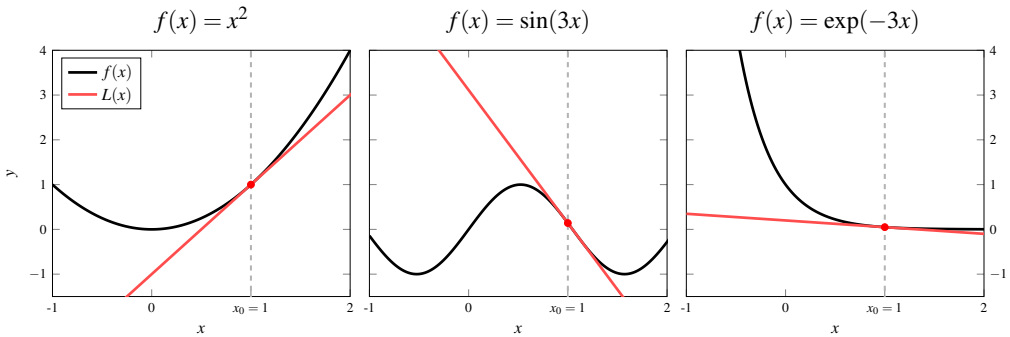


Figure 3.2: Linear approximations (red) to the three functions $f(x) = x^2$, $f(x) = \sin(3x)$ and $f(x) = \exp(-3x)$ at the point $x_0 = 1$. Observe that the linear approximations coincide with the *tangent lines* at $x_0 = 1$. That is, the linear approximation is the line whose slope coincides with the derivative of f at x_0.

easily generalises for multivariate functions $f\colon \mathbb{R}^d \to \mathbb{R}$ via the *gradient* and results in numerical methods that are straightforward to implement and understand.

Example 3.7. Let us form linear approximations to the functions

$$f_1(x) = -x^2, \quad f_2(x) = \sin(\tfrac{\pi}{2}x) \quad \text{and} \quad f_3(x) = e^x - x^2$$

at the point $x_0 = 1$. For f_1 we get

$$f_1(1) = -1 \text{ and } f_1'(1) = -2 \quad \Longrightarrow \quad L(x) = f(1) + f'(1)(x-1) = -2x + 1.$$

For f_2 we get [recall from trigonometry that $\sin(\frac{\pi}{2}) = 1$]

$$f_2(1) = 1 \text{ and } f_2'(1) = 0 \quad \Longrightarrow \quad L(x) = f(1) + f'(1)(x-1) = 1.$$

For f_3 we get

$$f_3(1) = e - 1 \text{ and } f_3'(1) = e - 2 \quad \Longrightarrow \quad L(x) = f(1) + f'(1)(x-1) = (e-2)x + 1.$$

Figure 3.2 plots linearisations at $x_0 = 1$ for three similar functions.

Example 3.7 and Figure 3.2 show that linearisation corresponds to drawing a line that (a) passes through $(x_0, f(x_0))$ and (b) whose slope equals the derivative $f'(x_0)$.

Theorem 3.8. *The linear approximation is the unique line that passes through $(x_0, f(x_0))$ and whose slope equals $f'(x_0)$.*

Proof. Every line is given by the equation $y = \alpha x + \beta$ for some $\alpha, \beta \in \mathbb{R}$, where α is the slope of the line. That a line passes through $(x_0, f(x_0))$ means that $f(x_0) = \alpha x_0 + \beta$. That the slope of the line equals $f'(x_0)$ means that $\alpha = f'(x_0)$. Therefore

$$f(x_0) = \alpha x_0 + \beta = f'(x_0)x_0 + \beta \quad \Longrightarrow \quad \beta = f(x_0) - f'(x_0)x_0.$$

Therefore any line specified in the theorem has $\alpha = f'(x_0)$ and $\beta = f(x_0) - f'(x_0)x_0$. That is, there is only one line that passes through $(x_0, f(x_0))$ and whose slope equals $f'(x_0)$. The equation

for this line is

$$y = \alpha x + \beta = f'(x_0)x + f(x_0) - f'(x_0)x_0 = f(x_0) + f'(x_0)(x - x_0),$$

which is exactly the linear approximation in (3.3). □

From Figure 3.2 it is apparent that linearisation gives an approximation that is good for x close to x_0 but that can degrade quickly in quality when one moves away from x_0. The following theorem states that the error of linear approximation has *quadratic* order as $x \to x_0$ if f is twice differentiable.

Theorem 3.9. *Suppose that $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and twice differentiable on (a,b). Let $x_0 \in (a,b)$. If there is a constant $L \geq 0$ such that $|f''(x)| \leq L$ for all $x \in (a,b)$, then*

$$f(x) - L(x) = O(|x - x_0|^2) \quad \text{as} \quad x \to x_0.$$

Proof. Because $L(x) = P_1(x)$, the claim follows by setting $n = 1$ in Corollary 3.4. □

3.3 Array programming

Array programming, or *vectorisation*, refers to a style of programming where mathematical operations are applied to an array of values at once, rather than to each value individually (i.e., via a loop). Array programming tends to be significantly more *concise* and *easier to read* than an equivalent construction based on loops. In many high-level programming languages, such as Python and MATLAB, array programming is also *significantly faster* than loops and should therefore be used whenever possible. Suppose that we want to square (i.e., compute x^2) for ten million x between 0 and 1 with Python. For array programming we use the Python library `numpy`.

```
>>> import numpy as np
>>> import time
>>> # an array of ten million numbers between 0 and 1
>>> a = np.linspace(start=0, stop=1, num=int(1e7))
>>> # use loop to square each number
>>> start1 = time.perf_counter()
>>> b1 = [x**2 for x in a]
>>> end1 = time.perf_counter()
>>> # square each number using array programming
>>> start2 = time.perf_counter()
>>> b2 = a**2
>>> end2 = time.perf_counter()
>>> # print elapsed times
>>> print(end1 - start1)
0.5299568176269531
>>> print(end2 - start2)
0.007016181945800781
```

The elapsed times depend on hardware and vary slightly from one run to the next. But what remains constant is that the second approach based on array programming is much faster (in this case by approximately two orders of magnitude).

Example 3.10. Monte Carlo integration is a straightforward technique to approximate the integral of a function. Suppose that we want to approximate the integral $\int_0^1 f(x)\,\mathrm{d}x$ for some integrand function $f\colon [0,1] \to \mathbb{R}$. In Monte Carlo integration we draw n numbers $x_1, \ldots, x_n$ randomly on $[0,1]$ and approximate the integral as the average of integrand evaluations at

these points:

$$\int_0^1 f(x)\,\mathrm{d}x \approx \frac{1}{n}\sum_{i=1}^{n} f(x_i). \tag{3.4}$$

In programming languages that support array programming, no loops are needed to compute the Monte Carlo approximation in (3.4).

In Python the Monte Carlo approximation ($n = 200$) to the integral of $f(x) = x^{3.4} + \sin(x)$ could be computed as follows:

```
>>> import numpy as np
>>> rng = np.random.default_rng()
>>> # draw n = 200 random numbers on [0, 1]
>>> n = 200
>>> samples = rng.random(size=n)
>>> # evaluate f at these point and compute average
>>> evals = np.power(samples, 3.4) + np.sin(samples)
>>> mc_approx = np.sum(evals) / n
>>> print(mc_approx)
0.692222065735476
```

The final result will vary from one run to the next because the 200 samples are random. In MATLAB this could be done as follows:

```
>>> # draw n = 200 random numbers on [0, 1]
>>> n = 200;
>>> samples = rand(n, 1);
>>> # evaluate f at these point and compute average
>>> evals = samples.^(3.4) + sin(samples);
>>> mc_approx = sum(evals) / n
mc_approx =

    0.6686
```

Here we used MATLAB's elementwise power signified by dot to compute $x^{3.4}$ for every sample x.

4 Numerical Integration

Recall that the *antiderivative* (or *indefinite integral*) of a function f is a function F such that

$$F'(x) = f(x).$$

Let $a < b$. The *definite integral* of f from a to b represents the area between the graph of f and the x-axis. The definite integral can be expressed in terms of the antiderivative as

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a)$$

by the fundamental theorem of calculus. Differentiation is easy. For example, if one is familiar with the chain rule and knows how to differentiate exponentials, polynomials and trigonometric functions, there is no difficulty in differentiating the function $f(x) = \exp(-x^2 + \sin(x))$. In contrast, integration tends to be difficult or outright impossible. The antiderivative of $f(x) = \exp(-x^2 + \sin(x))$ does not admit a closed form expression in terms of the familiar elementary functions[2]. As a result, numerical methods are necessary to compute most definite integrals. To compute an antiderivative one must use *symbolic computation* that is not discussed on this course. Note that a function whose antiderivative is not available in terms of elementary functions can nevertheless have a simple definite integral. For example,

$$\int_{-\infty}^{\infty} \exp(-x^2)\,\mathrm{d}x = \sqrt{\pi}.$$

It is useful to think of definite integrals as areas and *numerical integration* methods as methods that estimate areas. Such methods are called quadrature rules (or simply rules). The terminology comes from the Latin word *quadratus* for "square" (particularly apt for the midpoint rule of Section 4.1).

Definition 4.1 (Quadrature rule)**.** Let $n \in \mathbb{N}$. A *quadrature rule* with *weights* $w_1, \ldots, w_n \in \mathbb{R}$ and *nodes* $t_1, \ldots, t_n \in [a,b]$ is the following integral approximation:

$$\int_a^b f(x)\,\mathrm{d}x \approx \sum_{i=1}^n w_i f(t_i). \tag{4.1}$$

How the weights and nodes are selected determines how good a quadrature rule is. This chapter discusses three quadrature rules, the midpoint rule, the trapezoidal rule, and Simpson's rule.

4.1 Midpoint Rule

Let $x_i = a + (b-a)i/n$ for $i \in \{0, 1, \ldots, n\}$ be equispaced points $[a,b]$. Recall from the course on integral calculus that the integral of a sufficiently regular (i.e., Riemann integrable) function is obtained as the limit of a Riemann sum:

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \sum_{i=1}^n (x_i - x_{i-1}) f(t_i) = \lim_{n\to\infty} \frac{b-a}{n} \sum_{i=1}^n f(t_i), \tag{4.2}$$

where $t_i \in [x_{i-1}, x_i]$. Each term in the sums represents *the area of a rectangle* with base width $(x_i - x_{i-1}) = (b-a)/n$ and height $f(t_i)$. The heights of the rectangles can be determined using

[2] A function is typically considered *elementary* if it is composed out of finitely many constants, polynomials, exp, log, trigonometric and hyperbolic functions (sin, cos, tan, sinh, cosh, tanh and so on) and inverses of trigonometric and hyperbolic functions.

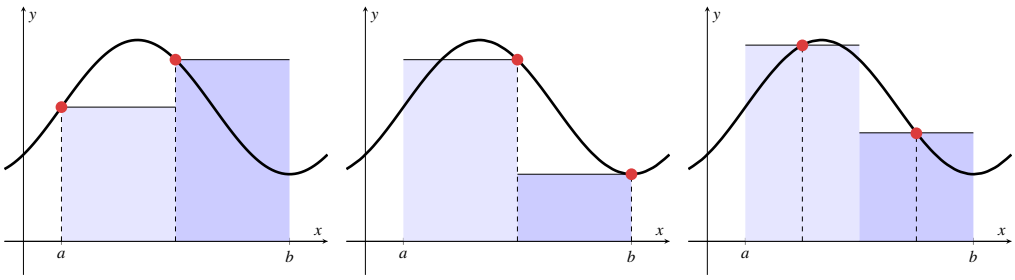


Figure 4.1: Approximations of the definite integral $\int_a^b f(x)\,\mathrm{d}x$ using two rectangles whose heights are calculated at the starting point (*left*), at the endpoint (*middle*), and at the midpoint (*right*) of each subinterval. The midpoint rule in Definition 4.2 corresponds to the rightmost figure.

any value of the function in the subintervals $[x_{i-1}, x_i]$. The simplest approach is to calculate the height at an endpoint of the interval, which translates to replacing $f(t_i)$ with $f(x_{i-1})$ or $f(x_i)$ in (4.2). By replacing $f(t_i)$ with $f(\frac{1}{2}(x_i + x_{i-1}))$, where $\frac{1}{2}(x_i + x_{i-1})$ is the *midpoint* of the subinterval $[x_{i-1}, x_i]$, and the limit in (4.2) with a finite n yields the midpoint rule, which is conceptually similar to the difference quotient approximation from Example 2.5 in that an approximation is constructed by "truncating a limiting process". The three approaches are illustrated in Figure 4.1.

Definition 4.2 (Midpoint rule)**.** Let $x_i = a + (b-a)i/n$ for $i \in \{0, 1, \ldots, n\}$. The *midpoint rule* (or *rectangle rule*) is the following integral approximation:

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{n} \sum_{i=1}^{n} f\left(\frac{x_i + x_{i-1}}{2}\right) = h \sum_{i=1}^{n} f\left(\frac{x_i + x_{i-1}}{2}\right), \tag{4.3}$$

where $h = x_{i-1} - x_i = (b-a)/n$ is the length of each subinterval $[x_{i-1}, x_i]$.

By comparing (4.3) to (4.1) we see that the midpoint rule is a quadrature rule with the nodes $t_i = \frac{1}{2}(x_i + x_{i-1})$ and uniform weights $w_i = (b-a)/n$. Figure 4.2 contains a graphical illustration of the midpoint rule with two (i.e., $n = 2$) subintervals.

Remark 4.3. We could have used non-equispaced points $a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b$ to construct the midpoint rule. The resulting rule is

$$\int_a^b f(x)\,\mathrm{d}x \approx \sum_{i=1}^{n} (x_i - x_{i-1}) f\left(\frac{x_i + x_{i-1}}{2}\right).$$

All quadrature rules defined in this section can be constructed for non-equispaced points. However, for simplicity we always use equispaced points.

Example 4.4. Let us use the midpoint rule with four subintervals (i.e., $n = 4$) to approximate the integral $\int_0^4 x^3\,\mathrm{d}x = 4^4/4 = 64$. To compute the right-hand side of (4.3) we need (a) the length of the subintervals, $h = (b-a)/n$, and (b) the midpoints $\frac{1}{2}(x_i + x_{i-1})$. Because $n = 4$, $b = 4$ and $a = 0$, we have

$$\frac{b-a}{n} = \frac{4-0}{4} = 1 \tag{4.4}$$

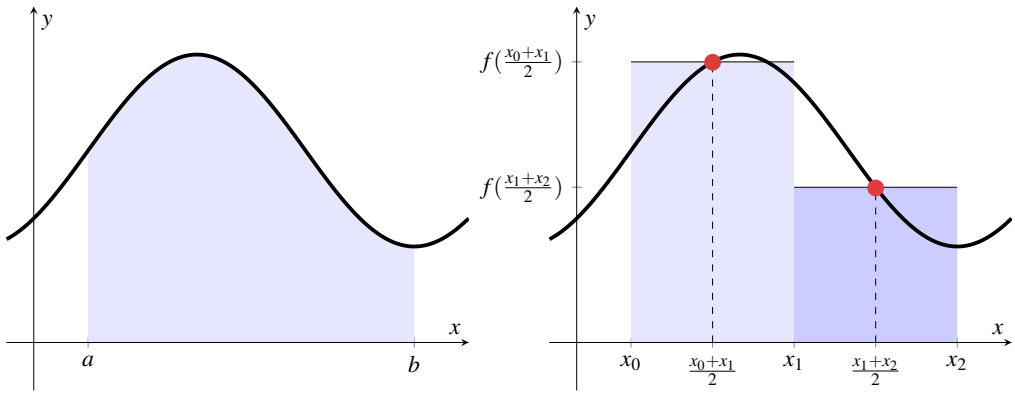


Figure 4.2: *Left*: The area corresponding to the definite integral $\int_a^b f(x)\,\mathrm{d}x$. *Right*: Its approximation using the midpoint rule with two (i.e., $n = 2$) subintervals.

and

$$x_i = a + \frac{(b-a)i}{n} = 0 + \frac{(4-0)i}{4} = i.$$

Therefore the midpoints are

$$\frac{x_i + x_{i-1}}{2} = \frac{i + (i-1)}{2} = i - \frac{1}{2}. \tag{4.5}$$

Inserting (4.4) and (4.5) in (4.3) gives

$$\begin{aligned}\int_0^4 x^3\,\mathrm{d}x \approx 1 \cdot \sum_{i=1}^{4} f\left(i - \frac{1}{2}\right) &= 1 \cdot f\left(\frac{1}{2}\right) + 1 \cdot f\left(\frac{3}{2}\right) + 1 \cdot f\left(\frac{5}{2}\right) + 1 \cdot f\left(\frac{7}{2}\right) \\ &= \left(\frac{1}{2}\right)^3 + \left(\frac{3}{2}\right)^3 + \left(\frac{5}{2}\right)^3 + \left(\frac{7}{2}\right)^3 \\ &= 62.\end{aligned}$$

Therefore the relative error (recall Definition 2.8) of the midpoint rule is

$$\left|\frac{\int_0^4 x^3\,\mathrm{d}x - 62}{\int_0^4 x^3\,\mathrm{d}x}\right| = \left|\frac{64-62}{64}\right| = \frac{1}{32} = 3.125\ \%.$$

A quadrature rule is said to be *exact* for a function f if the integral approximation coincides with the true integral.

Theorem 4.5. *The midpoint rule is exact for every polynomial of degree at most one.*

Proof. Every polynomial of degree at most one takes the form $f(x) = \alpha + \beta x$ for some constants $\alpha, \beta \in \mathbb{R}$. We want to show that

$$\int_a^b f(x)\,\mathrm{d}x = \frac{b-a}{n} \sum_{i=1}^{n} f\left(\frac{x_i + x_{i-1}}{2}\right)$$

for such f. The integral is

$$\int_a^b f(x)\,\mathrm{d}x = \int_a^b (\alpha + \beta x)\,\mathrm{d}x = (b-a)\alpha + \frac{(b^2 - a^2)\beta}{2}. \tag{4.6}$$

Because $x_i = a + (b-a)i/n$, the midpoints are

$$\frac{x_i + x_{i-1}}{2} = a + \frac{(b-a)i/n + (b-a)(i-1)/n}{2} = a + \frac{(b-a)i}{n} - \frac{b-a}{2n}.$$

Therefore the midpoint rule is

$$\begin{aligned}\frac{b-a}{n}\sum_{i=1}^{n} f\left(\frac{x_i + x_{i-1}}{2}\right) &= \frac{b-a}{n}\left(\alpha n + \beta \sum_{i=1}^{n}\left[a + \frac{(b-a)i}{n} - \frac{b-a}{2n}\right]\right) \\ &= (b-a)\alpha + \frac{b-a}{n}\beta\left(na - \frac{b-a}{2} + \frac{b-a}{n}\sum_{i=1}^{n} i\right) \\ &= (b-a)\alpha + (b-a)a\beta - \frac{(b-a)^2\beta}{2n} + \frac{(b-a)^2\beta}{n^2}\sum_{i=1}^{n} i.\end{aligned}$$

Recall then the arithmetic sum formula $\sum_{i=1}^{n} i = n(n+1)/2$, which yields

$$\begin{aligned}\frac{b-a}{n}\sum_{i=1}^{n} f\left(\frac{x_i + x_{i-1}}{2}\right) &= (b-a)\alpha + \frac{(b-a)a\beta}{n} - \frac{(b-a)^2\beta}{2n} + \frac{(b-a)^2\beta}{n^2}\cdot\frac{n(n+1)}{2} \\ &= (b-a)\alpha + (b-a)a\beta - \frac{(b-a)^2\beta}{2n} + (n+1)\frac{(b-a)^2\beta}{2n} \\ &= (b-a)\alpha + (b-a)a\beta + \frac{(b-a)^2\beta}{2} \\ &= (b-a)\alpha + (b-a)a\beta + \frac{(b^2 - 2ab + a^2)\beta}{2} \\ &= (b-a)\alpha + \frac{(2(b-a)a + b^2 - 2ab + a^2)\beta}{2} \\ &= (b-a)\alpha + \frac{(b^2 - a^2)\beta}{2}.\end{aligned}$$

This equals (4.6). Thus the midpoint rule is exact for all functions of the form $f(x) = \alpha + \beta x$. □

The midpoint rule is *not* exact for all polynomials of degree two. To show this it suffices to come up with one polynomial of degree two for which (4.3) is not an equality. Select $f(x) = x^2$ and consider the case $a = 0$ for simplicity. Then

$$\int_a^b f(x)\,\mathrm{d}x = \int_0^b x^2\,\mathrm{d}x = \frac{b^3}{3} \tag{4.7}$$

and (as in the proof of Theorem 4.5)

$$\begin{aligned}\frac{b-a}{n}\sum_{i=1}^{n} f\left(\frac{x_i + x_{i-1}}{2}\right) = \frac{b-a}{n}\sum_{i=1}^{n}\left(a + \frac{(b-a)i}{n} - \frac{b-a}{2n}\right)^2 &= \frac{b}{n}\sum_{i=1}^{n}\left(\frac{bi}{n} - \frac{b}{2n}\right)^2 \\ &= \frac{b^3}{n^3}\sum_{i=1}^{n}\left(i - \frac{1}{2}\right)^2 \\ &= \frac{b^3}{n^3}\sum_{i=1}^{n}\left(i^2 - i + \frac{1}{4}\right).\end{aligned}$$

Recall the summation formulae

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \quad \text{and} \quad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

which yield

$$\begin{aligned}\frac{b^3}{n^3}\sum_{i=1}^{n}\left(i^2-i+\frac{1}{4}\right) &= \frac{b^3}{n^3}\left(\frac{n(n+1)(2n+1)}{6}-\frac{n(n+1)}{2}+\frac{n}{4}\right)\\ &= \frac{b^3}{n^3}\cdot\frac{n}{12}[2n(n+1)(2n+1)-6n(n+1)+3n]\\ &= \frac{b^3(4n^3+6n^2+2n-6n^2-6n+3n)}{12n^2}\\ &= \frac{b^3(4n^2-1)}{12n^2}.\end{aligned}$$

Therefore

$$\frac{b-a}{n}\sum_{i=1}^{n} f\left(\frac{x_i+x_{i-1}}{2}\right) = \frac{b^3}{3}\left(1-\frac{1}{4n^2}\right),$$

which is not equal to the true integral in (4.7) for any $n \in \mathbb{N}$ because $1-1/(4n^2) \neq 1$.

Theorem 4.6. *Suppose that $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and twice differentiable on (a,b). If there is a constant $L \geq 0$ such that $|f''(x)| \leq L$ for all $x \in (a,b)$, then the midpoint rule satisfies*

$$\left|\int_a^b f(x)\,\mathrm{d}x - \sum_{i=1}^{n} h\cdot f\left(\frac{x_i+x_{i-1}}{2}\right)\right| \leq \frac{(b-a)^3L}{24n^2} = \frac{(b-a)L}{24}h^2,$$

where $h = x_i - x_{i-1} = (b-a)/n$ is the length of each subinterval $[x_{i-1}, x_i]$.

Proof. By breaking the integration interval into n subintervals we may write the error of the midpoint rule as

$$\int_a^b f(x)\,\mathrm{d}x - \sum_{i=1}^{n} h\cdot f\left(\frac{x_i+x_{i-1}}{2}\right) = \sum_{i=1}^{n}\left(\int_{x_{i-1}}^{x_i} f(x)\,\mathrm{d}x - h\cdot f\left(\frac{x_i+x_{i-1}}{2}\right)\right). \tag{4.8}$$

Let $t_i = \frac{1}{2}(x_i+x_{i-1})$ denote the ith midpoint. We have

$$\begin{aligned}\int_{x_{i-1}}^{x_i}[f(t_i)+f'(t_i)(x-t_i)]\,\mathrm{d}x &= f(t_i)(x_i-x_{i-1}) + \frac{f'(t_i)}{2}(x_i^2-x_{i-1}^2) - f'(t_i)t_i(x_i-x_{i-1})\\ &= (x_i-x_{i-1})\left[f(t_i)+\frac{f'(t_i)}{2}(x_i+x_{i-1}) - f'(t_i)t_i\right]\\ &= (x_i-x_{i-1})[f(t_i)-f'(t_i)t_i+f'(t_i)t_i]\\ &= (x_i-x_{i-1})f(t_i)\\ &= h\cdot f\left(\frac{x_i+x_{i-1}}{2}\right).\end{aligned}$$

Therefore (4.8) becomes

$$\int_a^b f(x)\,\mathrm{d}x - \sum_{i=1}^{n} h\cdot f\left(\frac{x_i+x_{i-1}}{2}\right) = \sum_{i=1}^{n}\int_{x_{i-1}}^{x_i}[f(x)-f(t_i)-f'(t_i)(x-t_i)]\,\mathrm{d}x.$$

Observe that $f(x)-f(t_i)-f'(t_i)(x-t_i) = f(x)-P_{1,t_i}(x) = R_{2,t_i}(x)$, where P_{1,t_i} and R_{2,t_i} are the Taylor polynomial of degree one and its remainder term from Section 3.1 with $x_0 = t_i$. From Taylor's theorem (Theorem 3.3) and Corollary 3.4 with $n+1=2$ it then follows that

$$\begin{aligned}\left|\int_a^b f(x)\,\mathrm{d}x - \sum_{i=1}^{n} h\cdot f\left(\frac{x_i+x_{i-1}}{2}\right)\right| = \left|\sum_{i=1}^{n}\int_{x_{i-1}}^{x_i} R_{2,t_i}(x)\,\mathrm{d}x\right| &\leq \sum_{i=1}^{n}\int_{x_{i-1}}^{x_i}\frac{L}{2!}(x-t_i)^2\,\mathrm{d}x\\ &= \frac{L}{2}\sum_{i=1}^{n}\int_{-h/2}^{h/2} y^2\,\mathrm{d}y,\end{aligned}$$

where the last equation used the change of variables $y = x - t_i$ and the fact that t_i is the midpoint of $[x_{i-1}, x_i]$. Therefore

$$\begin{aligned}\left|\int_a^b f(x)\,\mathrm{d}x - \sum_{i=1}^n h \cdot f\left(\frac{x_i + x_{i-1}}{2}\right)\right| = \frac{L}{2}\sum_{i=1}^n \int_{-h/2}^{h/2} y^2\,\mathrm{d}y &= \frac{L}{2}\sum_{i=1}^n \frac{1}{3}\cdot[(h/2)^3 - (-h/2)^3] \\ &= \frac{L}{24}\sum_{i=1}^n h^3 \\ &= \frac{L}{24} n h^3 \\ &= \frac{L}{24} n \left(\frac{b-a}{n}\right)^3 \\ &= \frac{(b-a)^3 L}{24 n^2}. \end{aligned}$$

□

With big-O notation from Section 2.4 we can express Theorem 4.6 as

$$\left|\int_a^b f(x)\,\mathrm{d}x - \frac{b-a}{n}\sum_{i=1}^n f\left(\frac{x_i + x_{i-1}}{2}\right)\right| = O(n^{-2}) \quad \text{as} \quad n \to \infty.$$

This gives information about the order of the integration error as n, the number of subintervals grows. Equivalently, we can write

$$\left|\int_a^b f(x)\,\mathrm{d}x - h\sum_{i=1}^n f\left(\frac{x_i + x_{i-1}}{2}\right)\right| = O(h^2) \quad \text{as} \quad h \to 0,$$

which describes the error as $h = (b-a)/n$, the length of each subinterval, tends to zero.

4.2 Trapezoidal Rule

The midpoint rule used rectangles to approximate an integral. The trapezoidal rule uses *trapezoids*, which generalise rectangles. Both pairs of sides of a rectangle are parallel; a trapezoid has at least one pair of parallel sides. Whereas a rectangle with base length h and height a has area $a \cdot h$, a trapezoid with base length h and heights b_1 and b_1 has area $\frac{1}{2}(b_1 + b_2)\cdot h$ (see Figure 4.3). By replacing rectangles whose heights are computed from the value of f at the midpoints of the subintervals, $\frac{1}{2}(x_i + x_{i-1})$, with trapezoids whose side heights are given by the values of f at the endpoints of the subintervals, x_{i-1} and x_i, we obtain the trapezoidal rule.

Definition 4.7 (Trapezoidal rule)**.** Let $x_i = a + (b-a)i/n$ for $i \in \{0, 1, \ldots, n\}$. The *trapezoidal rule* is the following integral approximation:

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{n}\sum_{i=1}^n \frac{f(x_i) + f(x_{i-1})}{2} = h\sum_{i=1}^n \frac{f(x_i) + f(x_{i-1})}{2}, \tag{4.9}$$

where $h = x_i - x_{i-1} = (b-a)/n$ is the length of each subinterval $[x_{i-1}, x_i]$.

By noting that $f(x_0)$ and $f(x_n)$ appear once and all other function evaluations twice in (4.9), we can write the trapezoidal rule as

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{h}{2}[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)]. \tag{4.10}$$

Therefore the trapezoidal rule is a quadrature rule [recall (4.1)] with nodes $t_i = x_i$ for $i \in \{0, \ldots, n\}$ and weights $w_0 = w_n = \frac{1}{2}h$ and $w_1 = \cdots = w_{n-1} = h$. The trapezoidal rule is illustrated in Figure 4.4.

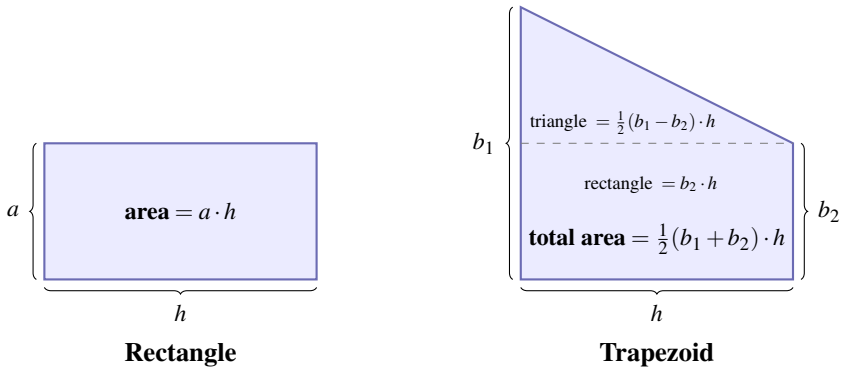


Figure 4.3: Areas of a rectangle and a trapezoid with two right angles.

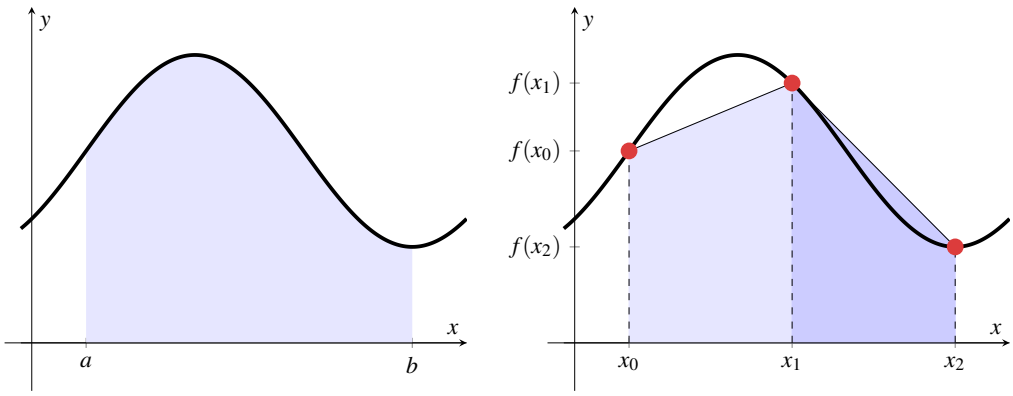


Figure 4.4: *Left*: The area corresponding to the definite integral $\int_a^b f(x)\,\mathrm{d}x$. *Right*: Its approximation using the trapezoidal rule with two (i.e., $n = 2$) subintervals.

Example 4.8. Let us use the trapezoidal rule with four subintervals (i.e., $n = 4$) to approximate the integral $\int_0^4 x^3\,\mathrm{d}x = 4^4/4 = 64$. To compute the right-hand side of (4.9) we need (a) the length of the subintervals, $h = (b-a)/n$, and (b) the points x_i. Because $n = 4$, $b = 4$ and $a = 0$, we have

$$h = \frac{b-a}{n} = \frac{4-0}{4} = 1 \tag{4.11}$$

and

$$x_i = a + \frac{(b-a)i}{n} = 0 + \frac{(4-0)i}{4} = i. \tag{4.12}$$

Inserting (4.11) and (4.12) in (4.10) gives

$$\begin{aligned}\int_0^4 x^3\,\mathrm{d}x &\approx \frac{h}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)] \\ &= \frac{1}{2}[f(0) + 2f(1) + 2f(2) + 2f(3) + f(4)] \\ &= \frac{1}{2}[0^3 + 2\cdot 1^3 + 2\cdot 2^3 + 2\cdot 3^3 + 4^3] \\ &= 68.\end{aligned}$$

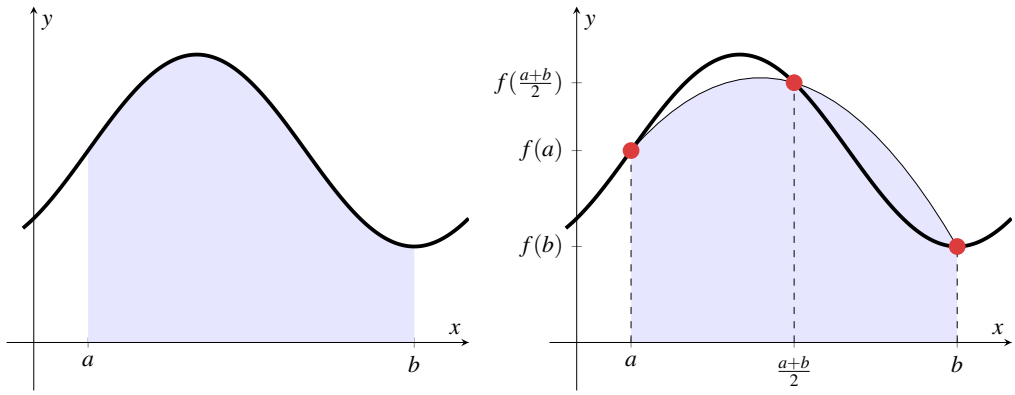


Figure 4.5: *Left*: The area corresponding to the definite integral $\int_a^b f(x)\,\mathrm{d}x$. *Right*: Its approximation using Simpson's rule in (4.14) with three points.

Therefore the relative error (recall Definition 2.8) of the trapezoidal rule is

$$\left|\frac{\int_0^4 x^3\,\mathrm{d}x - 68}{\int_0^4 x^3\,\mathrm{d}x}\right| = \left|\frac{64-68}{64}\right| = \frac{1}{16} = 6.25\ \%.$$

The trapezoidal rule behaves in a manner similar to the midpoint rule. You should compare the following theorems to Theorems 4.5 and 4.6.

Theorem 4.9. *The trapezoidal rule is exact for all polynomials of degree at most one.*

Proof. Similar to the proof of Theorem 4.5 and is left as an exercise. □

Theorem 4.10. *Suppose that $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and twice differentiable on (a,b). If there is a constant $L \geq 0$ such that $|f''(x)| \leq L$ for all $x \in (a,b)$, then the trapezoidal rule satisfies*

$$\left|\int_a^b f(x)\,\mathrm{d}x - \sum_{i=1}^n h \cdot \frac{f(x_i)+f(x_{i-1})}{2}\right| \leq \frac{(b-a)^3 L}{12n^2} = \frac{(b-a)L}{12}h^2,$$

where $h = x_i - x_{i-1} = (b-a)/n$ is the length of each subinterval $[x_{i-1}, x_i]$.

Proof. We omit the proof, which is similar to that of Theorem 4.6. □

4.3 Simpson's Rule

The midpoint rule uses rectangles and the trapezoidal rule trapezoids to approximate an integral. The midpoint rule is constructed using a single value of f on each subinterval $[x_{i-1}, x_i]$ (i.e., the value of f at the midpoint) while the trapezoidal rule uses two values (the values of f at the endpoints). Simpson's rule uses three values on each subinterval: the endpoints and the midpoint. Three values of f at given points determine a unique quadratic polynomial (i.e., a quadratic *interpolant*) that coincides with f at those points and whose integral (i.e., the area under the graph of the polynomial) can be used to approximate the integral of f.

Suppose that we have only one subinterval, which is simply $[a,b]$. The midpoint is $(a+b)/2$. The quadratic polynomial $\alpha+\beta x+\gamma x^2$ that coincides with f at the endpoints a and b and at the midpoint $c=(a+b)/2$ must satisfy the three equations

$$\begin{aligned}\alpha+\beta c+\gamma c^2 &= f(c),\\ \alpha+\beta a+\gamma a^2 &= f(a),\\ \alpha+\beta b+\gamma b^2 &= f(b).\end{aligned} \tag{4.13}$$

Because we have three equations for three unknowns (α, β and γ), it is straightforward to solve (the computations are omitted) the unique coefficients for which these three equations hold. Because

$$\int_a^b (\alpha+\beta x+\gamma x^2)\,\mathrm{d}x = (b-a)\alpha+\frac{(b^2-a^2)\beta}{2}+\frac{(b^3-a^3)\gamma}{3}$$

for any $\alpha,\beta,\gamma\in\mathbb{R}$, we can use the coefficients just computed to obtain the integral approximation

$$\int_a^b f(x)\,\mathrm{d}x \approx (b-a)\alpha+\frac{(b^2-a^2)\beta}{2}+\frac{(b^3-a^3)\gamma}{3}.$$

For the coefficients that solve the equations in (4.13) the right-hand side ends up simplifying considerably, so that we finally obtain

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{b-a}{6}\left[f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right]. \tag{4.14}$$

This approximation is illustrated in Figure 4.5. To construct an approximation that uses more than three evaluations of f we can stitch together multiple approximations of this type. Namely, let n be even and $x_i=a+hi$ for $h=(b-a)/n$. Then

$$\int_a^b f(x)\,\mathrm{d}x = \sum_{k=1}^{\frac{n}{2}}\int_{x_{2k-2}}^{x_{2k}} f(x)\,\mathrm{d}x \approx \sum_{k=1}^{\frac{n}{2}}\frac{x_{2k}-x_{2k-2}}{6}[f(x_{2k-2})+4f(x_{2k-1})+f(x_{2k})],$$

where we applied (4.14) with $a=x_{2k-2}$ and $b=x_{2k}$ to each of the integrals and used the fact that in this case $\frac{1}{2}(a+b)=x_{2k-1}$ (i.e., now *two* subintervals are used for *one* quadratic interpolation). Grouping terms corresponding to odd and even indices gives the full Simpson's rule.

Definition 4.11 (Simpson's rule)**.** Let $n\in\mathbb{N}$ be even and $x_i=a+(b-a)i/n$ for $i\in\{0,1,\ldots,n\}$. *Simpson's rule* is the following integral approximation:

$$\int_a^b f(x)\,\mathrm{d}x \approx \frac{h}{3}[f(x_0)+4f(x_1)+2f(x_2)+\cdots+2f(x_{n-2})+4f(x_{n-1})+f(x_n)], \tag{4.15}$$

where $h=x_i-x_{i-1}=(b-a)/n$ is the length of each subinterval $[x_{i-1},x_i]$.

We can use summation notation to express Simpson's rule as

$$\int_a^b f(x)\,\mathrm{d}x = \frac{h}{3}\left(f(x_0)+f(x_n)+4\sum_{i=1}^{\frac{n}{2}} f(x_{2i-1})+2\sum_{i=1}^{\frac{n}{2}-1} f(x_{2i})\right).$$

From this we see that the endpoints of the interval $[a,b]$ receive weights $\frac{1}{3}h$, odd points (x_{2i-1}) receive weights $\frac{4}{3}h$ and even points (x_{2i}) receive weights $\frac{2}{3}h$.

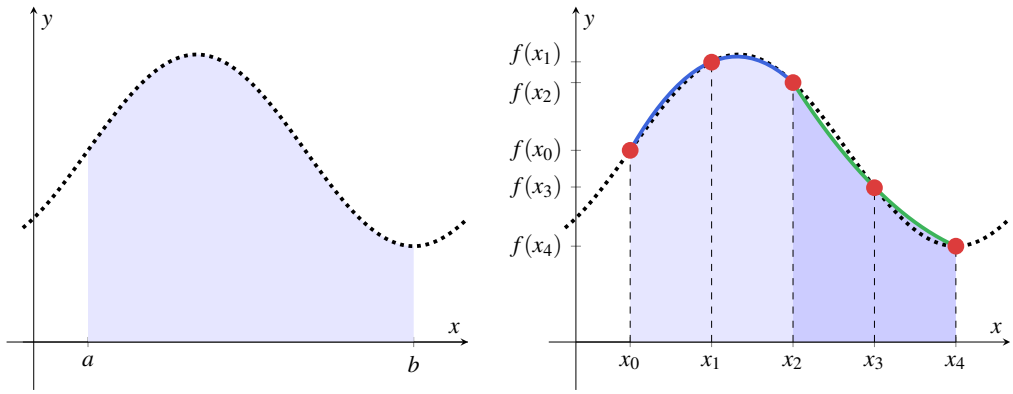


Figure 4.6: *Left*: The area corresponding to the definite integral $\int_a^b f(x)\,\mathrm{d}x$. *Right*: Its approximation using Simpson's rule in (4.12) with $n = 4$. Note that one quadratic polynomial is fitted on the subintervals $[x_0, x_1] \cup [x_1, x_2] = [x_0, x_2]$ and another on the subintervals $[x_2, x_3] \cup [x_3, x_4] = [x_2, x_4]$.

Example 4.12. Let us use Simpson's rule with $n = 4$ to approximate the integral $\int_0^4 x^3\,\mathrm{d}x = 4^4/4 = 64$. To compute the right-hand side of (4.9) we need (a) the length of the subintervals, $h = (b-a)/n$, and (b) the points x_i. Because $n = 4$, $b = 4$ and $a = 0$, we have

$$h = \frac{b-a}{n} = \frac{4-0}{4} = 1 \tag{4.16}$$

and

$$x_i = a + \frac{(b-a)i}{n} = 0 + \frac{(4-0)i}{4} = i. \tag{4.17}$$

Inserting (4.16) and (4.17) in (4.12) gives

$$\begin{aligned}\int_0^4 x^3\,\mathrm{d}x &\approx \frac{h}{3}[f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \cdots + 4f(x_{n-1}) + f(x_n)] \\ &= \frac{1}{3}[f(0) + 4f(1) + 2f(2) + 4f(3) + f(1)] \\ &= \frac{1}{3}[0^3 + 4\cdot 1^3 + 2\cdot 2^3 + 4\cdot 3^3 + 4^3] \\ &= 64.\end{aligned}$$

Therefore the relative error (recall Definition 2.8) of Simpson's rule is

$$\left|\frac{\int_0^4 x^3\,\mathrm{d}x - 64}{\int_0^4 x^3\,\mathrm{d}x}\right| = \left|\frac{64-64}{64}\right| = 0 = 0\,\%.$$

It is no coincidence that Simpson's rule was exact in Example (4.12).

Theorem 4.13. *Simpson's rule is exact for all polynomials of degree at most three.*

Proof. Similar to the proof of Theorem 4.5 but more complicated. □

Let

$$E_n(f) = \left|\int_a^b f(x)\,\mathrm{d}x - \frac{h}{3}\left(f(x_0) + f(x_n) + 4\sum_{i=1}^{\frac{n}{2}} f(x_{2i-1}) + 2\sum_{i=1}^{\frac{n}{2}-1} f(x_{2i})\right)\right|$$

denote the absolute error of Simpson's rule.

Theorem 4.14. *Suppose that $f\colon [a,b] \to \mathbb{R}$ is continuous on $[a,b]$ and four times differentiable on (a,b). If there is a constant $L \geq 0$ such that $|f^{(4)}(x)| \leq L$ for all $x \in (a,b)$, then Simpson's rule satisfies*

$$E_n(f) \leq \frac{(b-a)^5 L}{180 n^4} = \frac{(b-a)L}{180} h^4,$$

where $h = x_i - x_{i-1} = (b-a)/n$ is the length of each subinterval $[x_{i-1}, x_i]$.

Proof. Omitted. □

If f is sufficiently smooth (i.e., has sufficiently many derivatives), Theorems 4.6, 4.10 and 4.14 show that the errors of the midpoint and trapezoidal rules are of order h^2 while the error of Simpson's rule is of order h^4. Therefore Simpson's rule may converge faster than the midpoint or trapezoidal rule if f is sufficiently smooth.

5 Solutions to Systems of Equations

Problems encountered while studying mathematics typically admit precise closed-form solutions. This might give the impression that every equation can be solved exactly. However, the opposite is true: most equations cannot be solved in closed form in terms of a finite number of elementary functions. This is almost always the case when the equation involves different types of elementary functions. For example, the solution to

$$e^x = 2$$

is $x = \log 2 \approx 0.693$. But the solution to

$$e^x + x = 2$$

is $x = 2 - W(e^2) \approx 0.442$, where W is a special function called the *Lambert W function*. Finally, the solution to

$$e^x + x + \sin(x) = 2,$$

which is $x \approx 0.316$, lacks an expression in terms of any standard elementary or special functions. Solutions to equations that cannot be solved analytically are obtained using *numerical root-finding methods*.

If f and g are real-valued functions of one real variable, an $x^* \in \mathbb{R}$ is *a solution to the equation*

$$f(x) = g(x)$$

if $f(x^*) = g(^*)$. Similarly, if $\mathbf{f}, \mathbf{g} \colon \mathbb{R}^m \to \mathbb{R}^n$, an $\mathbf{x}^* \in \mathbb{R}^m$ is a solution to

$$\mathbf{f}(\mathbf{x}) = \mathbf{g}(\mathbf{x})$$

if $\mathbf{f}(\mathbf{x}^*) = \mathbf{g}(\mathbf{x}^*)$. A solution to an equation is said to *solve the equation*.

Example 5.1. Consider the equation $x^2 + 1 = 2$, so that $f(x) =^2 +1$ and $g(x) = 2$. The real $x = -1$ is a solution to the equation $x^2 + 1 = 2$ because

$$f(-1) = (-1)^2 + 1 = 1 + 1 = 2 = g(-1),$$

but $x = 2$ is not a solution because

$$f(2) = 2^2 + 1 = 4 + 1 = 5 \neq 2 = g(2).$$

Graphically the solutions of an equation correspond to the points on the x-axis where the graphs of the functions f and g in the equation $f(x) = g(x)$ intersect. For example, the solutions of the equation $x^2 + 1 = 2$ are $x_0 = \pm 1$ because at these points the graphs of the functions $f(x) = x^2 + 1$ and $g(x) = 2$ are at the same height. The functions f and g can also be chosen differently, as equations can be equivalently rewritten in different forms. For instance, the equation $x^2 + 1 = 2$ can be rewritten as $x^2 - 1 = 0$, in which case $f(x) = x^2 - 1$ and $g(x) = 0$. This leads to the equivalence between solutions and roots. An $x^* \in \mathbb{R}$ is a *root* (or *zero*) of $f \colon \mathbb{R} \to \mathbb{R}$ if $f(x^*) = 0$. Graphically, roots are those points on the x-axis at which the function f intersects with the x-axis. Similarly, $\mathbf{x}^* \in \mathbb{R}^n$ is a root of $f \colon \mathbb{R}^m \to \mathbb{R}^n$ if $\mathbf{f}(\mathbf{x}^*) = \mathbf{0}$.

Theorem 5.2. *Let* $\mathbf{f}, \mathbf{g} \colon \mathbb{R}^n \to \mathbb{R}^m$. *An* $\mathbf{x}^* \in \mathbb{R}^n$ *is a solution to the equation* $\mathbf{f}(\mathbf{x}) = \mathbf{g}(\mathbf{x})$ *if and only if it is the root of the function* $\mathbf{f} - \mathbf{g}$.

Proof. Suppose that $\mathbf{x}^*$ solves the equation. Therefore $\mathbf{f}(\mathbf{x}^*) = \mathbf{g}(\mathbf{x}^*)$, which is equivalent to $\mathbf{f}(\mathbf{x}^*) - \mathbf{g}(\mathbf{x}^*) = \mathbf{0}$. This shows that $\mathbf{x}^*$ is a root of $\mathbf{f} - \mathbf{g}$. Conversely, if $\mathbf{x}^*$ is a root of $\mathbf{f} - \mathbf{g}$, it satisfies $\mathbf{f}(\mathbf{x}^*) - \mathbf{g}(\mathbf{x}^*) = \mathbf{0}$. This is equivalent to $\mathbf{f}(\mathbf{x}^*) = \mathbf{g}(\mathbf{x}^*)$, so that $\mathbf{x}^*$ solves the equation. □

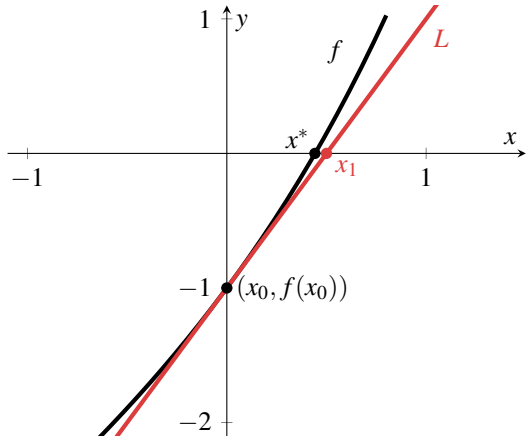


Figure 5.1: *The idea behind Newton's method*: The root x_1 of the linear approximation (i.e., tangent line) L_{x_0} for the function $f(x) = e^x + x - 2$ serves as an approximation for the root x^* of the function provided that the expansion point (i.e., point of tangency) x_0 is close to the actual root.

Therefore solving an equation is equivalent to finding roots of a function obtained by taking the difference of the sides of the original equation.

A function can have zero, one, or multiple roots. Equivalently, an equation can have zero, one, or multiple solutions. For example, the equation in Example 5.1 had two solutions ($x^* \pm 1$). The equation $x^2 = 0$ has one solution ($x^* = 0$) and the equation $x^2 + 1 = 0$ no solutions[3] (because x^2 is always non-negative). One can use Bolzano's theorem from basic analysis to ascertain if a continuous function has at least one solution.

Theorem 5.3 (Bolzano's theorem)**.** *Suppose that $f\colon [a,b] \to \mathbb{R}$ is continuous. If $f(a)$ and $f(b)$ have different signs, then $f(c) = 0$ for some $c \in [a,b]$.*

That $f(a)$ and $f(b)$ have different signs means that either (i) $f(a) < 0$ and $f(b) > 0$ or (ii) $f(a) > 0$ and $f(b) < 0$. These can be combined in the single condition $f(a)f(b) < 0$ because the product of a positive and a negative number is negative. Note that there may be roots on the interval $[a,b]$ even if the condition $f(a)f(b) < 0$ does not hold. Moreover, $f(a)f(b) < 0$ guarantees only that there is *at least one root* (i.e., there can be either one or multiple roots).

5.1 Newton's Method

Let us construct a root-finding method named after Newton. Recall from Section 3.2 that the linear approximation to a function f (or the Taylor polynomial of order one) at x_0 is the tangent line

$$L_{x_0}(x) = f(x_0) + f'(x_0)(x - x_0). \tag{5.1}$$

By Taylor's theorem (see Theorem 3.3 and Corollary 3.4), the linear approximation is a good approximation to f for x that are close to the expansion point x_0. Consequently, if f has a root near x_0, also the linear approximation has a root near x_0 and this root can be used to approximate the root of f. This is illustrated in Figure 5.1 for the function $f(x) = e^x + x - 2$ when $x_0 = 0$. If, in contrast to the previous case, the point of tangency x_0 is *not* close to the root of the function f, the root x_1 of the tangent line is still often closer to the root of the function than x_0. This means

[3] Unless one considers complex numbers, in which case the solutions are $x^* = \pm i$. We do not use complex numbers on this course.

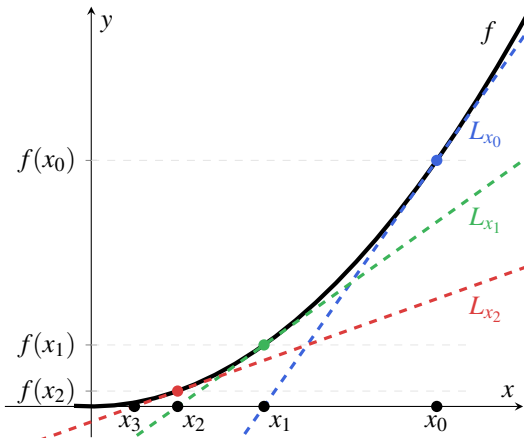


Figure 5.2: The roots x_k of the tangent lines of the function f approach the root of the function at $x = 0$.

that if tangent lines are iteratively formed at the root of the previous tangent line, these roots x_1, x_2, x_3, ... often approach the root of the function f. This is illustrated in Figure 5.2.

The roots of x_k of the tangent lines generally serve as increasingly accurate approximations for the root of the function. The resulting method is known as Newton's method. Let us derive an iteration formula that allows us to compute the next approximation, x_{k+1}, from the previous one, x_k. The root x_1 of the linear approximation in (5.1) formed at the point x_0 is given by

$$L_{x_0}(x_1) = 0 \quad \Longleftrightarrow \quad f(x_0) + f'(x_0)(x_1 - x_0) = 0 \quad \Longleftrightarrow \quad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

By similar reasoning, the root x_2 of the tangent line formed at the point x_1 is given by

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)},$$

and, more generally, the root x_{k+1} of the tangent line formed at the point x_k is given by

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}. \tag{5.2}$$

Note that this formula assumes that $f'(x_k) \neq 0$. If $f'(x_k) = 0$, the linear approximation $L_{x_k}(x) = f(x_k) + f'(x_k)(x - x_k) = f(x_k)$ corresponds to a horizontal line that does not have a root, which means that one cannot proceed with the iteration as there is no next root x_{k+1} to construct a linear approximation at. With (5.2), new approximations can be calculated recursively once x_0 is selected. The choice of x_0 is often nothing but an informed guess. Accordingly, x_0 is called the initial guess. The closer the initial guess is to a root x^* of f, the more reliably the method works and the faster the approximations converge to the solution.

Definition 5.4 (Newton's method)**.** Let $f\colon \mathbb{R} \to \mathbb{R}$ be a differentiable function and $x_0 \in \mathbb{R}$ an *initial guess*. *Newton's method* is a sequence of approximations for a root of f given by the iteration

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)} \tag{5.3}$$

for $k \geq 0$.

Example 5.5. Let us use Newton's method with initial guess $x_0 = 4$ to compute approximations x_1 and x_2 to the root of the equation $x^2 = 0$, which is $x^* = 0$. Because $f(x) = x^2$ and $f'(x) = 2x$, the iteration formula (5.3) for Newton's method is

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_{k+1})} = x_k - \frac{x_k^2}{2x_k}. \tag{5.4}$$

Therefore

$$\begin{aligned} x_0 &= 4, \\ x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} = 4 - \frac{4^2}{2 \cdot 4} = 2, \\ x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} = 2 - \frac{2^2}{2 \cdot 2} = 1. \end{aligned}$$

In this case the iteration formula (5.4) simplifies to

$$x_{k+1} = x_k - \frac{x_k^2}{2x_k} = x_k - \frac{1}{2}x_k = \frac{1}{2}x_k,$$

so that

$$x_{k+1} = \frac{1}{2}x_k = \frac{1}{2} \cdot \frac{1}{2}x_{k-1} = \cdots = \frac{1}{2^{k+1}x_0}.$$

That is, in this case one could have computed each x_k directly from x_0 without iteration. *This happens very rarely in practice.*

The process described in the previous example, where the *same steps* are repeated over and over again, is called *iteration*. A single step of iteration is called an *iteration* or *iteration step*. The values calculated by iteration, such as x_k above, are called *iterates*. Therefore, Newton's method is based on iteration, and the formula in (5.3) is referred to as the Newton's method formula, the recursive formula, and also the *iteration formula*.

Newton's method does not work always. There are several potential issues:

- Although the iterates converge, they do not necessarily converge to a root.
- Geometrically it is evident that it can happen that the iterates keep repeating two consecutive values. This is the case, for example, with the function $f(x) = x^2 + x + 1$ when the initial guess or some subsequent iterate is 0 or -1.
- The iteration formula clearly fails if the derivative needs to be calculated at a point where it is zero.
- The iteration formula fails if the derivative needs to be calculated at a point where it does not exist, such as with the function $f(x) = |x^2 - 1| - 1$ at the points $x = \pm 1$. Less obviously, problems can also arise in the vicinity of such points, as with the function $f(x) = \sqrt[3]{x}$ around $x = 0$.
- In some cases, the iterates diverge if the initial guess is too far from the root, as in the case of the function $f(x) = x/(x^2 + 1)$.

The following theorem provides sufficient conditions to guarantee that Newton's method works.

Theorem 5.6. *Let $f(c) = 0$ and $f'(c) \neq 0$. Suppose that f is twice continuously differentiable on the interval $[c-a, c+a]$ for some $a > 0$. Then the iteration formula of Newton's method converges to c if the initial guess x_0 is sufficiently close to c.*

Proof. Omitted. □

Note that the theorem does not cover all functions for which Newton's method works. For example, for the function $f(x) = x^2$ we have $f'(x) = 0$, so that the assumptions of the theorem are not satisfies, but as seen above in Example 5.5, Newton's method nevertheless works for this function.

The theorem does not provide the exact value of c, but the information that the initial guess must be sufficiently close is still useful. For example, the function $f(x) = x/(x^2+1)$ satisfies the assumptions of the theorem, so that Newton's method will definitely work for it, as long as the initial guess is chosen appropriately. On the other hand, $f(x) = \sqrt[3]{x}$ does not satisfy the assumptions (why not?), so Newton's method may not work, even if the initial guess arbitrarily close to a root.

If Newton's method does not work on the first try, there is no reason to give up; instead, a different initial guess should be tried. The root can also be roughly determined first using another root-finding method. For this reason, more advanced algorithms, such as MATLAB's `fzero` function, apply multiple methods.

5.2 Newton's Method for Systems of Equations

Newton's method for systems of equations is almost identical to the single-variable case when formulated using vector-valued functions. For this purpose, one must be familiar with the concept of the *Jacobian matrix*, which is a generalization of the gradient for vector-valued functions. Before introducing the Jacobian matrix, we will also review other necessary concepts.

5.2.1 Nonlinear Systems of Equations

A system of equations is linear if it has the form

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \tag{5.5}$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \tag{5.6}$$
$$\vdots \tag{5.7}$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m, \tag{5.8}$$

where a_{ij} and b_i are constant and x_j variables. In other cases, a system of equations is *nonlinear*. A system of equations is considered nonlinear if it contains at least one term that is neither a constant nor of the form constant · variable. For example, the system of equations

$$\begin{aligned} 4xy + y &= 1 \\ x - 2y &= 3 \end{aligned}$$

is nonlinear due to the term $4xy$ and the system of equations

$$\begin{aligned} 4x + y &= 1 \\ x - 2\sin(y) &= 3 \end{aligned}$$

due to the term $-2\sin(y)$.

Linear systems of equations can be written in the form $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ is a constant matrix, and they always have exactly 1, 0, or infinitely many solutions. They can all be solved using

Gaussian elimination, and there also exist numerical solution methods that are more efficient than Gaussian elimination and Newton's method.

Nonlinear systems of equations cannot be written in the form $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ is a constant matrix, and the number of their solutions can be other than 1, 0, or infinite. Solving them is significantly more difficult than solving linear systems of equations.

5.2.2 Roots of a Multivariate Function

The roots of a single-variable function are the points on the x-axis where the function's graph, which is often a curve, intersects the x-axis. These are typically distinct points and are often finite in number.

The roots of a real-valued function of two variables are the points in the xy-plane where the function's graph, which is often a surface, intersects the xy-plane. These usually form a curve, meaning that there are infinitely many such points. For example the roots of the function $f(x,y) = x^2 + y^2 - 1$ satisfy

$$\begin{aligned} f(x,y) &= 0 \\ x^2 + y^2 - 1 &= 0 \\ x^2 + y^2 &= 1, \end{aligned}$$

so they form a circle of radius 1 centered at the origin. In the case of two variables, instead of referring to roots, one may also speak of level curves given by $f(x,y) = 0$, which represent the curves in the xy-plane where the function's height is zero (compare level curves to contour lines on maps).

The roots of a real-valued function of three variables generally form surfaces. For example, the zeroes of the function $f(x,y,z) = x^2 + y^2 + z^2 - 1$ form a sphere of radius 1 centered at the origin, which is given by $x^2 + y^2 + z^2 = 1$. In the case of three variables one may also refer to level surfaces given by $f(x,y,z) = 0$ instead of roots.

5.2.3 Tangent Plane

For functions of two variables, the concept corresponding to the tangent line is the tangent plane. The linear approximation, or tangent plane, or first-degree Taylor polynomial, of a function f at a point (x_0, y_0) is given by

$$L(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x - x_0) + f_y(x_0,y_0)(y - y_0).$$

The roots of a function of two variables can be approximated by the zeroes of the tangent plane formed near the actual zero, similarly to how the zeroes of a single-variable function are approximated by the zeroes of the tangent line. However, the difference is that only a portion of the zeroes of the tangent plane, which form a line, provide a good approximation of some of the roots. The accompanying figure shows the function $f(x,y) = 3 - x^2 - y^2$ along with the tangent plane $L(x,y) = 1 - 2(x-1) - 2(y-1)$ at the point $(1,1)$ and its roots. Note that only some of the points on the line provide a good approximation of some of the function's roots.

5.2.4 Vector-Valued Functions

Vector-valued functions are functions of one or more variables whose values are vectors. For example,

$$\mathbf{f}\colon \mathbb{R} \to \mathbb{R}^3, \quad \mathbf{f}(t) = \begin{bmatrix} t^2 \\ t^2 \\ 3 \end{bmatrix}$$

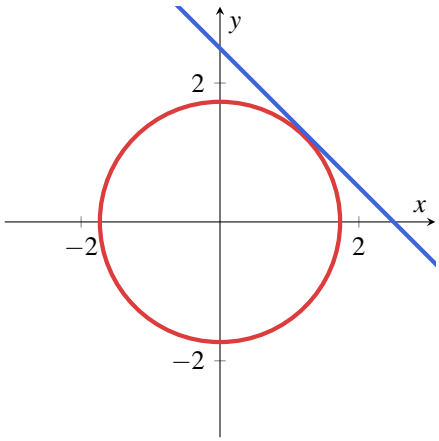


Figure 5.3: The function $f(x,y) = 3 - x^2 - y^2$ along with the tangent plane $L(x,y) = 1 - 2(x-1) - 2(y-1)$ at the point $(1,1)$ and its roots.

is a vector-valued function of one variable whose values are three-dimensional vectors. Such a function could, for instance, describe the (three-dimensional) position of an object as a function of time t. On the other hand, the function

$$\mathbf{f}\colon \mathbb{R}^3 \to \mathbb{R}^3, \quad \mathbf{f}(x,y,z) = \begin{bmatrix} xz \\ y+1 \\ x-2z+\sin(y) \end{bmatrix}$$

is a vector-valued function of three variables whose values are three-dimensional vectors. Such a function could, for example, represent the wind velocity vector as a function of position (x,y,z).

In general, vector-valued functions have the form

$$\mathbf{f}\colon D \to \mathbb{R}^n,\ D \subset \mathbb{R}^m, \quad \mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_n(\mathbf{x}) \end{bmatrix},$$

and the functions f_i are called *component functions*. We can also write $\mathbf{f}(x_1, x_2, \ldots, x_m)$, but using vector notation for variables is clearly more concise in the case of multiple variables.

The roots of a vector-valued function refer to the set of input values for which $\mathbf{f}(\mathbf{x}) = \mathbf{0}$ holds. From this it follows that solving a system of equations can equivalently be interpreted as solving for the roots of a vector-valued function. For example, we can write

$$\begin{aligned} x^2 + y^2 &= 3 \\ (x-2)^2 + (y-1)^2 &= 2, \end{aligned}$$

which is equivalent to

$$\begin{aligned} x^2 + y^2 - 3 &= 0 \\ (x-2)^2 + (y-1)^2 - 2 &= 0 \end{aligned}$$

which is equivalent to

$$f(x,y) = 0$$

$$g(x,y) = 0,$$

and define the function

$$\mathbf{f}(x,y) = \begin{bmatrix} f(x,y) \\ g(x,y) \end{bmatrix} = \begin{bmatrix} x^2 + y^2 - 3 \\ (x-2)^2 + (y-1)^2 - 2 \end{bmatrix},$$

so that the original pair of equations can be viewed as the equation $\mathbf{f}(x,y) = \mathbf{0}$.

When generalizing Newton's method for systems of equations, we think of solving for the roots of a vector-valued function. The iteration formula becomes almost identical to the single-variable case. In this case, x_k and x_{k+1} are replaced by vectors $\mathbf{x}_k$ and $\mathbf{x}_{k+1}$, and the real-valued function f is replaced by the vector-valued function $\mathbf{f}$.

5.2.5 Jacobian Matrix

The *gradient* of a real-valued function $f\colon D \to \mathbb{R}$, $D \subset \mathbb{R}^2$ of two variables is the function

$$\nabla f(x,y) = \begin{bmatrix} f_x(x,y) \\ f_y(x,y) \end{bmatrix},$$

which generalizes the derivative. Generally, the gradient of a function $f\colon D \to \mathbb{R}$, $D \subset \mathbb{R}^n$ of multiple variables is the function

$$\nabla f(\mathbf{x}) = \begin{bmatrix} f_{x_1}(\mathbf{x}) \\ f_{x_2}(\mathbf{x}) \\ \vdots \\ f_{x_n}(\mathbf{x}) \end{bmatrix}.$$

The *Jacobian matrix* of a vector-valued function

$$\mathbf{f}\colon D \to \mathbb{R}^2,\ D \subset \mathbb{R}^2, \quad \mathbf{f}(x,y) = \begin{bmatrix} f(x,y) \\ g(x,y) \end{bmatrix},$$

of two variables is the matrix-valued function

$$\mathrm{J}_{\mathbf{f}}(x,y) = \begin{bmatrix} f_x(x,y) & f_y(x,y) \\ g_x(x,y) & g_y(x,y) \end{bmatrix}.$$

For example, the function

$$\mathbf{f}(x,y) = \begin{bmatrix} x^2 y \\ x + 2y^3 \end{bmatrix}$$

has the Jacobian matrix

$$\mathrm{J}_{\mathbf{f}}(x,y) = \begin{bmatrix} 2xy & x^2 \\ 1 & 6y^2 \end{bmatrix}.$$

Generally, the Jacobian matrix of a vector-valued function

$$\mathbf{f}\colon D \to \mathbb{R}^m,\ D \subset \mathbb{R}^n, \quad \mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix},$$

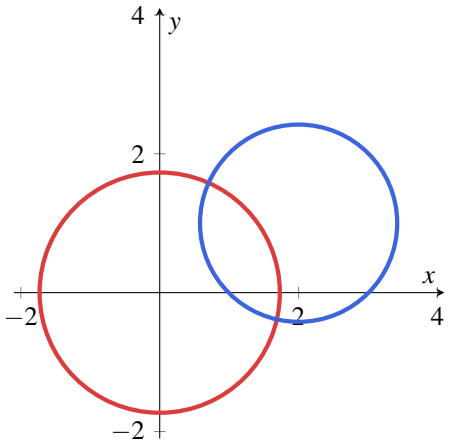


Figure 5.4: Equations $x^2+y^2=3$ and $(x-2)^2+(y-1)^2=2$ hold simultaneously only when the corresponding curves intersect.

of multiple variables is the matrix-valued function

$$\mathrm{J}_{\mathbf{f}}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

When Newton's method is later extended to systems of equations, the division by the derivative $f'(x_k)$ in the single-variable iteration formula $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$ is replaced by multiplication by the inverse of the Jacobian matrix $\mathrm{J}_{\mathbf{f}}(\mathbf{x})^{-1}$.

5.2.6 Geometric Idea of Newton's Method for a Pair of Equations

Solving a system of two equations involves finding the intersection points of level curves. For example, for the pair of equations

$$x^2+y^2=3 \tag{5.9}$$

$$(x-2)^2+(y-1)^2=2 \tag{5.10}$$

the solutions are the intersection points of the curves in the accompanying figure. To understand why Newton's method can also be applied to solving pairs of equations, we need to look at the pair from a slightly different perspective.

The previous pair of equations can be written as

$$x^2+y^2-3=0$$
$$(x-2)^2+(y-1)^2-2=0,$$

where the left-hand sides of the equations can be interpreted as the functions $f(x,y)=x^2+y^2-3$ and $g(x,y)=(x-2)^2+(y-1)^2-2$. Interpreted in this way, the pair

$$f(x,y)=0$$

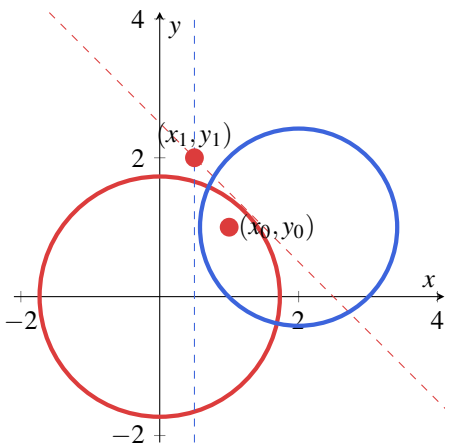


Figure 5.5: The intersection point of the lines formed by the roots of the tangent planes of the functions $f(x,y) = x^2 + y^2 - 3$ and $g(x,y) = (x-2)^2 + (y-1)^2 - 2$ can be used to approximate a common root of the functions f and g.

$$g(x,y) = 0$$

seeks the common roots of the bivariate functions f and g.

The roots of a function of two variables can be approximated by the roots of the tangent plane formed at a point (x_0, y_0) near the root, similarly to how the roots of a single-variable function are approximated by the roots of the tangent line. Thus, the common roots of functions f and g can be approximated by the intersection points of the roots of the tangent planes formed for the functions. The accompanying figure shows the roots of the functions f and g as well as the roots of the tangent planes formed at the point $(x_0, y_0) = (1,1)$. The intersection point of the lines, (x_1, y_1), roughly approximates the upper root of the system of equations (5.9).

When new tangent planes are formed at the intersection point of the lines (x_1, y_1) from the previous figure, and the intersection point (x_2, y_2) of the new lines formed by the roots of these tangent planes is calculated, a better approximation for the solution of the system of equations is obtained. This process can be repeated indefinitely, and often the intersection points (x_k, y_k) converge towards one of the solutions of the system. This is the geometric idea of Newton's method for a system of two equations. Next, let us move on to examine the problem computationally.

5.2.7 The Formula of Newton's Method for Systems of Equations

Let us derive the formula for Newton's method in the case of a system of two equations. All systems of two equations can be written in the form

$$\begin{aligned} f(x,y) &= 0 \\ g(x,y) &= 0. \end{aligned}$$

We approximate the solution of the system of equations using tangent planes in the way described in the previous section. We begin by defining the tangent planes at the point (x_0, y_0). The equations of the tangent planes are

$$L_f(x,y) = f(x_0,y_0) + f_x(x_0,y_0)(x - x_0) + f_y(x_0,y_0)(y - y_0)$$

and

$$L_g(x,y) = g(x_0,y_0) + g_x(x_0,y_0)(x - x_0) + g_y(x_0,y_0)(y - y_0).$$

The intersection lines of these planes with the xy-plane are given by $L_f(x,y) = 0$ and $L_g(x,y) = 0$. The solution of the system of equations can be approximated by the intersection point of these lines. This can be solved from the system of equations

$$\begin{aligned} L_f(x,y) &= 0 \\ L_g(x,y) &= 0 \end{aligned}$$

which is equivalent to

$$\begin{aligned} f(x_0,y_0) + f_x(x_0,y_0)(x-x_0) + f_y(x_0,y_0)(y-y_0) &= 0 \\ g(x_0,y_0) + g_x(x_0,y_0)(x-x_0) + g_y(x_0,y_0)(y-y_0) &= 0, \end{aligned}$$

which becomes

$$\begin{aligned} f_x(x_0,y_0)x + f_y(x_0,y_0)y &= f_x(x_0,y_0)x_0 + f_y(x_0,y_0)y_0 - f(x_0,y_0) \\ g_x(x_0,y_0)x + g_y(x_0,y_0)y &= g_x(x_0,y_0)x_0 + g_y(x_0,y_0)y_0 - g(x_0,y_0), \end{aligned}$$

when the variables and constants are moved to their respective sides. In these equations, x and y are the unknowns and all other coefficients and terms are constants.

The system of equations could be solved directly, but before doing so, let us note that the system appears to be highly regular in a certain way way. This regularity becomes clearer when we write the system using matrices:

$$\begin{bmatrix} f_x(x_0,y_0) & f_y(x_0,y_0) \\ g_x(x_0,y_0) & g_y(x_0,y_0) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} f_x(x_0,y_0) & f_y(x_0,y_0) \\ g_x(x_0,y_0) & g_y(x_0,y_0) \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} - \begin{bmatrix} f(x_0,y_0) \\ g(x_0,y_0) \end{bmatrix}.$$

This can be written even more conveniently if we adopt the notation

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad \mathbf{x}_0 = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \quad \text{and} \quad \mathbf{f}(\mathbf{x}) = \begin{bmatrix} f(x,y) \\ g(x,y) \end{bmatrix}.$$

We also observe that the coefficient matrix is actually the Jacobian matrix of the function $\mathbf{f}$, meaning that we have

$$\begin{bmatrix} f_x(x_0,y_0) & f_y(x_0,y_0) \\ g_x(x_0,y_0) & g_y(x_0,y_0) \end{bmatrix} = \mathbf{J_f}(\mathbf{x}).$$

With this notation the matrix equation becomes

$$\mathbf{J_f}(\mathbf{x}_0)\mathbf{x} = \mathbf{J_f}(\mathbf{x}_0)\mathbf{x}_0 - \mathbf{f}(\mathbf{x}_0),$$

and if we assume that the Jacobian matrix is invertible, the solution of the system of equations is

$$\mathbf{x} = \mathbf{x}_0 - \mathbf{J_f}^{-1}(\mathbf{x}_0)\mathbf{f}(\mathbf{x}_0).$$

The iterate proceeding the initial guess $\mathbf{x}_0$ is thus $\mathbf{x}_1 = \mathbf{x}_0 - \mathbf{J_f}^{-1}(\mathbf{x}_0)\mathbf{f}(\mathbf{x}_0)$ and the next one $\mathbf{x}_2 = \mathbf{x}_1 - \mathbf{J_f}^{-1}(\mathbf{x}_1)\mathbf{f}(\mathbf{x}_1)$, and so on. In general we thus have $\mathbf{x}_{k+1} = \mathbf{x}_k - \mathbf{J_f}^{-1}(\mathbf{x}_k)\mathbf{f}(\mathbf{x}_k)$.

The reasoning above generalizes in a similar way to $n \times n$ systems of equations.

Definition 5.7 (Newton's method for systems of equations)**.** In Newton's method, approximate solutions to the system of equations $\mathbf{f}(\mathbf{x}) = \mathbf{0}$ are calculated using the iteration

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \mathbf{J_f}^{-1}(\mathbf{x}_k)\mathbf{f}(\mathbf{x}_k),$$

starting from a user-specified vector $\mathbf{x}_0$, which is called the initial guess.

Note that the formula for Newton's method for systems of equations is almost identical to the case of a single variable, the only difference being that the functions and variables are now vectors, and the derivative is replaced by the Jacobian matrix. Specifically, it is important to remember that the multiplication by the Jacobian matrix must happen *from the left*.

Example 5.8. Let us use Newton's method to find an approximation $\mathbf{x}_1$ to the solution of the system of equations

$$\begin{aligned} x^2 + y^2 &= 3 \\ (x-2)^2 + (y-1)^2 &= 2 \end{aligned}$$

when the initial guess is $\mathbf{x}_0 = (1,1)$. Information necessary for iteration:

$$\mathbf{f}(x,y) = \begin{bmatrix} f(x,y) \\ g(x,y) \end{bmatrix} = \begin{bmatrix} x^2 + y^2 - 3 \\ (x-2)^2 + (y-1)^2 - 2 \end{bmatrix}.$$

$$\mathbf{J_f}(x,y) = \begin{bmatrix} 2x & 2y \\ 2(x-2) & 2(y-1) \end{bmatrix}$$

$$\mathbf{J_f}^{-1}(x,y) = \frac{1}{4x(y-1) - 4y(x-2)} \begin{bmatrix} 2(y-1) & -2y \\ -2(x-2) & 2x \end{bmatrix}.$$

First iterate:

$$\begin{aligned} \mathbf{x}_1 &= \mathbf{x}_0 - \mathbf{J_f}^{-1}(\mathbf{x}_0)\mathbf{f}(\mathbf{x}_0) \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{4(1-1) - 4(1-2)} \begin{bmatrix} 2(1-1) & -2 \\ -2(1-2) & 2 \end{bmatrix} \begin{bmatrix} 1^2 + 1^2 - 3 \\ (1-2)^2 + (1-1)^2 - 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 0 & -2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \frac{1}{4} \begin{bmatrix} 2 \\ -4 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} \\ 2 \end{bmatrix}. \end{aligned}$$

5.3 Fixed Point Method

Fixed point method is numerical equation-solving method similar to Newton's method.

5.3.1 Fixed Point

Usually a function maps an input to an output that does not equal the input. For example, the function $f(x) = x^2$ maps $x = 2$ to the $f(x) = 4$. This function also has two variable values that remain unchanged under the mapping, namely $x = 0$ and $x = 1$. These are the fixed points of the function f.

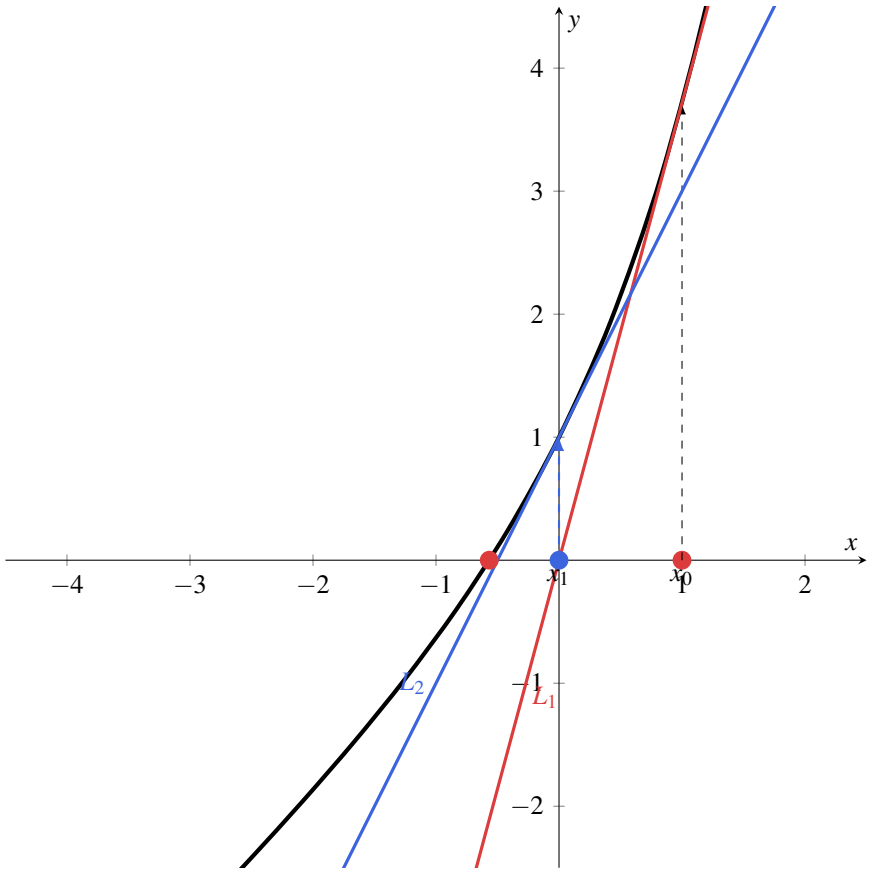
Definition 5.9 (Fixed point)**.** The fixed-points of a function $f\colon D \to \mathbb{R}$, where $D \subset \mathbb{R}$, are points $c \in D$ for which $f(c) = c$.

A function can have multiple fixed points, as in the previous example, or it may have none at all. The geometric interpretation of a fixed point is the x-coordinate of the intersection point between the line $y = x$ and the graph of the function f (which is, in turn, the same as the y-coordinate of that point). The fixed points of a function can be determined by solving the equation $f(x) = x$.

5.3.2 Geometric Intuition

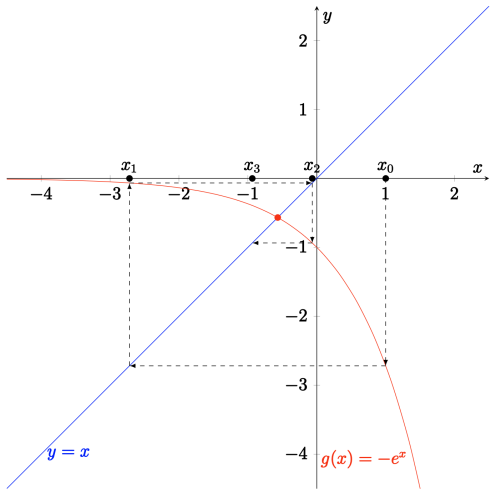
Just like Newton's method, the fixed point method is based on a highly intuitive geometric idea. To begin, let us note for comparison that the equation $e^x = -x$ is solved graphically using Newton's method with the following steps, which are illustrated in the accompanying figure:

1. The equation is rewritten in the form $e^x + x = 0$, meaning that all terms are moved to the same side.
2. The graph of the function $f(x) = e^x + x$ is drawn, so that the solutions of the original equation correspond to the roots of the function f.
3. An initial guess x_0 is chosen as the first approximation to the root of the function f.
4. A tangent is drawn for the function f at the point x_0.
5. The zero of the tangent line, denoted as x_1, is determined. This becomes the second approximation to the root of the function f.
6. The process of drawing tangent lines is repeated as described in steps 4 and 5, always replacing the previous iterate x_k with the next one, x_{k+1}.
7. In certain cases, the iterates converge towards the root of the function f.



The following stesps, illustrated in the accompanying figure, are used to solve the equation $e^x = -x$ graphically with the fixed point method:

1. The equation is rewritten in the form $x = -e^x$, so that one side contains only x. There are different ways to do this. We will discuss these later.
2. A line $y = x$ and the graph of the function $g(x) = -e^x$ (i.e., graphs of both sides of the equation from step 2) are drawn. Then the solutions of the original equation are the x-coordinates of the intersection points of these curves.
3. An initial guess x_0 is chosen as the first approximation to the solution.
4. At point x_0, move vertically to the height $g(x_0)$.
5. Move horizontally at height $g(x_0)$ to the line $y = x$, that is, to the point $(g(x_0), g(x_0))$. The x-coordinate of the point becomes a new approximation x_1 to the solution to the equation.
6. Repeat the vertical and horizontal shifts of steps 4 and 5, replacing the previous iterate x_k with the new one, x_{k+1}.
7. In certain cases, the iterates converge to the solution of the equation.



When using the fixed point method above, the equation $e^x = -x$ could also have been solved in the form $x = \ln(-x)$, where the considered function would have been $g(x) = \ln(-x)$. This is an example of how the function g is not unique, as it can actually be chosen in infinitely many different ways. For example, the above can be written as

$$\begin{aligned} -x &= e^x \\ ax - x &= e^x + ax \\ x &= \frac{e^x + ax}{a-1} \end{aligned}$$

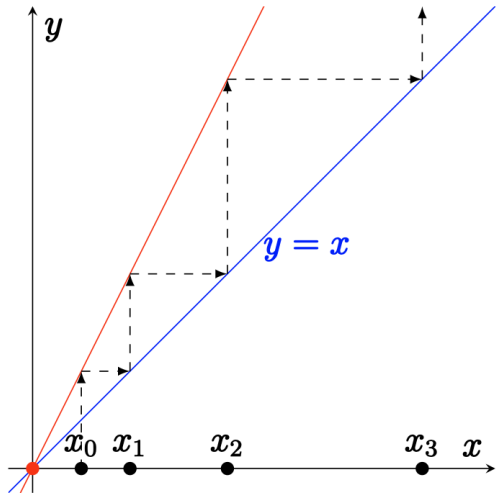
so that each value of the parameter a corresponds to the function

$$g(x) = \frac{e^x + ax}{a-1}.$$

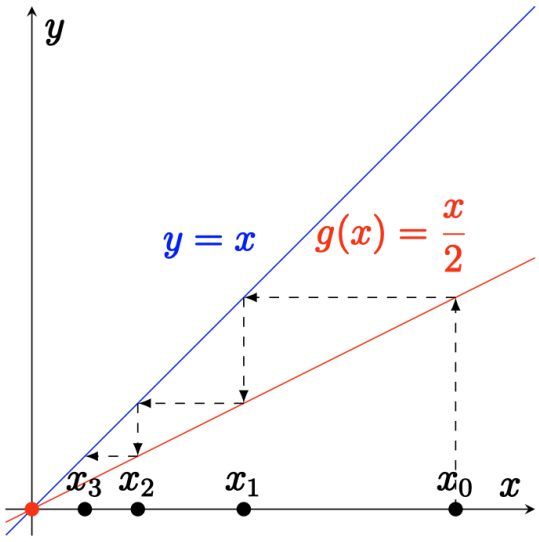
It was noted above that the iterates converge to the solution *in certain situations*. The fixed point method does not always work, just as the Newton method does not always work. Solving the equation $-x = e^x$ with the choice $g(x) = \ln(-x)$ fails because it requires examining the function g at points where it is not defined (Draw a graph and try different initial guesses). However, iteration can also fail even if g is defined everywhere. For example, if the equation

$$x - \frac{x}{2} = 0$$

is solved by first writing it as $x = 2x$, so that $g(x) = 2x$, the iterates diverge as shown in the accompanying figure.



If we instead try to solve the same equation by first writing it as $x = \frac{x}{2}$, where $g(x) = \frac{x}{2}$, then the iterates converge as shown in the accompanying figure.



An obvious difference between the functions g that behaved differently above is at least that the first one is steeper than the second. By repeating similar analyses, it can be observed that the steepness of the function g appears to be a critical factor for convergence. The fixed point method can also fail because the iterates do not converge to the correct value. However, this can happen only when the function g is not continuous.

Theorem 5.10. *Let g be a continuous function. If the iterates of the fixed point iteration $x_{k+1} = g(x_k)$ converge, they converge to a fixed point.*

Proof. Omitted. □

The following function serves as a counter-example for the necessity of continuity of g:

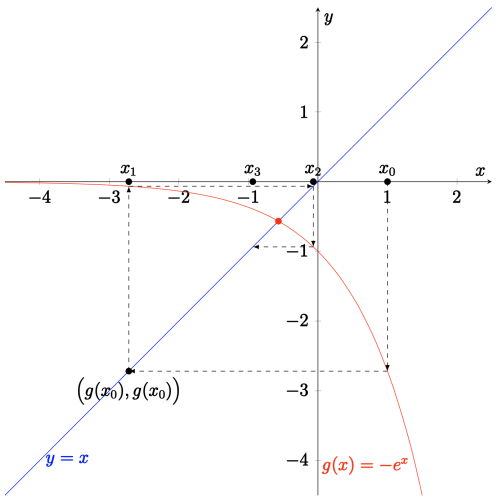
$$g(x) = \begin{cases} \frac{x}{2}, & x \neq 0 \\ -1, & x = 0 \end{cases}.$$

5.3.3 *Iteration Formula*

Let us derive the iteration formula for the fixed point method. The goal is to deduce how x_{k+1} can be computed from on the previous value x_k (or from on multiple previous values such as x_k and x_{k-1} if this proves necessary). For comparison, recall that in Newton's method, the iteration formula was of the form

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}.$$

We begin by examining the first few iterations as shown in the figure shown earlier.



The initial guess x_0 is chosen arbitrarily. Then, we move to the height $g(x_0)$, move to the line $y = x$ at the point $(g(x_0), g(x_0))$, and take the horizontal coordinate of this point as the new iterate x_1. Thus, we have $x_1 = g(x_0)$. By similar reasoning, we observe that $x_2 = g(x_1)$, $x_3 = g(x_2)$, and so on. That is, in general $x_{k+1} = g(x_k)$.

Definition 5.11 (Fixed point method)**.** In the *fixed point method*, a root of the function $x - g(x)$ is approximated with the iteration formula

$$x_{k+1} = g(x_k),$$

which is initialized with the initial guess x_0.

Let us take a simple but illustrative example that has already been examined graphically above.

Example 5.12. Let us solve the equation

$$x - \frac{x}{2} = 0$$

with the fixed point method. An obvious choice is to rewrite the equation in the form $x = \frac{x}{2}$,

which gives the iteration formula $x_{k+1} = \frac{x_k}{2}$, so that $g(x) = \frac{x}{2}$. If the initial guess is chosen as $x_0 = 4$, we get

$$\begin{aligned}
x_0 &= 4 \\
x_1 &= g(x_0) = g(4) = \frac{4}{2} = 2 \\
x_2 &= g(x_1) = g(2) = \frac{2}{2} = 1 \\
x_3 &= g(x_2) = g(1) = \frac{1}{2} \\
&\vdots
\end{aligned}$$

We obtain a sequence that seems to converge to zero. By substituting or solving, we can check that this is the correct solution. Also, refer to the figure from the previous chapter or draw your own figure.

Another obvious choice is to rewrite the equation in the form $x = 2x$, which gives the iteration formula $x_{k+1} = 2x_k$. In this case, the iteration diverges for any initial guess that is not zero. For example, in the case $x_0 = 4$, we get

$$\begin{aligned}
x_0 &= 4 \\
x_1 &= g(x_0) = g(4) = 2 \cdot 4 = 8 \\
x_2 &= g(x_1) = g(8) = 2 \cdot 8 = 16 \\
x_3 &= g(x_2) = g(16) = 2 \cdot 16 = 32 \\
&\vdots
\end{aligned}$$

A quick examination of the fixed-point method can be done, similar to Newton's method, by storing the initial guess in the variable `ans` and repeatedly executing the command `g(ans)`. A more careful examination is best done using a loop.

Newton's method can actually be interpreted as a fixed point method, where

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

This allows for the treatment of the theory of Newton's method using the theory of the fixed point method, which is not addressed in this course.

5.3.4 On Convergence

It was noted above that the fixed point iteration seems to converge when the graph of the function g rises or falls sufficiently gently. The following theorem provides a precise condition.

Theorem 5.13. *Let c be a fixed point of a function g. Suppose that g is differentiable at the point c. If $\left|g'(c)\right| < 1$, then the fixed point iteration converges, provided that x_0 is chosen sufficiently close to the fixed point. In this case, the fixed point is called attractive or attracting. On the other hand, if $\left|g'(c)\right| > 1$, then the fixed point iteration converges only in the case where $x_n = c$. In this case, the fixed point is called repulsive or repelling.*

Proof. Omitted. □

We can illustrate the connection between the change in iterates and the derivative. For the iteration to converge, the horizontal shifts seen in the figures should decrease. This means that

we should have

$$\frac{|x_{k+2} - x_{k+1}|}{|x_{k+1} - x_k|} < K$$

for some $K \in (0,1)$, so that the next horizontal move is smaller than the previous one. We can rewrite this condition as

$$\left| \frac{g(x_{k+1}) - g(x_k)}{x_{k+1} - x_k} \right| < K,$$

which means that the average rate of change of the interval (x_k, x_{k+1}) (or (x_{k+1}, x_k) if $x_k > x_{k+1}$) must be less than one in absolute value. If $|g'(c)| < 1$, so that the instantaneous rate of change at c is less than one, then it is possible to find an interval containing c such that the average rate of change on that interval is less than one.

5.4 Jacobi Method for Linear Systems

The solutions to systems of *linear equations* fall into three categories: a unique solution, multiple solutions, or no solution. Regardless of which of these three types the solution belongs to, it can always be determined straightforwardly using Gaussian elimination. Because of this, one might think that solving systems of linear equations does not require any special numerical theory as one can simply let a computer determine the solution using Gaussian elimination. On the other hand, solving systems of linear equations is computationally expensive, with the computational effort growing *cubically with the size of the system*. This is what creates the need for numerical solution methods. For sufficiently large matrices, computational budget inevitably becomes insufficient. Another reason is that rounding errors can significantly affect the final result. This necessitates the development of more efficient and accurate methods for solving systems of linear equations.

A numerical method finds only one solution at a time for an equation or a system of equations. Therefore, it cannot determine the multiple solutions of a linear system that includes free parameters. Instead, the method merely outputs a collection of decimal numbers approximating the solution. For this reason we focus only on systems that have a unique solution. In practice, this means $n \times n$ systems, since in the case of a low-rank matrix, the solution can never be unique (too few equations), and in the case of a high-rank matrix, a unique solution is quite rare (too many equations). Thus, unless otherwise stated, the matrix that determines the linear system can be assumed to be a square matrix in this chapter.

5.4.1 Direct and Iterative Methods

Newton's method does not immediately yield a good approximation to the solution of an equation. Instead, a sequence of approximations is generated, which (under certain conditions) converge to the exact solution. In contrast, when solving a system of linear equations numerically using Gaussian elimination, only a single approximation to the solution is obtained (which may be exact in exceptional cases). Gaussian elimination is not an iterative process but an example of a *direct method*, whereas Newton's method is an example of an *iterative method*. A direct method is thus a non-iterative method that, in theory, provides an exact solution, but in practice outputs in an approximation due to factors such as rounding errors and the limited precision of real number representation in computers.

There are both direct and iterative methods for solving systems of linear equations. This course focuses on iterative methods. Direct methods are also covered to provide a better overall understanding of the topic, but a more detailed treatment of a significant direct method, the LU decomposition, is reserved for other courses. Systems of linear equations can be solved not only using the Gaussian elimination algorithm and its variations but also by using the Gaussian matrix and Cramer's rule. All of these are direct methods, and although they theoretically always

yield an exact solution, in practice, when computed with floating-point arithmetic, the solution is merely an approximation.

5.4.2 The Jacobi Method

Let us try to solve the system of equations

$$\begin{aligned} 4x_1 + 2x_2 + x_3 &= 8 \\ x_1 + 5x_2 + 3x_3 &= 12 \\ x_1 + x_2 + 3x_3 &= 8 \end{aligned}$$

using the fixed point method. The goal is to rewrite the system of equations in the form $\mathbf{x} = \mathbf{g}(\mathbf{x})$. There are infinitely many ways to do this, but for reasons that will become clear later, we choose to solve for the variable corresponding to the diagonal element in each row:

$$\begin{aligned} 4x_1 + 2x_2 + x_3 &= 8 \\ x_1 + 5x_2 + 3x_3 &= 12 \\ x_1 + x_2 + 3x_3 &= 8 \end{aligned}$$

which is equivalent with

$$\begin{aligned} 4x_1 &= 8 - 2x_2 - x_3 \\ 5x_2 &= 12 - x_1 - 3x_3 \\ 3x_3 &= 8 - x_1 + x_2 \end{aligned}$$

which is equivalent with

$$\begin{aligned} x_1 &= \frac{8}{4} - \frac{2}{4}x_2 - \frac{1}{4}x_3 \\ x_2 &= \frac{12}{5} - \frac{1}{5}x_1 - \frac{3}{5}x_3 \\ x_3 &= \frac{8}{3} - \frac{1}{3}x_1 - \frac{1}{3}x_2. \end{aligned}$$

This choice gives the following iteration formula for the fixed point method:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{k+1} = \begin{bmatrix} 2 \\ \frac{12}{5} \\ \frac{8}{3} \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{2} & -\frac{1}{4} \\ \frac{1}{5} & 0 & -\frac{3}{5} \\ \frac{1}{3} & -\frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_k .$$

When the formula is tested, it is observed that the iterates converge to the correct solution $\mathbf{x} = \begin{bmatrix} 1 & 1 & 2 \end{bmatrix}^\mathsf{T}$ regardless of the initial guess. For example, in the case $\mathbf{x}_0 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^\mathsf{T}$ we have

$$\mathbf{x}_1 \approx \begin{bmatrix} 2.0000 \\ 2.4000 \\ 2.6667 \end{bmatrix}, \quad \mathbf{x}_2 \approx \begin{bmatrix} 0.1333 \\ 0.4000 \\ 1.2000 \end{bmatrix}, \quad \ldots, \quad \mathbf{x}_{35} \approx \begin{bmatrix} 1.0000 \\ 1.0000 \\ 2.0000 \end{bmatrix}.$$

A Convergence Criterion When the system of equations discussed earlier is written in the form $\mathbf{x} = \mathbf{g}(\mathbf{x})$ in a different way than previously, the fixed point method may no longer converge. For example, if we use the simplest choice

$$4x_1 + 2x_2 + x_3 = 8$$

$$\begin{aligned} x_1 + 5x_2 + 3x_3 &= 12 \\ x_1 + x_2 + 3x_3 &= 8 \end{aligned}$$

which is equivalent with

$$\begin{aligned} x_3 &= 8 - 4x_1 - 2x_2 \\ x_1 &= 12 - 5x_2 - 3x_3 \\ x_2 &= 8 - x_1 - 3x_3 \end{aligned}$$

which is equivalent with

$$\begin{aligned} x_1 &= 12 - 5x_2 - 3x_3 \\ x_2 &= 8 - x_1 - 3x_3 \\ x_3 &= 8 - 4x_1 - 2x_2 \end{aligned}$$

which gives the iteration formula

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{k+1} = \begin{bmatrix} 12 \\ 8 \\ 8 \end{bmatrix} + \begin{bmatrix} 0 & -5 & -3 \\ -1 & 0 & -3 \\ -4 & -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_k ,$$

then the iterates diverge for the initial guess $\mathbf{x}_0 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^\mathsf{T}$:

$$\mathbf{x}_1 = \begin{bmatrix} 12 \\ 8 \\ 8 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} -52 \\ -28 \\ -56 \end{bmatrix}, \quad \ldots, \quad \mathbf{x}_3 = \begin{bmatrix} 320 \\ 228 \\ 272 \end{bmatrix}.$$

This happens with this iteration formula for any initial guess. On the other hand, solving for the variables corresponding to the diagonal elements, which was found to work with the previous system of equations, does not always work. For example, for the linear system of equations

$$\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

we obtain the iteration formula

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{k+1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_k$$

which diverges for every initial guess $\mathbf{x}_0$.

These observations raise the question about the cases in which the iteration converges. If for each row the absolute value of the diagonal element is greater than the sum of the absolute values of the other coefficients in that row, then the iteration formula solved for the diagonal elements will converge. Such a matrix is called *diagonally dominant*.

Definition 5.14 (Diagonally dominant matrix)**.** An $n \times n$ matrix $\mathbf{A}$ is said to be *diagonally dominant* if its elements satisfy

$$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^{n} |a_{ij}|$$

for every $i = 1, 2, \ldots, n$.

Theorem 5.15. *If an $n \times n$ matrix $\mathbf{A}$ is diagonally dominant and $\mathbf{x}_{k+1} = \mathbf{g}(\mathbf{x})$ is the iteration formula solved with respect to the diagonal elements using the fixed-point method, then the iteration converges to the solution of the system for every initial guess $\mathbf{x}_0$.*

Proof. Omitted. □

The theorem above does not cover all possible cases where the iteration can converge, but only provides a sufficient condition for convergence. There may therefore be other types of systems for which the iteraton converges either for all or certain initial guesses.

Note that even if the coefficient matrix of the system is not diagonally dominant as such, it can in some cases be transformed into such a matrix by swapping rows. For example, for the system of equations considered earlier,

$$\begin{aligned} 4x_1 + 2x_2 + x_3 &= 8 \\ x_1 + 5x_2 + 3x_3 &= 12 \\ x_1 + x_2 + 3x_3 &= 8 \end{aligned}$$

the coefficient matrix is diagonally dominant, but this is no longer the case if the rows of the system are rearranged into any of the other six possible orders. The order of the columns in the system could also be changed, as long as corresponding changes are made to the variables' order, but this is not useful if the issue could not be resolved by rearranging the rows. To convince yourself of this, consider the following system of equations:

$$\begin{aligned} x_1 + 2x_2 + 4x_3 &= 8 \\ 2x_1 + 5x_2 + 2x_3 &= 12 \\ 3x_1 + 2x_2 + 7x_3 &= 8 \end{aligned}$$

The theorem cannot hold when the coefficient matrix is not invertible because in that case the system does not have a unique solution. It can be shown that a diagonally dominant matrix is invertible. This proof, as well as the proof of the previous theorem, is omitted.

Iteration Formula Next we derive the iteration formula of the Jacobi method in the general case. This allows the iteration formula corresponding to each system of equations to be written directly using the information obtained from the coefficient matrix A and the constant vector $\mathbf{b}$.

We attempt to solve the system of equations $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ is diagonally dominant, with respect to the variables corresponding to the diagonal. These can be accessed in matrix form by decomposing the coefficient matrix into the sum of a lower triangular matrix, a diagonal matrix, and an upper triangular matrix:

$$\begin{aligned} \mathbf{Ax} &= \mathbf{b} \\ (\mathbf{L} + \mathbf{D} + \mathbf{U})\mathbf{x} &= \mathbf{b} \\ \mathbf{Lx} + \mathbf{Dx} + \mathbf{Ux} &= \mathbf{b} \\ \mathbf{Dx} &= \mathbf{b} - \mathbf{Lx} - \mathbf{Ux} \\ \mathbf{x} &= \mathbf{D}^{-1}\mathbf{b} - \mathbf{D}^{-1}(\mathbf{L} + \mathbf{U})\mathbf{x}. \end{aligned}$$

Definition 5.16 (Jacobi method)**.** The iteration formula of the *Jacobi method* is of the form

$$\mathbf{x}_{k+1} = \mathbf{Bx}_k + \mathbf{c},$$

where $\mathbf{B} = -\mathbf{D}^{-1}(\mathbf{L}+\mathbf{U})$ and $\mathbf{c} = \mathbf{D}^{-1}\mathbf{b}$.

At first glance, it might seem that such an iterative method for solving a system of equations is completely unnecessary, since it requires computing the coefficient matrix of size $n \times n$, whereas the inverse matrix $\mathbf{A}^{-1}$ of size $n \times n$ would provide a direct solution. However, $\mathbf{D}$ is a diagonal matrix, so its inverse matrix can be obtained directly by taking the reciprocals of the diagonal elements, which is a computationally cheap operation.

In the implementation of the solution algorithm, it is advisable to compute the matrix $\mathbf{B}$ and the vector $\mathbf{c}$ before entering the loop to avoid unnecessary computations.

Illustration of the Convergence Criterion Let us illustrate why diagonal dominance of the coefficient matrix suffices to ensure the convergence of the iteration. It is enough to show that $\mathbf{x}_n$ converges in the case of such a system of equations.

We begin by writing out the expressions for the iterates to derive an expression for the iterate $\mathbf{x}_n$ that depends on the index n. This would allow us to analyze what happens as $n \to \infty$. Based on the iteration formula $\mathbf{x}_{k+1} = \mathbf{B}\mathbf{x}_k + \mathbf{c}$ we have

$$\begin{aligned}
\mathbf{x}_1 &= \mathbf{B}\mathbf{x}_0 + \mathbf{c} \\
\mathbf{x}_2 &= \mathbf{B}\mathbf{x}_1 + \mathbf{c} = \mathbf{B}(\mathbf{B}\mathbf{x}_0 + \mathbf{c}) + \mathbf{c} = \mathbf{B}^2\mathbf{x}_0 + \mathbf{B}\mathbf{c} + \mathbf{c} \\
\mathbf{x}_3 &= \mathbf{B}\mathbf{x}_2 + \mathbf{c} = \mathbf{B}(\mathbf{B}^2\mathbf{x}_0 + \mathbf{B}\mathbf{c} + \mathbf{c}) + \mathbf{c} = \mathbf{B}^3\mathbf{x}_0 + \mathbf{B}^2\mathbf{c} + \mathbf{B}\mathbf{c} + \mathbf{c}.
\end{aligned}$$

For the nth iteration we get

$$\mathbf{x}_n = (\mathbf{I} + \mathbf{B} + \mathbf{B}^2 + \cdots + \mathbf{B}^{n-1})\mathbf{c} + \mathbf{B}^n\mathbf{x}_0.$$

We observe that the coefficient of the vector $\mathbf{c}$ forms an expression corresponding to a geometric sum. For real numbers it holds that

$$1 + b + b^2 + \cdots + b^{n-1} = \frac{1-b^n}{1-b}$$

(Multiply the equation with the denominator and you will see why). For matrices it analogously holds that

$$\mathbf{I} + \mathbf{B} + \mathbf{B}^2 + \cdots + \mathbf{B} + \mathbf{B}^{-1} = (\mathbf{I} - \mathbf{B})^{-1}(\mathbf{I} - \mathbf{B}^n),$$

where the inverse matrix can appear on either side because, exceptionally in this particular case,

$$(\mathbf{I} + \mathbf{B} + \mathbf{B}^2 + \cdots + \mathbf{B} + \mathbf{B}^{-1})(\mathbf{I} - \mathbf{B}) = (\mathbf{I} - \mathbf{B})(\mathbf{I} + \mathbf{B} + \mathbf{B}^2 + \cdots + \mathbf{B} + \mathbf{B}^{-1}).$$

Therefore

$$\mathbf{x}_n = (\mathbf{I} - \mathbf{B})^{-1}(\mathbf{I} - \mathbf{B}^n)\mathbf{c} + \mathbf{B}^n\mathbf{x}_0.$$

For real numbers, it is well known that $b^n \to 0$ as $n \to \infty$ if $|b| < 1$ (What happens in the cases $|b| > 1$, $b = 1$, and $b = -1$?). A corresponding result holds for matrices, but this year we do not discuss matrix norms on this years. It can be shown that this always holds for the matrix $\mathbf{B}$ in the iteration formula. For example, in the case discussed above,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{k+1} = \begin{bmatrix} 2 \\ \frac{12}{5} \\ \frac{8}{3} \end{bmatrix} + \begin{bmatrix} 0 & \frac{2}{4} & -\frac{1}{4} \\ \frac{1}{5} & 0 & -\frac{3}{5} \\ \frac{1}{3} & -\frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_k,$$

where the coefficient matrix is diagonally dominant, we have

$$\mathbf{B} = \begin{bmatrix} 0 & \frac{2}{4} & -\frac{1}{4} \\ \frac{1}{5} & 0 & -\frac{3}{5} \\ \frac{1}{3} & -\frac{1}{3} & 0 \end{bmatrix}, \quad \mathbf{B}^2 = \begin{bmatrix} \frac{1}{60} & \frac{1}{12} & -\frac{3}{10} \\ -\frac{1}{5} & \frac{3}{10} & -\frac{1}{20} \\ -\frac{1}{15} & \frac{1}{6} & \frac{7}{60} \end{bmatrix} \quad \text{and} \quad \mathbf{B}^3 = \begin{bmatrix} -\frac{1}{12} & \frac{13}{120} & -\frac{13}{240} \\ -\frac{13}{300} & -\frac{1}{12} & -\frac{13}{100} \\ \frac{13}{180} & -\frac{13}{180} & -\frac{1}{12} \end{bmatrix}.$$

This illustrates that diagonal dominance ensures that the elements of the matrix $\mathbf{B}$ are sufficiently small, so that they decrease and approach zero when raised to higher powers. On the other hand, in the case discussed above,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_{k+1} = \begin{bmatrix} 12 \\ 8 \\ 8 \end{bmatrix} + \begin{bmatrix} 0 & -5 & -3 \\ -1 & 0 & -3 \\ -4 & -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}_k,$$

where the coefficient matrix is not diagonally dominant, we have

$$\mathbf{B} = \begin{bmatrix} 0 & -5 & -3 \\ -1 & 0 & -3 \\ -4 & -2 & 0 \end{bmatrix}, \quad \mathbf{B}^2 = \begin{bmatrix} 17 & 6 & 15 \\ 12 & 11 & 3 \\ 2 & 20 & 18 \end{bmatrix} \quad \text{and} \quad \mathbf{B}^3 = \begin{bmatrix} -66 & -115 & -69 \\ -23 & -66 & -69 \\ -92 & -46 & -66 \end{bmatrix},$$

so that $\mathbf{B}$ blows up when raised to higher powers.

For diagonally dominant matrices it thus holds that

$$\mathbf{x}_n = (\mathbf{I}-\mathbf{B})^{-1}(\mathbf{I}-\mathbf{B}^n)\mathbf{c} + \mathbf{B}^n\mathbf{x}_0 \xrightarrow{n\to\infty} (\mathbf{I}-\mathbf{B})^{-1}\mathbf{c},$$

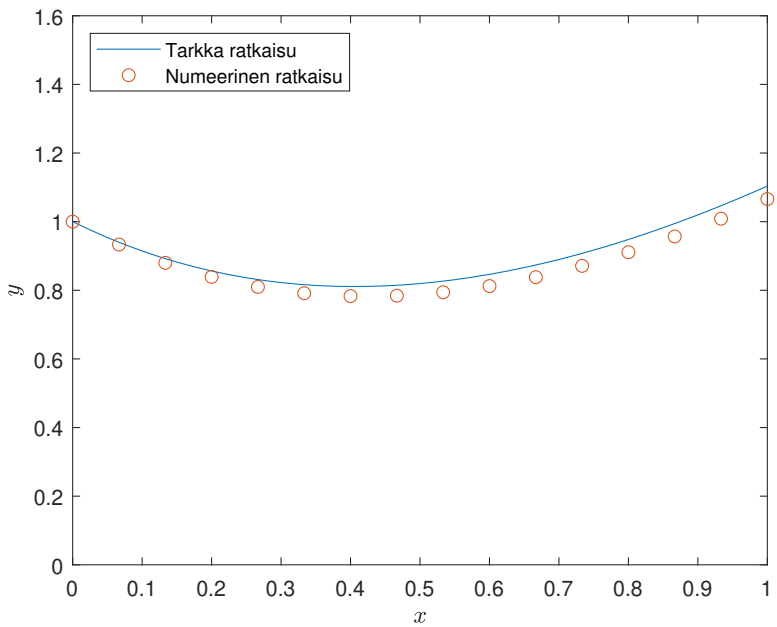
which means that the iteration converges to $(\mathbf{I}-\mathbf{B})^{-1}\mathbf{c}$. We also observe that the initial guess $\mathbf{x}_0$ does not affect whether or not the iteration converges.

One might wonder why the solution is not computed directly using the formula $(\mathbf{I}-\mathbf{B})^{-1}\mathbf{c}$. The reason is the same as why the solution is not computed using the formula $\mathbf{A}^{-1}\mathbf{b}$: calculating the inverse matrix is computationally expensive.

6 Differential Equations

Differential equations are generally very difficult to solve. Even though there are many different solution techniques for first-order differential equations, most of them cannot be solved analytically. All linear first-order differential equations can be solved analytically, provided the integrals required in the solution formula can be calculated. However, for nonlinear cases, a solution is only found in certain special cases. As we move to higher-order differential equations and systems of differential equations, analytical solutions are available for an even fewer number of cases.

When a typical equation, such as $e^x = x + 1$, cannot be solved analytically, it must be solved numerically, which gives an approximate solution. Similarly, when an initial value problem cannot be solved analytically, it must be solved numerically. Since the solution is not a single number but a whole function, the solution to the initial value problem is not a single approximation but several approximations over the considered interval. This is illustrated in the accompanying figure.



This chapter examines a few numerical methods for solving initial value problems. The aim is not to cover the topic comprehensively, but rather to become familiar with the basic ideas behind these methods. For example, the reason why higher-order differential equations must be written as systems of equations for use with the ODE solver will become clear.

6.1 Euler's Method

The initial value problem

$$y' = \sin(ty), \quad y(0) = 1,$$

cannot be solved analytically. Let us settle for a slightly less ambitious goal and attempt to find a rough approximation for the value of the solution y at $t = 1$. Based on the initial condition, we know that $y(0) = 1$. Using this information, we can also calculate the value of the function's derivative at that point:

$$y'(0) = \sin\Big(0 \cdot y(0)\Big) = \sin(0 \cdot 1) = 0.$$

Based on the derivative, we can estimate how the function behaves in the neighborhood of $t = 0$. The linear approximation developed at $t = 0$ is

$$L_0(t) = y(0) + y'(0)(t-0) = 1 + 0(t-0) = 1.$$

(The subscript 0 has been added to distinguish this from the linear approximations that will be formed at different points later.) In the neighbourhood of $t = 0$ we get the approximation

$$y(1) \approx L_0(1) = 1.$$

The previous approximation would be more accurate if the linear approximation were made closer to $t = 1$. The problem here is that we do not know the values of the function y and its derivative y' at a closer point. However, we can use the previous linear approximation to compute an estimate for this purpose. For example, at $t = 0.5$ we have

$$y(0.5) \approx L_0(0.5) = 1$$

and thus

$$y'(0.5) = \sin\Big(0.5 \cdot 1\Big) \approx 0.48.$$

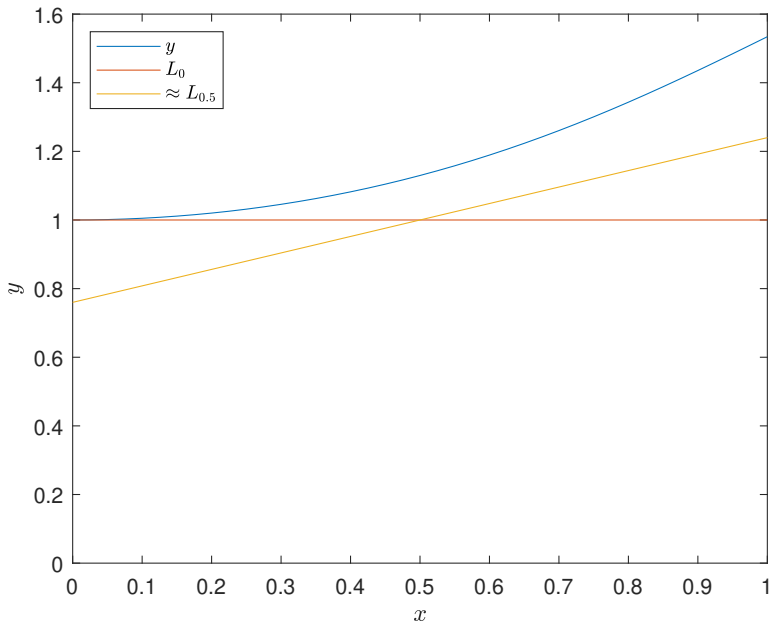
Using these values, an *estimate* for the linear approximation of the function y at $t = 0.5$ is obtained as

$$L_{0.5}(t) = y(0.5) + y'(0.5)(t-0.5) \approx 1 + \sin(0.5)(t-0.5) \approx 0.48t + 0.76.$$

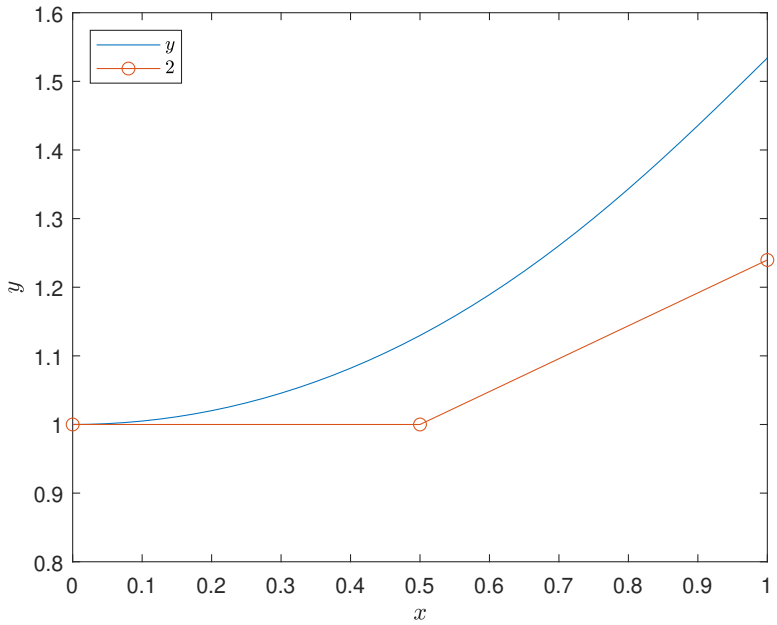
With this we can approximate $y(1)$ as

$$y(1) \approx L_{0.5}(1) \approx 0.48 \cdot 1 + 0.76 = 1.24.$$

This is a better approximation than the previous one, as illustrated ibythe accompanying figure. Note that the figure also clearly shows that $0.48t + 0.76$ is only an estimate for the linear approximation $L_{0.5}$. In reality, the corresponding line should be the tangent at $t = 0.5$.

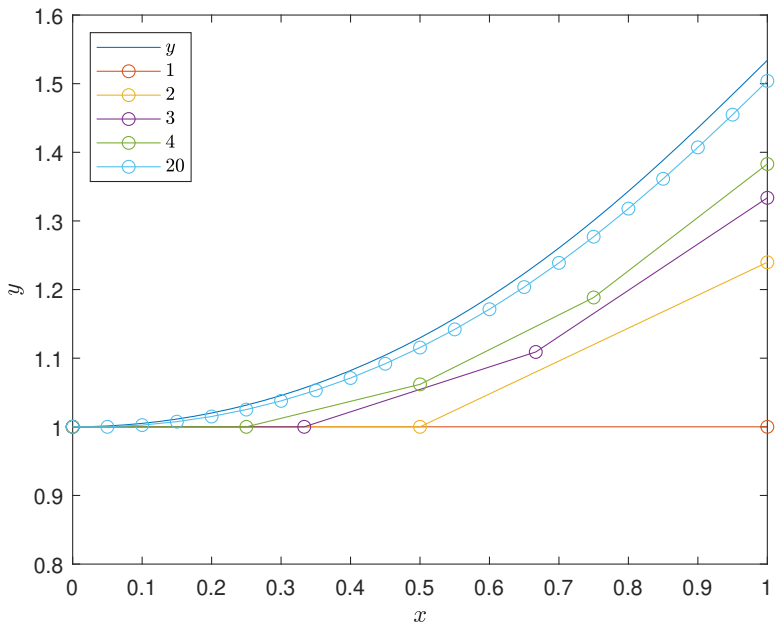


For the sake of generalization, it is clearer to present the previous figure in the following form.



In this case, the tangent lines are drawn only over the intervals where they are used as approximations for the function value.

A better estimate than the previous two-linear-approximation case can be obtained by using three or more linear approximations. The following figure illustrates this for the cases of 1, 2, 3, 4, and 20 linear approximations.



Based on the figure, it is clear that a method has been found in which an arbitrarily accurate estimate for $y(1)$ can be obtained by simply using enough linear approximations. It is also significant that, in the process, estimates for other values of the function y are obtained as well. For example, in the case of four linear approximations, estimates for the values of the function at $y(0.25)$, $y(0.5)$, and $y(0.75)$ are also obtained. These estimates can be considered as the *numerical solution* of the differential equation over the interval $[0,1]$. Thus, a method has been found to solve first-order initial value problems numerically.

Before moving on to a more general treatment of the topic, we will show that computing the next estimate from the previous one in the preceding example can be simplified into a rather straightforward process. For this purpose, we will use the notation t_i for the points where the linear approximations are formed, and the corresponding estimates for the function values will be denoted as y_1. We thus have

$$y_0 = y(t_0), \quad y_1 \approx y(t_1), \quad y_2 \approx y(t_2), \quad \ldots, \quad y_n \approx y(t_n).$$

With this notation the quantities in the previous example are $y_1 = y_0 + y'(t_0)(t_1 - t_0)$, $y_2 = y_1 + y'(t_1)(t_2 - t_1)$ and more generally

$$y_{i+1} = y_i + y'(t_i)(t_{i+1} - t_i).$$

Given the initial condition, we can thus compute the estimate y_1, and based on this, we can further compute the estimate y_2, and so on.

The method presented above for numerically solving the differential equation is an example of a *step method*. The idea behind these methods is as follows:

1. Move from the initial value point t_0 to the point t_1.
2. Calculate an approxmation to the value of the solution function at t_1, i.e., $y_1 \approx y(t_1)$.
3. Move to the next point and calculate an approximation at this point.
4. Repeat the previous step until the value at the endpoint t_n of the considered interval is calculated.
5. The approximations y_1, y_2, $\ldots$, y_n constitute a numerical solution to the initial value problem.

The transitions $t_i \to t_{i+1}$ are called *steps*, and the differences $t_{i+1} - t_i$ are called *step sizes* or *step lengths*. In the methods of this course, the step size is always constant, as in the previous examples, and is denoted by h. It does not necessarily have to be constant; for instance, it might be reasonable to take shorter steps when the function grows rapidly, i.e., when the derivative is large in absolute value. However, this complicates the formulation of the general algorithm.

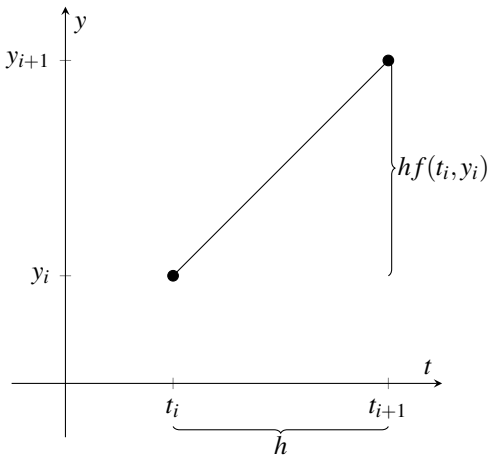
Each estimate y_{i+1} is computed based on the previously computed estimates. The formula used to obtain the new estimate is called the *recurrence formula*. In different methods, the recurrence formula takes different forms. In the method discussed above, the formula is of the form

Definition 6.1 (Euler's method)**.**

$$y_{i+1} = y_i + hf(t_i, y_i),$$

where $f(t,y) = y'(t)$.

This method is called *Euler's method*. In Euler's method the next estimate is obtained from the previous one using a linear approximation, as illustrated in the accompanying figure.



Example 6.2. Let us solve the initial value problem

$$y' = xy, \quad y(0) = 1,$$

at $x = 1$ with Euler's method and (a) one (b) two (c) one hundred steps. In all parts the function is $f(x,y) = xy$ and the interval is $[0,1]$.

(a) $n = 1$, so that the step size is

$$h = \frac{1-0}{1} = 1.$$

0th step: $x_0 = 0$, $y_0 = 1$. **1st step:**

$$x_1 = x_0 + h = 0 + 1 = 1,$$

$$y_1 = y_0 + hf(x_0, y_0) = 1 + 1 \cdot f(0,1) = 1 + 1 \cdot 0 \cdot 1 = 1.$$

For comparison, it is worth mentioning that the equation can be solved by separating the variables. The solution is $y(x) = e^{\frac{1}{2}x^2}$, so that $y(1) = e^{\frac{1}{2}} \approx 1.65$.

(b) $n = 2$, so that the step size is

$$h = \frac{1-0}{2} = \frac{1}{2}.$$

0th step: $x_0 = 0$, $y_0 = 1$. **1st step:**

$$x_1 = x_0 + h = 0 + \frac{1}{2} = \frac{1}{2},$$

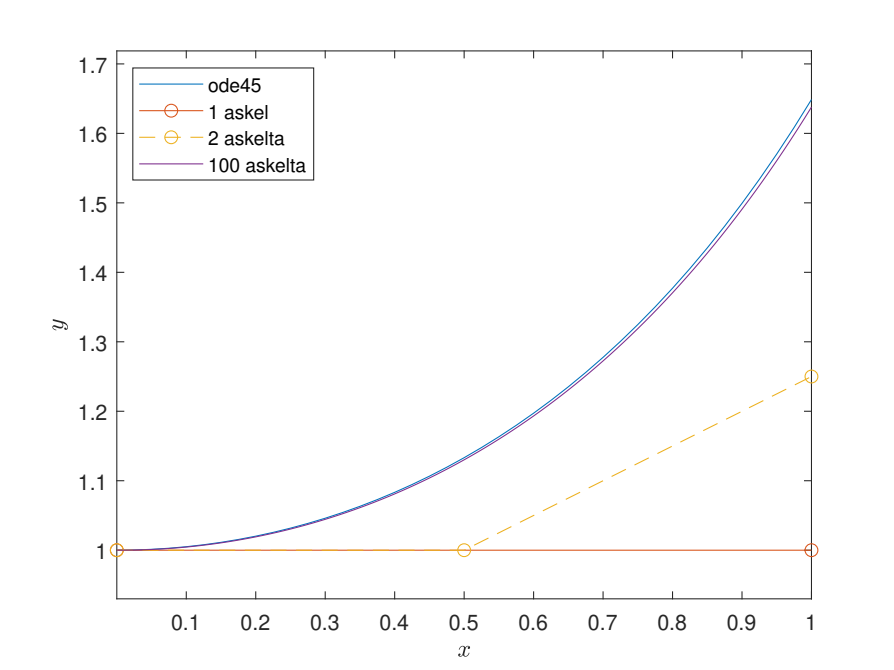
$$y_1 = y_0 + hf(x_0, y_0) = 1 + \frac{1}{2} f(0,1) = 1 + \frac{1}{2} \cdot 0 \cdot 1 = 1.$$

2nd step:

$$x_2 = x_1 + h = \frac{1}{2} + \frac{1}{2} = 1,$$

$$y_2 = y_1 + hf(x_1, y_1) = 1 + \frac{1}{2} f(\tfrac{1}{2}, 1) = 1 + \frac{1}{2} \cdot \frac{1}{2} \cdot 1 = \frac{5}{4}.$$

(c) Problems like this are done on a computer. To allow comparison let us draw a figure that includes all the cases.



The following theorem provides information about the error of Euler's method.

Theorem 6.3. *Suppose that y is continuous on $[t_0,t_n]$, twice differentiable on (t_0,t_n). If there are constants $M,L \geq 0$ such that $|y''(x)| \leq M$ for all $x \in (t_0,t_n)$ and f is L-Lipschitz in its second argument, meaning that $|f(t,y)-f(t,\tilde{t})| \leq L|y-\tilde{y}|$ for all $y,\tilde{y} \in \mathbb{R}$, then Euler's method with uniform step-size $h=(t_n-t_0)/n$ satisfies*

$$|y(t_n)-y_n| \leq \frac{Mh}{2L}(e^{L(t_n-t_0)}-1).$$

Proof. The recurrence formula for Euler's method is $y_{i+1}=y_i+hf(t_i,y_i)$ while Taylor's theorem states that

$$y(t_{i+1})=y(t_i)+hf(t_i,y(t_i))+\frac{h^2}{2}y''(c)$$

for some $c \in [t_i,t_{i+1}]$. Therefore the error ε_{i+1} satisfies

$$\begin{aligned}
\varepsilon_{i+1}=|y(t_{i+1})-y_{i+1}| &= \left|y(t_i)+hf(t_i,y(t_i))+\frac{h^2}{2}y''(c)-y_i-hf(t_i,y_i)\right| \\
&= \left|\varepsilon_i+h[f(t_i,y(t_i))-f(t_i,y_i)]+\frac{h^2}{2}y''(c)\right| \\
&\leq \varepsilon_i+h|f(t_i,y(t_i))-f(t_i,y_i)|+\frac{M}{2}h^2 \\
&\leq \varepsilon_i+Lh|y_{t_i}-y_i|+\frac{M}{2}h^2 \\
&= (1+Lh)\varepsilon_i+\frac{M}{2}h^2
\end{aligned}$$

for any $i \leq n-1$, where the first inequality used the assumption on the boundedness of the second derivative of y and the second the Lipschitz assumption. By iterating this inequality we obtain

$$\begin{aligned}
\varepsilon_n &\leq (1+Lh)\varepsilon_{n-1} + \frac{M}{2}h^2 \\
&\leq (1+Lh)\left[(1+LH)\varepsilon_{n-2} + \frac{M}{2}h^2\right] + \frac{M}{2}h^2 \\
&= (1+Lh)^2\varepsilon_{n-2} + (1+Lh)\frac{M}{2}h^2 + \frac{M}{2}h^2 \\
&\vdots \\
&\leq (1+Lh)^n\varepsilon_0 + \frac{M}{2}h^2\sum_{i=0}^{n-1}(1+Lh)^i \\
&= \frac{M}{2}h^2\sum_{i=0}^{n-1}(1+Lh)^i
\end{aligned}$$

since $\varepsilon_0 = |y(t_0) - y_0| = 0$. The geometric summation formula $\sum_{i=0}^{n-1} r^i = (1-r^n)/(1-r)$ for $r \neq 1$ then gives

$$\varepsilon_i \leq \frac{M}{2}h^2\sum_{i=0}^{n-1}(1+Lh)^k = \frac{M}{2}h^2 \cdot \frac{1-(1+Lh)^n}{1-(1+Lh)} = \frac{M}{2}h^2 \cdot \frac{(1+Lh)^n - 1}{Lh}.$$

From $1+x \leq e^x$ (recall the Taylor series of the exponential function) for all $x \geq 0$, we get

$$\varepsilon_n \leq \frac{M}{2}h^2 \cdot \frac{(1+Lh)^n - 1}{Lh} \leq \frac{M}{2}h^2 \cdot \frac{e^{Lhn} - 1}{Lh} = \frac{M}{2}h^2 \cdot \frac{e^{L(t_n - t_0)} - 1}{Lh} = \frac{Mh}{2L}(e^{L(t_n - t_0)} - 1)$$

because $h = (t_n - t_0)/n$. □

6.2 Midpoint Method

Previously, Euler's method was introduced using tangent lines. It can also be approached through integration. Next we will proceed this way, as it provides a straightforward way to derive other numerical methods.

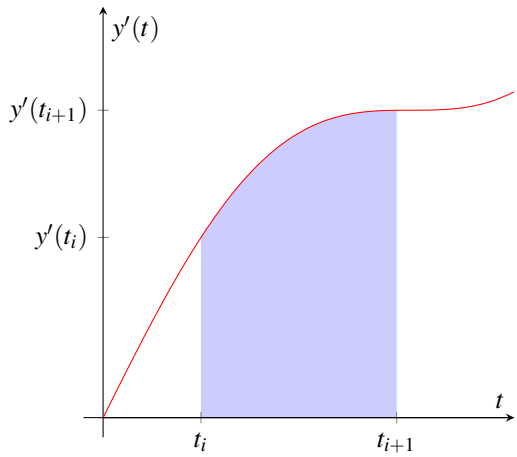
A definite integral satisfies

$$F(b) - F(a) = \int_a^b f(x)\,\mathrm{d}x.$$

This equation can also be written as

$$f(b) - f(a) = \int_a^b f'(x)\,\mathrm{d}x,$$

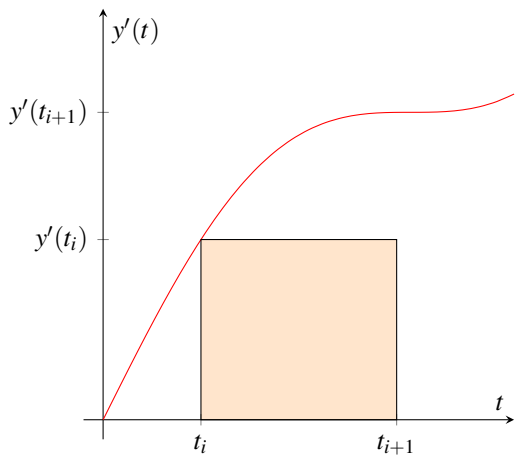
where f corresponds to the function F in the previous equation. Based on this formulation, the colored area in the figure



satisfies

$$y(t_{i+1}) - y(t_i) = \int_{t_i}^{t_{i+1}} y'(t)\,\mathrm{d}t.$$

It is essential to note that the vertical axis is not y as in the previous examination of the topic using tangent lines, but rather y'. The area can be roughly approximated by the rectangle in the figure below.



This rectangle has length $t_{i+1} - t_i = h$ and its height equals the value of the derivative at the starting point of the interval, $y'(t_i)$. Thus

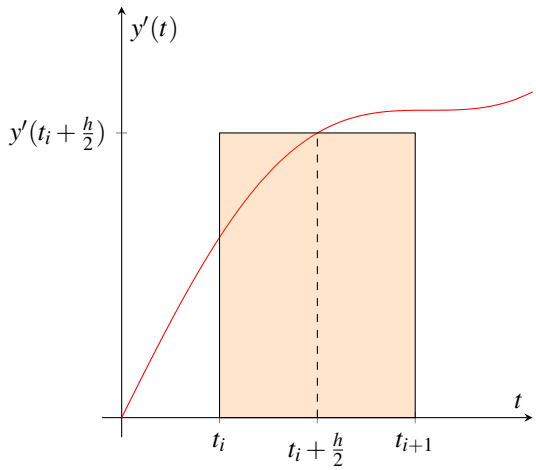
$$y(t_{i+1}) - y(t_i) \approx hy'(t_i).$$

By using the approximations $y(t_{i+1}) \approx y_{i+1}$ and $y(t_i) \approx y_i$ on the left-hand side and writing $y'(t_k) = f(t_i, y_i)$, we obtain

$$y_{i+1} - y_i \approx hf(t_i, y_i),$$

that corresponds to the recurrence formula of Euler's method. The term $hf(t_i, y_i)$ can thus be interpreted as a rectangle that approximates the integral $\int_{t_i}^{t_{i+1}} y'(t)\,\mathrm{d}t$ (What was its interpretation in the previous chapter?). Note that in this case, the estimate $f(t_i, y_i)$ is used for the height of the rectangle instead of the exact value $f\Big(t_i, y(t_i)\Big)$, so the rectangle used in Euler's method is merely an approximation of the rectangle in the previous figure.

The approximation used above for the integral $\int_{t_i}^{t_{i+1}} y'(t)\,\mathrm{d}t$ is clearly too small, as evidenced by the figure. A better approximation could be obtained, for instance, by choosing the function value at the midpoint of the interval $[t_i, t_{i+1}]$, i.e., at $t_i + \frac{h}{2}$, as the height of the rectangle.



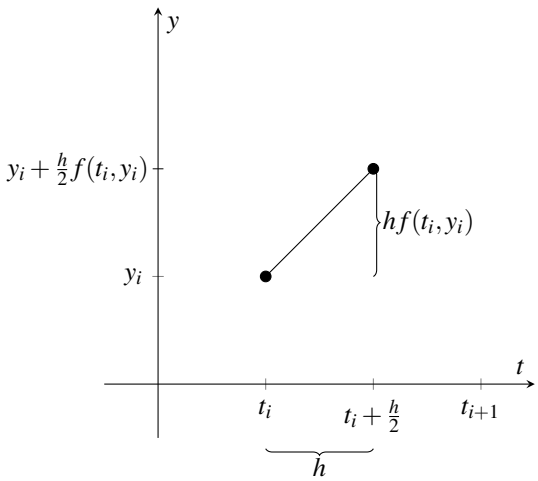
Then the recurrence formula becomes

$$y_{i+1} = y_i + hf\left(t_i + \frac{h}{2}, y\left(t_i + \frac{h}{2}\right)\right).$$

The problem with this formula is that the value $y(t_i + \frac{h}{2})$ is not known. But it can be approximates using a tangent line as in the previous chapter. This gives the approximation

$$y(t_i + \tfrac{h}{2}) \approx y(t_i) + \tfrac{h}{2} f(t_i, y_i),$$

which is illustrated in the accompanying figure. (Note that in this figure the vertical axis is y rather than y'.)



Due to this approximation, the used rectangle is, like in Euler's method, only an approximation to the rectangle drawn earlier. This gives a method whose recurrence formula is

Definition 6.4 (Midpoint method)**.**

$$y_{i+1} = y_i + hf\left(t_i + \frac{h}{2}, y_i + \frac{h}{2} f(t_i, y_i)\right).$$

This method is called the *midpoint method*.

Example 6.5. Let us solve the initial value problem

$$y' = xy, \quad y(0) = 1,$$

at $x = 1$ with the midpoint method and (a) one (b) two steps. In both parts $f(x,y) = xy$ and the considered interval is $[0,1]$.

(a) $n = 1$, so that the step size is

$$h = \frac{1-0}{1} = 1.$$

0th step: $x_0 = 0$, $y_0 = 1$. **1st step:**

$$x_1 = x_0 + h = 0 + 1 = 1,$$

$$\begin{aligned} y_1 &= y_0 + hf\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} f(x_0, y_0)\right) \\ &= 1 + 1 \cdot f\left(0 + \frac{1}{2}, 1 + \frac{1}{2} f(0,1)\right) \\ &= 1 + f\left(\frac{1}{2}, 1 + \frac{1}{2} \cdot 0 \cdot 1\right) \\ &= 1 + f\left(\frac{1}{2}, 1\right) \\ &= 1 + \frac{1}{2} \cdot 1 \\ &= \frac{3}{2}. \end{aligned}$$

(b) $n = 2$, so that the step size is

$$h = \frac{1-0}{2} = \frac{1}{2}.$$

0th step: $x_0 = 0$, $y_0 = 1$. **1st step:**

$$x_1 = x_0 + \frac{1}{2} = 0 + \frac{1}{2} = \frac{1}{2},$$

$$\begin{aligned} y_1 &= y_0 + hf\left(x_0 + \frac{h}{2}, y_0 + \frac{h}{2} f(x_0, y_0)\right) \\ &= 1 + \frac{1}{2} f\left(0 + \frac{1}{4}, 1 + \frac{1}{4} f(0,1)\right) \\ &= 1 + \frac{1}{2} f\left(\frac{1}{4}, 1\right) \\ &= \frac{9}{8}. \end{aligned}$$

2nd step:

$$x_2 = x_1 + h = \frac{1}{2} + \frac{1}{2} = 1,$$

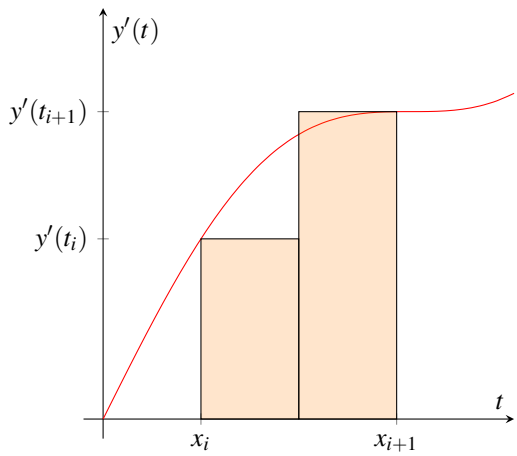
$$
\begin{aligned}
y_2 &= y_1 + hf\left(x_1 + \frac{h}{2}, y_1 + \frac{h}{2} f(x_1, y_1)\right) \\
&= \frac{9}{8} + \frac{1}{2} f\left(\frac{1}{2} + \frac{1}{4}, \frac{9}{8} + \frac{1}{4} f(\tfrac{1}{2}, \tfrac{9}{8})\right) \\
&= \frac{9}{8} + \frac{1}{2} f(\tfrac{3}{4}, \tfrac{81}{64}) \\
&= \frac{9}{8} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{81}{64} \\
&= \frac{819}{512}.
\end{aligned}
$$

6.3 Runge–Kutta Methods

The formulas for Euler's method and the midpoint method are obtained by approximating the integral

$$y_{i+1} \approx y_i + \int_{t_i}^{t_{i+1}} y'(t)\,\mathrm{d}t$$

appearing in the approximation using a rectangle. Nothing prevents us from approximating it in some other way. In particular, we could approximate the integral using *two* rectangles. This is illustrated in the accompanying figure.



This interpretation allows the generalization to a method based on an estimate constructed out of two arbitrary rectangles. Such methods are called *two-stage Runge–Kutta methods*, and their recurrence formulas have the form

Definition 6.6 (Two-stage Runge–Kutta)**.**

$$y_{i+1} = y_i + h(b_1 k_1 + b_2 k_2), \quad b_1 + b_2 = 1,$$

where

$$k_1 = f(t_i, y_i)$$

and

$$k_2 = f(t_i + ch, y_i + ahk_1), \quad a, c \in (0, 1].$$

In the formula, $b_1 h$ and $b_2 h$ are the widths of the rectangles, and k_1 and k_2 are their heights. The height k_1 is always calculated at the starting point of the interval $[t_i, t_{i+1}]$, and k_2 at some

other point within the interval. The case $b_1 = b_2 = \frac{1}{2}$, $c = a = 1$ corresponds to a method called *Heun's method*. The case $b_1 = 0$, $b_2 = 1$, $c = a = \frac{1}{2}$ corresponds to the midpoint method, since its recurrence formula can also be written as

$$y_{i+1} = y_i + hk_2,$$

$$\begin{aligned} k_1 &= f(t_i, y_i), \\ k_2 &= f\left(t_i + \frac{h}{2}, y_i + \frac{h}{2}k_1\right). \end{aligned}$$

Here, the area of the first rectangle is $0 \cdot hk_1 = 0$. Euler's method is not a two-stage Runge–Kutta method but a one-stage method, because it does not require the height k_2.

In an n-stage Runge–Kutta method, the integral $\int_{t_i}^{t_{i+1}} y'(t)\,\mathrm{d}t$ is approximated by n rectangles. In this case, the recurrence formula has the form

Definition 6.7 (n-stage Runge–Kutta)**.**

$$y_{i+1} = y_i + h\sum_{i=1}^{n} b_i k_i, \quad \sum_{i=1}^{n} b_i = 1,$$

$$\begin{aligned} k_1 &= f(t_i, y_i), \\ k_i &= f\left(t_i + c_i h, y_i + h\sum_{m=j}^{j-1} a_{jm} k_m\right), \quad j = 2, 3, \ldots, n. \end{aligned}$$

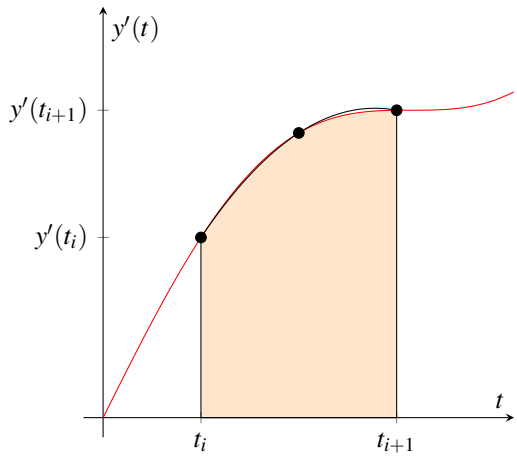
Perhaps the most well-known a Runge–Kutta method is the *classical Runge–Kutta method* (RK4), whose recurrence formula has the form

Definition 6.8 (Classical Runge–Kutta — RK4)**.**

$$y_{i+1} = y_i + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4),$$

$$\begin{aligned} k_1 &= f(t_i, y_i), \\ k_2 &= f(t_i + \tfrac{h}{2}, y_i + \tfrac{h}{2}k_1), \\ k_3 &= f(t_i + \tfrac{h}{2}, y_i + \tfrac{h}{2}k_2), \\ k_4 &= f(t_i + h, y_i + hk_3). \end{aligned}$$

This is a four-stage method, and the four-rectangle estimate used actually corresponds to Simpson's method, where the upper (or lower) boundary of the area being estimated is approximated by a parabola passing through the points $\left(t_i, y(t_i)\right)$, $\left(t_i + \frac{h}{2}, y(t_i + \frac{h}{2})\right)$, and $\left(t_{i+1}, y(t_{i+1})\right)$ This is illustrated in the accompanying figure.



The previous figure corresponds to the ideal situation, and the actual situation is its estimate. Two different estimates are used for the derivative value at the midpoint of the interval, $f\Big(t_i + \frac{h}{2}, y(t_i + \frac{h}{2})\Big)$: in one, $y(t_i + \frac{h}{2}) \approx y_i + \frac{h}{2}k_1$, and in the other, $y(t_i + \frac{h}{2}) \approx y_i + \frac{h}{2}k_2$. These are, in the rectangle interpretation, the heights of the two middle rectangles.

Example 6.9. Let us solve the initial value problem

$$y' = xy, \quad y(0) = 1,$$

at $x = 1$ with the classical Runge–Kutta method and one step. The function is $f(x,y) = xy$ and the considered interval $[0,1]$. $n = 1$, so that the step-size is

$$h = \frac{1-0}{1} = 1.$$

0th step $x_0 = 0$, $y_0 = 1$. **1st step**

$$x_1 = x_0 + h = 0 + 1 = 1,$$

$$k_1 = f(x_0, y_0) = f(0,1) = 0 \cdot 1 = 0,$$

$$k_2 = f(x_0 + \tfrac{h}{2}, y_0 + \tfrac{h}{2}k_1) = f(0 + \tfrac{1}{2}, 1 + \tfrac{1}{2} \cdot 0) = \frac{1}{2} \cdot 1 = \frac{1}{2},$$

$$k_3 = f(x_0 + \tfrac{h}{2}, y_0 + \tfrac{h}{2}k_2) = f(0 + \tfrac{1}{2}, 1 + \tfrac{1}{2}\tfrac{1}{2}) = \frac{1}{2} \cdot \frac{5}{4} = \frac{5}{8}$$

$$k_4 = f(x_0 + h, y_0 + hk_3) = f(0 + 1, 1 + 1 \cdot \tfrac{5}{8}) = 1 \cdot \frac{13}{8} = \frac{13}{8},$$

$$y_1 = y_0 + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) = 1 + \frac{1}{6}\left(0 + 2 \cdot \frac{1}{2} + 2 \cdot \frac{5}{8} + \frac{13}{8}\right) = \frac{79}{48}.$$

Let us summarize the solutions produced by different methods with one step:

$$\begin{aligned} y_{\text{exact}}(1) &\approx 1.649 \\ y_{\text{Euler}}(1) &\approx 1.000 \\ y_{\text{midpoint}}(1) &\approx 1.500 \\ y_{\text{RK4}}(1) &\approx 1.646. \end{aligned}$$

RK4 clearly outperforms the other methods, but it requires most computations.

6.4 Numerical Solutions to Systems of Differential Equations

The previous methods can also be used for systems of first-order differential equations. Let us illustrate this with an example in the case of Euler's method.

Example 6.10. Let us solve the initial value problem

$$\begin{matrix} y_1' = y_1 + y_2 \\ y_2' = y_1^2 \end{matrix}, \quad \begin{matrix} y_1(0) = -1 \\ y_2(0) = 2 \end{matrix},$$

at $t = 1$ using Euler's method with step-size $h = 0.5$. Let us start by modifying the notation so that subscripts have only one meaning in the calculations:

$$\begin{matrix} y' = y + u \\ u' = y^2 \end{matrix}, \quad \begin{matrix} y(0) = -1 \\ u(0) = 2 \end{matrix}.$$

Denote $f(t,y,u) = y + u$ and $g(t,y,u) = y^2$ **0th step:** $t_0 = 0$, $y_0 = -1$, $u_0 = 2$. **1st step:**

$$t_1 = t_0 + h = 0 + \frac{1}{2} = \frac{1}{2},$$

$$y_1 = y_0 + hf(t_0, y_0, u_0) = -1 + \frac{1}{2} \cdot f(0,-1,2) = -1 + \frac{1}{2} \cdot (-1+2) = -\frac{1}{2},$$

$$u_1 = u_0 + hg(t_0, y_0, u_0) = 2 + \frac{1}{2} \cdot g(0,-1,2) = 2 + \frac{1}{2}(-1)^2 = \frac{5}{2}.$$

2nd step:

$$t_2 = t_1 + h = \frac{1}{2} + \frac{1}{2} = 1,$$

$$y_2 = y_1 + hf(t_1, y_1, u_1) = -\frac{1}{2} + \frac{1}{2} f(\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{5}{2}) = -\frac{1}{2} + \frac{1}{2}\left(-\frac{1}{2} + \frac{5}{2}\right) = \frac{1}{2},$$

$$u_2 = u_1 + hg(t_1, y_1, u_1) = \frac{5}{2} + \frac{1}{2} g(\tfrac{1}{2}, -\tfrac{1}{2}, \tfrac{5}{2}) = \frac{5}{2} + \frac{1}{2}\left(-\frac{1}{2}\right)^2 = \frac{21}{8}.$$

We obtain the approximation

$$\begin{matrix} y(1) \approx \frac{1}{2} \\ u(1) \approx \frac{21}{8} \end{matrix}.$$

6.5 Implicit Methods

In Euler's method $y_{k+1} = y_k + hf(t_k, y_k)$ the integral $\int_{t_k}^{t_{k+1}} y'(t)\,\mathrm{d}t$ is approximated by a rectangle, the height $y'(t)$ of which is calculated using at the starting point of the interval $[t_k, t_{k+1}]$, and its width is $h = t_{k+1} - t_k$. If the height of the rectangle is instead calculated at the endpoint of the interval, the formula becomes

$$y_{k+1} = y_k + hf(t_{k+1}, y_{k+1}).$$

For example, in the case of the differential equation $y'(x) = xy$, this would result in the equation

$$y_{k+1} = y_k + ht_{k+1} y_{k+1}.$$

In this equation, the value on the left-hand side cannot be obtained by substituting known values on the right-hand side, as in the methods discussed earlier. However, we can solve for y_{k+1}:

$$\begin{aligned} y_{k+1} &= y_k + ht_{k+1}y_{k+1} \\ y_{k+1} &= \frac{y_k}{1 - ht_{k+1}}. \end{aligned}$$

This is an example of an *implicit method*, whereas the methods discussed earlier are called *explicit methods*. A method is called explicit if its formula defines the value of y_{k+1} explicitly, i.e., directly, and implicit if its formula defines the value of y_{k+1} implicitly, i.e., indirectly.

In an implicit method, y_{k+1} must be solved separately from the method's formula. In the previous example, it was solved quite straightforwardly. Generally, this is not the case, and, for example, in the case of the function $f(t,y) = e^y$, the equation would not be solvable manually. In such cases, it must be solved numerically, for instance, using Newton's method.

7 Optimisation

In this section we discuss *optimisation*, which refers to finding points at which the value of a function is either maximal or minimal.[4] We focus on the *gradient descent* algorithm, variants of which underlay much of machine learning.

7.1 Motivation: Training Neural Networks

Let us begin by motivating the need for optimisation.

7.1.1 Linear Regression

Suppose that we have observations (or outputs) $y_1, \dots, y_n \in \mathbb{R}$ at some distinct locations (or inputs) $x_1, \dots, x_n \in \mathbb{R}$. This is our *training data*. To name a few examples, the locations could describe spatial coordinates of some physical sensors, they could be time instances (in which case we are working with a *time series*), or they could be different parameters that an experiment is performed with. Given these data, we want to predict what the observation might be at a location that is not included in the dataset. To do this, we can postulate that observations have linear relationship to the locations:

$$y_i = a + bx_i + \varepsilon_i \quad \text{for} \quad i \in \{1, \dots n\} \tag{7.1}$$

where a and b determine the relationship between x_i and y_i and the residual ε_i accounts for observation noise or influence on y_i of sources other than x_i. Our task is to find a "good" or "optimal" values for a and b. This is called *linear regression*. One way to achieve this is to select a and b that minimise the *sum of squared residuals*

$$L(a,b) = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - a - bx_i)^2. \tag{7.2}$$

The function L is an important example of a *loss function*, or *cost function*. The selection of a loss function is arbitrary (e.g., we could alternatively try to minimise $\sum_{i=1}^{n} |\varepsilon_i|$), but the *quadratic* loss function in (7.2) is mathematically extremely convenient, yielding the *ordinary least squares* method. The values a^* and b^* of a and b that minimise $L(a,b)$ are obtained from the equation (we omit the derivation of this result)

$$\begin{bmatrix} a^* \\ b^* \end{bmatrix} = (\mathbf{X}^\intercal \mathbf{X})^{-1} \mathbf{X}^\intercal \mathbf{y}, \quad \text{where} \quad \mathbf{X} = \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \in \mathbb{R}^{n \times 2} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \tag{7.3}$$

This solution to the minimisation proble is *unique*, in that $L(a^*, b^*) < L(a,b)$ for every pair $(a,b) \neq (a^*, b^*)$. The *linear fits* $f(x) = a^* + bx^*$ produced by linear regression are depicted in Figure 7.1 for two rather different datasets. The figure shows that the linear fit can be quite useful if the data indeed exhibit a linear trend, as is the case for Dataset 1. That is, for an input x_0 not contained in the training data it seems plausible that $f(x_0)$ would be close to the corresponding output, y_0. However, the fit is useless if no such trend is present, as is the case for Dataset 2. This is because the relationship (7.1) between x_i and y_i that we have postulated is not particularly expressive, having only two parameters, a and b.

[4] The theory and practice of optimisation will be covered in much more detail on the 2nd year course *Optimization* (BM20A8201).

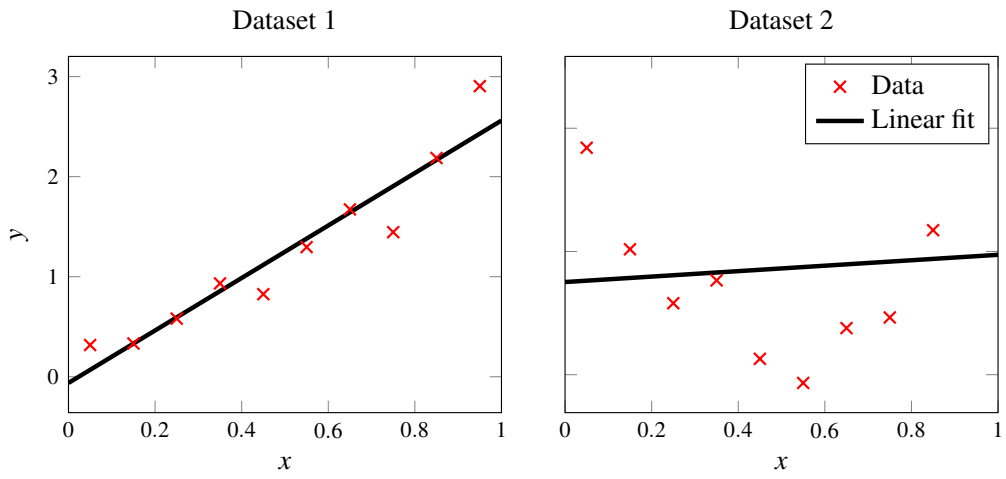


Figure 7.1: The linear fits $f(x) = a^* + bx^*$ for two different datasets. Here a^* and b^* are computed from (7.3). The fit is quite good for the first dataset but completely useless for the second.

7.1.2 Neural Networks

At their core, *neural networks* consist of nothing more than (i) a flexible postulate like (7.1) for the relationship between the inputs and outputs; (ii) selection of some parameters by minimising a loss function; and (iii) computation of a fit that is then used to predict the output at unobserved inputs. Let us consider a simple two-layer network based on the *sigmoid activation function*

$$\sigma(x) = \frac{1}{1+e^{-x}}. \tag{7.4}$$

Let $w_1, w_2 \in \mathbb{R}$ be *weights* and $b_1, b_2 \in \mathbb{R}$ *biases*. Inspired by (7.1), we postulate that the training outputs $0 \leq y_i \leq 1$ are related to the inputs x_i via the equation

$$y_i = \sigma(w_2\sigma(w_1x_i + b_1) + b_2) + \varepsilon_i \quad \text{for} \quad i \in \{1, \ldots, n\}. \tag{7.5}$$

Define

$$F(x \mid w_1, w_2, b_1, b_2) = \sigma(w_2\sigma(w_1x + b_1) + b_2), \tag{7.6}$$

so that (7.5) becomes $y_i = F(x_i \mid w_1, w_2, b_1, b_2) + \varepsilon_i$. As in the case of linear regression, we may define a quadratic loss function

$$L(w_1, w_2, b_1, b_2) = \sum_{i=1}^{n} \varepsilon_i^2 = \sum_{i=1}^{n} (y_i - F(x_i \mid w_1, w_2, b_1, b_2))^2 \tag{7.7}$$

and select the weights w_1, w_2 and biases b_1, b_2 by minimising this loss function. But we hit a roadblock. For, unlike in the case of ordinary least squares method that provides the simple expression (7.3) for the minimisers of the loss function (7.2), no such convenient formula is available now. Instead, we have to device an *optimisation method* to minimise (7.7). Because in real problems there are thousands of weights and biases and the training data sets are huge, it is essential that this method be efficient, in that it should call the loss function as few times as possible. In this section we shall learn how to use *derivatives* and *gradients* to describe the direction and rate of change of functions and how to implement the *gradient descent* algorithm, the workhorse of machine learning. Results of selecting the parameters w_1, w_2, b_1 and b_2 using gradient descent are displayed in Figure 7.2.

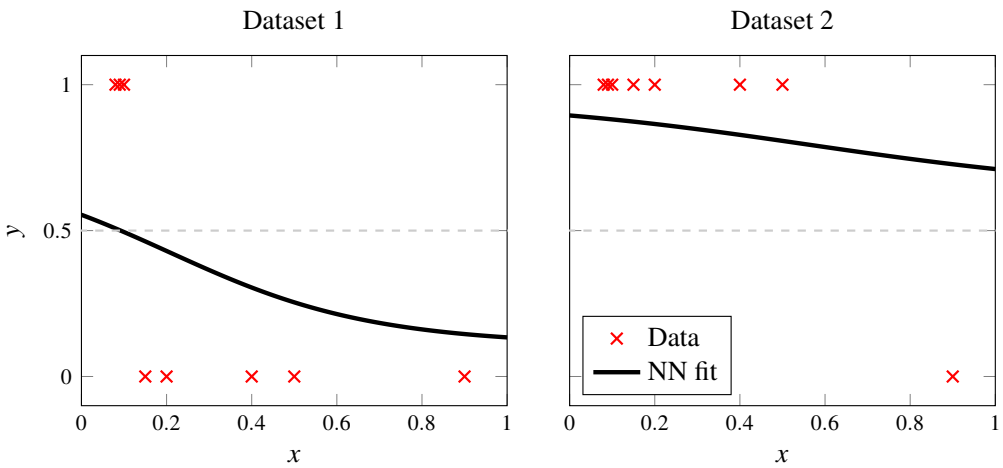


Figure 7.2: The neural network (NN) fit $f(x) = F(x \mid w_1^*, w_2^*, b_1^*, b_2^*) = \sigma(w_2^* \sigma(w_1^* x + b_1^*) + b_2^*)$ for two different datasets. Here w_1^*, w_2^*, b_1^* and b_2^* have been computed using gradient descent. The datasets consist of $n = 8$ *labels* $y \in \{0, 1\}$ (i.e., points labelled $y = 1$ are in a category A and points those labelled $y = 0$ in category B). After computing the neural network fit f, we may classify a new point by, for example, saying it is in category A if $f(x) > \frac{1}{2}$ and in category B if $f(x) \leq \frac{1}{2}$. This *decision boundary* is plotted as a dashed grey line. Such classification works for Dataset 1 (at least barely), but not for Dataset 2. This is understandable, as the neural network we have used is extremely simple. *In reality far more than two functions* σ *would appear in* (7.5).

7.2 Univariate Gradient Descent

The goal of optimisation is to find point (or points) x^* at which a function f of interest attains its smallest or largest value. In machine learning f is usually a loss function that is to be minimised (recall the examples in Section 7.1). We therefore focus on minimisation. Note that this comes at no loss of generality because

$$\text{maximising } f \quad \text{is equivalent to} \quad \text{minimising } -f.$$

Definition 7.1 (Minima)**.** Let $f \colon \mathbb{R} \to \mathbb{R}$ be a function. A point $x^* \in \mathbb{R}$ is a

- *global minimum point* of f if $f(x^*) \leq f(x)$ for every $x \in \mathbb{R}$;
- *local minimum point* of f if there exists $\varepsilon > 0$ such that $f(x^*) \leq f(x)$ for every point x in the interval $[x^* - \varepsilon, x^* + \varepsilon]$ of length 2ε centered at x^*.

If x^* is a global minimum point, $f(x^*)$ is called the *global minimum*. If x^* is a local minimum point, $f(x^*)$ is called a *local minimum*.

Global and local maxima are defined analogously, with "$\geq$" replacing "$\leq$" in the definitions. Minima and maxima are collectively known as *extrema* and minimum and maximum points as *extremal points*. A function attains its smallest possible value at a global minimum point. At a local minimum point a function attains a value that is smaller or equal to its "nearby" values. The definition of a local minimum point says simply that there is *some* interval around x^* on which $f(x^*) \leq f(x)$. Note that a global minimum point is also a local minimum point. Figure 7.3 shows global and local minima of a particular function. We focus on local optimisation as opposed to *global optimisation*. That is, we are content if we can find a local minimum point. Finding a global minimum point can be significantly more challenging. Points at which the derivative of a function is zero and local minimum and maximum points are intimately connected.

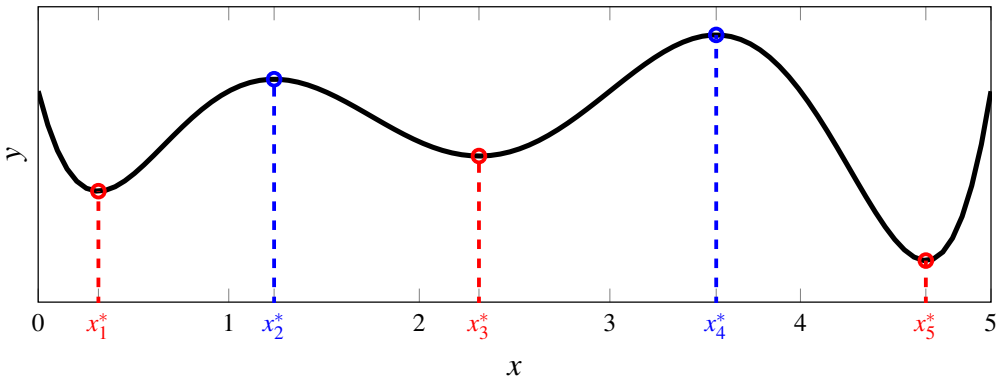


Figure 7.3: The local minimum points (red; x_1^*, x_3^*, and x_5^*) and maximum points (blue; x_2^* and x_4^*) of a certain function. The local minimum point x_5^* is also a *global* minimum point. Although this cannot be determined from the figure, this function has no global maximum point as it tends to ∞ as $x \to -\infty$ and $x \to \infty$.

Definition 7.2 (Critical point)**.** A point $x^* \in \mathbb{R}$ is a *critical point*, or a *stationary point*, of a differentiable function f if $f'(x^*) = 0$.

Example 7.3. Consider the functions $f(x) = \cos(x)$ and $g(x) = \frac{1}{3}x^3 - x$. Then $f'(x) = -\sin(x)$. From trigonometry we know that $\sin(x) = 0$ if and only if $x = \pi n$ for n an integer (i.e., $n \in \mathbb{Z}$). Therefore $f'(x) = 0$ if and only if $x = \pi n$ for an integer n. These are thus the critical points of $f(x) = \cos(x)$. That is, the cosine has an infinite number of critical points. For g we have $g'(x) = x^2 - 1$, so that $g'(x) = 0$ if and only if $x = 1$ or $x = -1$. Therefore the polynomial $g(x) = \frac{1}{3}x^3 - x$ has two critical points, $x^* = -1$ and $x^* = 1$.

Theorem 7.4 (Fermat's theorem)**.** *Every local extremal point x^* of a differentiable function f is a critical point, in that $f'(x^*) = 0$*

Proof. Suppose that x^* is a local extremal point. Assume that it is a local minimum point (the proof is completely analogous for a local maximum point). By the definition of a local minimum point (Definition 7.1), there is $\varepsilon > 0$ such that

$$f(x^*) \leq f(x) \quad \text{for all} \quad x \in [x^* - \varepsilon, x^* + \varepsilon]. \tag{7.8}$$

Recall that the derivative $f'(x^*)$ is defined as the limit

$$f'(x^*) = \lim_{h \to 0} \frac{f(x^* + h) - f(x^*)}{h} \tag{7.9}$$

of the difference quotient. For every $h \in (0, \varepsilon)$ the difference quotient satisfies

$$\frac{f(x^* + h) - f(x^*)}{h} \leq \frac{f(x^*) - f(x^*)}{h} = 0 \tag{7.10}$$

since, in this case, $f(x^* + h) \leq f(x^*)$ by (7.8). For $h \in (-\varepsilon, 0)$ we similarly obtain

$$\frac{f(x^* + h) - f(x^*)}{h} = \frac{f(x^*) - f(x^* + h)}{|h|} \geq \frac{f(x^*) - f(x^*)}{|h|} = 0, \tag{7.11}$$

where we have used the fact that $h = -|h|$ if $h < 0$. Equations (7.10) and (7.11) show that the difference quotient (a) is non-positive if h is positive and close to zero (b) non-negative if h negative and close to zero. Because the derivative is the limit of the difference quotient as $h \to 0$ and the only number that is simultaneously non-positive and non-negative is 0, we conclude that $f(x^*) = 0$. □

We can see Theorem 7.4 in action in Figure 7.3: at each extremal point the tangent line would be completely horizontal and the derivative hence zero. This result furnishes the basic principle behind local optimisation methods: *find a point at which the derivative is zero*. However, Theorem 7.4 only says that extremal points are critical points, not that every critical point is an extremal point. Whether or not a critical point is an extremal point can be usually determined by examining the derivative between critical points.

Theorem 7.5 (Classification of critical points)**.** *Let f be a differentiable function. Suppose that $x_1^* < x_2^* < x_3^*$ are critical points of f.*

1. *If $f'(x) > 0$ for all $x_1^* < x < x_2^*$ and $f'(x) > 0$ for all $x_2^* < x < x_3^*$, then x_2^* is* not an extremal point.
2. *If $f'(x) < 0$ for all $x_1^* < x < x_2^*$ and $f'(x) < 0$ for all $x_2^* < x < x_3^*$, then x_2^* is* not an extremal point.
3. *If $f'(x) > 0$ for all $x_1^* < x < x_2^*$ and $f'(x) < 0$ for all $x_2^* < x < x_3^*$, then x_2^* is a* local maximum point.
4. *If $f'(x) < 0$ for all $x_1^* < x < x_2^*$ and $f'(x) > 0$ for all $x_2^* < x < x_3^*$, then x_2^* is a* local minimum point.

Proof. Omitted. □

Recall that the sign of its derivative tells us whether a function is increasing or decreasing. In Cases 1 and 2 of Theorem 7.5 the function is increasing (Case 1) or decreasing (Case 2) both to the left and right of the critical point x_2^*. Therefore x_2^* cannot be an extremal point. In Case 3 the function is increasing to the left of x_2^* and decreasing to the right of it, meaning that it must attain a local maximum at x_2^*. In Case 4 the function is decreasing to the left and increasing to the right, so that it must have a local minimum at x_2^*. The following example and Figure 7.4 illustrate this.

Example 7.6. Let us classify the critical points of the function

$$f(x) = x^3 \exp(-\tfrac{1}{2}x^2), \tag{7.12}$$

which has the derivative

$$f'(x) = (3x^2 - x^4)\exp(-\tfrac{1}{2}x^2). \tag{7.13}$$

Since the function $\exp(-\frac{1}{2}x^2)$ is everywhere positive, $f'(x) = 0$ if and only if $3x^2 - x^4 = 0$. The equation

$$3x^2 - x^4 = x^2(3 - x^2) = 0 \tag{7.14}$$

has the three solutions $x = 0$, $x = \sqrt{3}$, and $x = -\sqrt{3}$. The critical points of f are therefore

$$x_1^* = -\sqrt{3}, \quad x_2^* = 0, \quad \text{and} \quad x_3^* = \sqrt{3}. \tag{7.15}$$

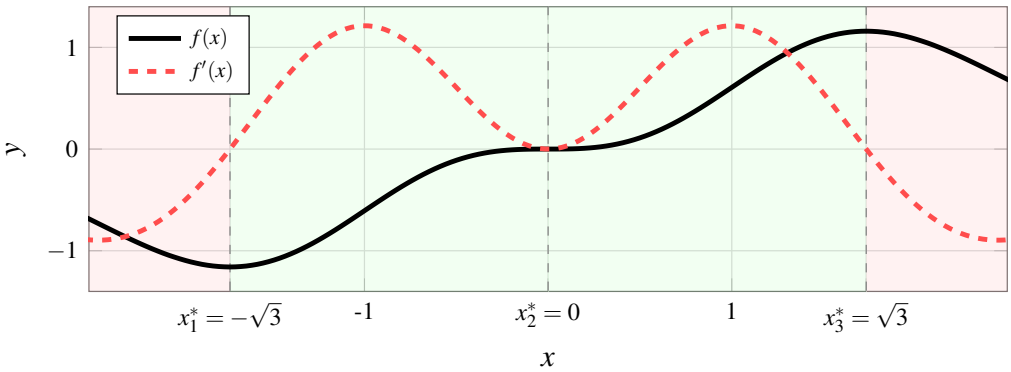


Figure 7.4: The function $f(x) = x^3 \exp(-\frac{1}{2}x^2)$ and its derivative in (7.13). The function is increasing (i.e., derivative is positive) on the green interval and decreasing (i.e., derivative is negative) on the two red intervals. The derivative vanishes at x_1^*, x_2^*, and x_3^*, which means that these are critical points of f. The critical points x_1^* and x_3^* are extremal points, but the critical point $x_2 = 0$ is not because f is increasing both the left and right of x_2^*.

Let us begin with x_2^*. The sign of the derivative (7.13) is equal to the sign of $3x^2 - x^4$. By writing this as $x^2(3 - x^2)$, we observe that

$$f'(x) > 0 \quad \text{for all} \quad -\sqrt{3} < x < 0 \quad \text{and} \quad 0 < x < \sqrt{3}. \tag{7.16}$$

Therefore f' does *not* change sign at $x_2^* = 0$ and thus x_2^* is not an extremal point by Case 1 of Theorem 7.5. For $x_1^* = -\sqrt{3}$ and $x_3^* = \sqrt{3}$ we note that

$$f'(x) < 0 \quad \text{for all} \quad x < -\sqrt{3} \quad \text{and } x > \sqrt{3}. \tag{7.17}$$

Therefore f is decreasing to the left of x_1^* and increasing to the right of it, meaning that x_1^* is a local (in fact, global) minimum point. Conversely, f is increasing to the left of x_3^* and increasing to the right of it, meaning that x_3^* is a local (in fact, global) maximum point. Figure 7.4 shows what is going on.

When minimising a function it is natural to move to a direction in which the function decreases. This basic idea is behind the following *gradient descent* method. In the multivariate version of this algorithm (Section 7.3) derivatives are replaced by gradients, their generalisations to higher dimensions.

Definition 7.7 (Univariate gradient descent)**.** Let $\eta > 0$ be the *learning rate* and $x_0 \in \mathbb{R}$ an initial point. The *gradient descent* finds approximations x_k to a minimum point of a differentiable function $f\colon \mathbb{R} \to \mathbb{R}$ via the iteration

$$x_{k+1} = x_k - \eta f'(x_k). \tag{7.18}$$

The iteration is terminated after sufficiently iterations or when $|f'(x_k)|$ is sufficiently small.

In this very basic form the gradient descent algorithm is quite simple. A loop is executed until the derivative becomes smaller than some user-specified tolerance (e.g., $|f(x_k)| \leq 10^{-6}$) or a maximum number of iterations is reached (which is to protect against infinite loops). Iteration formula (7.39) is the heart of the algorithm: at each iteration we move a step towards the direction *opposite* to the direction the derivative is pointing. For example, if $f'(x_n) > 0$, the function is increasing at x_n, which means that we want to move to the left (i.e., take x_{n+1} smaller than x_n).

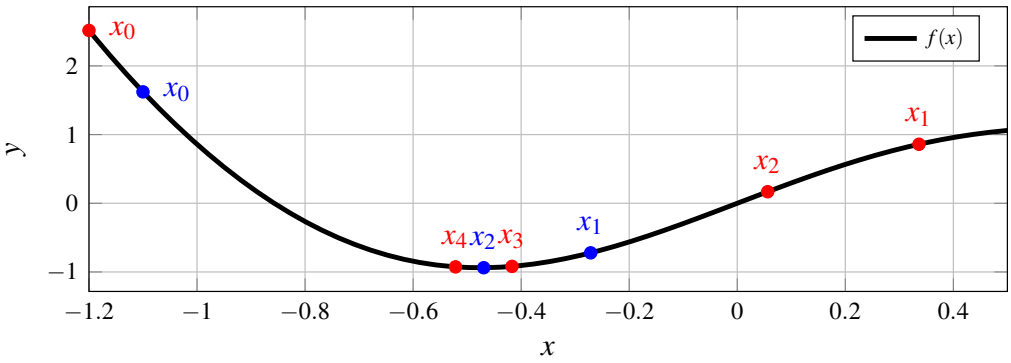


Figure 7.5: Gradient descent for the function $f(x) = x^4 + \sin(3x)$ with global minimum point $x^* \approx -0.47562$. The sequence of red points was produced by gradient descent with $x_0 = -1.2$ and $\eta = 0.16$; that of blue points with $x_0 = -1.1$ and $\eta = 0.1$. Observe that the point x_{n+1} is always to the right of x_n if f is decreasing at x_n and to the left of x_{n+1} if f is increasing at x_n (e.g., x_2 is to the left of x_1 since f is increasing at x_1).

The *learning rate*, or *step size*, η controls how large steps we take at each iteration. If η is too large, the algorithm may not work properly, "jumping around" too much. If the function has a local minimum, $f'(x_n)$ will hopefully get closer and closer to zero, so that the steps taken [i.e., $-\eta f'(x_n)$] become smaller and smaller.

Since the derivative is close to zero at the point returned by the gradient descent algorithm, this point is close to a critical point. Of course, a critical point is not necessarily a local extremal point, as we recall from Theorem 7.5 and Example 7.6. Figure 7.5 shows gradient descent in action.

Remark 7.8. Critical points and Theorem 7.4 are useful only when f is differentiable. Consider the absolute value function $f(x) = |x|$, which one cannot differentiate at $x = 0$. However, it is clear that f has a global minimum point $x^* = 0$.

7.3 Multivariate Gradient Descent

We next generalise gradient descent for minimising multivariate functions $f\colon \mathbb{R}^d \to \mathbb{R}$.

7.3.1 Gradient

The gradient is a vector of partial derivatives.

Definition 7.9 (Gradient)**.** The *gradient* of a function $f\colon \mathbb{R}^d \to \mathbb{R}$ is the function $\nabla f\colon \mathbb{R}^d \to \mathbb{R}^d$ given by

$$\nabla f(\mathbf{x}) = \nabla f(x_1, \ldots, x_d) = \nabla_{\mathbf{x}} f(\mathbf{x}) = \frac{\mathrm{d}}{\mathrm{d}\mathbf{x}} f(\mathbf{x}) = \begin{bmatrix} \partial_1 f(\mathbf{x}) \\ \vdots \\ \partial_d f(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \dfrac{\partial}{\partial x_1} f(\mathbf{x}) \\ \vdots \\ \dfrac{\partial}{\partial x_d} f(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^d. \quad (7.19)$$

The upside-down Δ (delta) in the notation for the gradient in (7.19) is called *nabla*. Here gradient is a column vector, a convention far from universal. Here notations such as $\mathrm{d}f(\mathbf{x})/\mathrm{d}\mathbf{x}$ in which one differentiates with respect to a vector are to be interpreted in a coordinate-wise sense:

one forms a vector by differentiating $f(\mathbf{x})$ with respect to each coordinate of $\mathbf{x}$. We consider a function $f\colon \mathbb{R}^d \to \mathbb{R}$ differentiable if all its partial derivative exist, meaning that the gradient exists as well.

Example 7.10. Consider the function $f(\mathbf{x}) = f(x_1, x_2) = x_1 + x_2 - x_1 x_2^3$. The partial derivatives of this function are

$$\partial_1 f(x_1, x_2) = 1 - x_2^3 \quad \text{and} \quad \partial_2 f(x_1, x_2) = 1 - 3x_1 x_2^2. \tag{7.20}$$

Therefore the gradient is given by

$$\nabla f(x_1, x_2) = \begin{bmatrix} \partial_1 f(x_1, x_2) \\ \partial_2 f(x_1, x_2) \end{bmatrix} = \begin{bmatrix} 1 - x_2^3 \\ 1 - 3x_1 x_2^2 \end{bmatrix}. \tag{7.21}$$

Example 7.11. Consider the *Gaussian function*

$$f(\mathbf{x}) = f(x_1, x_2, x_3) = \exp(-\tfrac{1}{2}\|\mathbf{x}\|^2) = \exp\left(-\frac{x_1^2 + x_2^2 + x_3^2}{2}\right). \tag{7.22}$$

This function has the partial derivatives

$$\partial_i f(x_1, x_2, x_3) = -x_i \exp(-\tfrac{1}{2}\|\mathbf{x}\|^2) \quad \text{for} \quad i \in \{1, 2, 3\}. \tag{7.23}$$

Consequently,

$$\nabla f(x_1, x_2, x_3) = \begin{bmatrix} \partial_1 f(x_1, x_2, x_3) \\ \partial_2 f(x_1, x_2, x_3) \\ \partial_3 f(x_1, x_2, x_3) \end{bmatrix} = \begin{bmatrix} -x_1 \exp(-\frac{1}{2}\|\mathbf{x}\|^2) \\ -x_2 \exp(-\frac{1}{2}\|\mathbf{x}\|^2) \\ -x_3 \exp(-\frac{1}{2}\|\mathbf{x}\|^2) \end{bmatrix} = -\exp(-\tfrac{1}{2}\|\mathbf{x}\|^2)\,\mathbf{x}. \tag{7.24}$$

Note that in the last expression we multiply the vector $\mathbf{x} \in \mathbb{R}^3$ by the scalar $-\exp(-\frac{1}{2}\|\mathbf{x}\|^2)$.

Example 7.12. It is often convenient to use the operations of linear algebra to express multivariate functions. Let

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 3 & 7 \end{bmatrix} \in \mathbb{R}^{3\times 3} \quad \text{and} \quad \mathbf{a} = \begin{bmatrix} 8 \\ 1 \\ -3 \end{bmatrix} \in \mathbb{R}^3. \tag{7.25}$$

Consider the function $f\colon \mathbb{R}^3 \to \mathbb{R}$ given by

$$f(\mathbf{x}) = \mathbf{a}^\intercal \mathbf{A}\mathbf{x}. \tag{7.26}$$

We have

$$\mathbf{a}^\intercal \mathbf{A} = \begin{bmatrix} 8 & 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 3 & 7 \end{bmatrix} = \begin{bmatrix} 9 & -17 & -11 \end{bmatrix}, \tag{7.27}$$

so that

$$f(\mathbf{x}) = \begin{bmatrix} 9 & -17 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 9x_1 - 17x_2 - 11x_3. \tag{7.28}$$

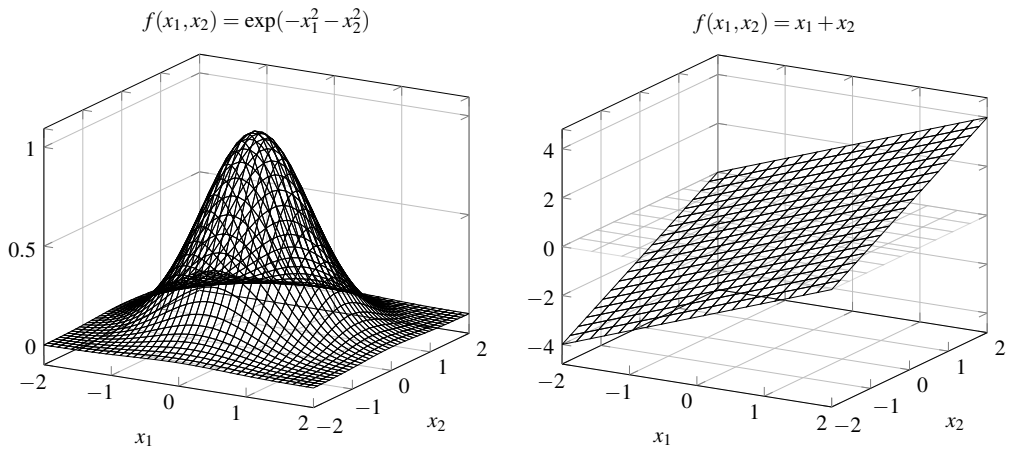


Figure 7.6: The two functions from $\mathbb{R}^2$ to $\mathbb{R}$ used in Examples 7.13 and 7.14.

The partial derivatives of f are

$$\partial_1 f(\mathbf{x}) = 9, \quad \partial_2 f(\mathbf{x}) = -17, \quad \text{and} \quad \partial_3 f(\mathbf{x}) = -11. \tag{7.29}$$

Recall that the derivative gives the *direction* (if positive, the function increases; if negative, the function decreases) and the *rate of change* of a function. The gradient has a similar property. Namely,

the gradient $\nabla f(\mathbf{x})$ gives the direction and the rate of fastest increase of f at $\mathbf{x}$.

The following example and Figure 7.7 illustrate what this means.

Example 7.13. Let us consider the Gaussian function

$$f(\mathbf{x}) = f(x_1,x_2) = \exp(-\tfrac{1}{2}\|\mathbf{x}\|^2) = \exp\left(-\frac{x_1^2+x_2^2}{2}\right) \tag{7.30}$$

of two variables; see Figure 7.6. We have used the three-dimensional version in Example 7.11 but two-dimensional functions are somewhat easier to plot than three-dimensional ones. Exactly the same computation as that in Example 7.11 provides us the gradient

$$\nabla f(\mathbf{x}) = \nabla f(x_1,x_2) = -\exp(-\tfrac{1}{2}\|\mathbf{x}\|^2)\,\mathbf{x}. \tag{7.31}$$

The gradient at $\mathbf{x}$ is thus a vector that points towards the origin. Now, observe that the function f depends only on the norm $r = \|\mathbf{x}\|$ (i.e., the distance of $\mathbf{x}$ from the origin). That is, for any vectors $\mathbf{x},\mathbf{z} \in \mathbb{R}^2$ we have $f(\mathbf{x}) = f(\mathbf{z})$ if $\|\mathbf{x}\| = \|\mathbf{z}\|$. Moreover, f is a decreasing function of $r = \|\mathbf{x}\|$. Therefore the direction in which f increases the fastest is the direction in which $\|\mathbf{x}\|$ decreases the fastest. The fastest decrease in $\|\mathbf{x}\|$ is obtained by moving directly towards the origin, which is precisely what the the gradient tells us.

Example 7.14. Consider the function $f(x_1,x_2) = x_1 + x_2$, which defines a plane depicted in

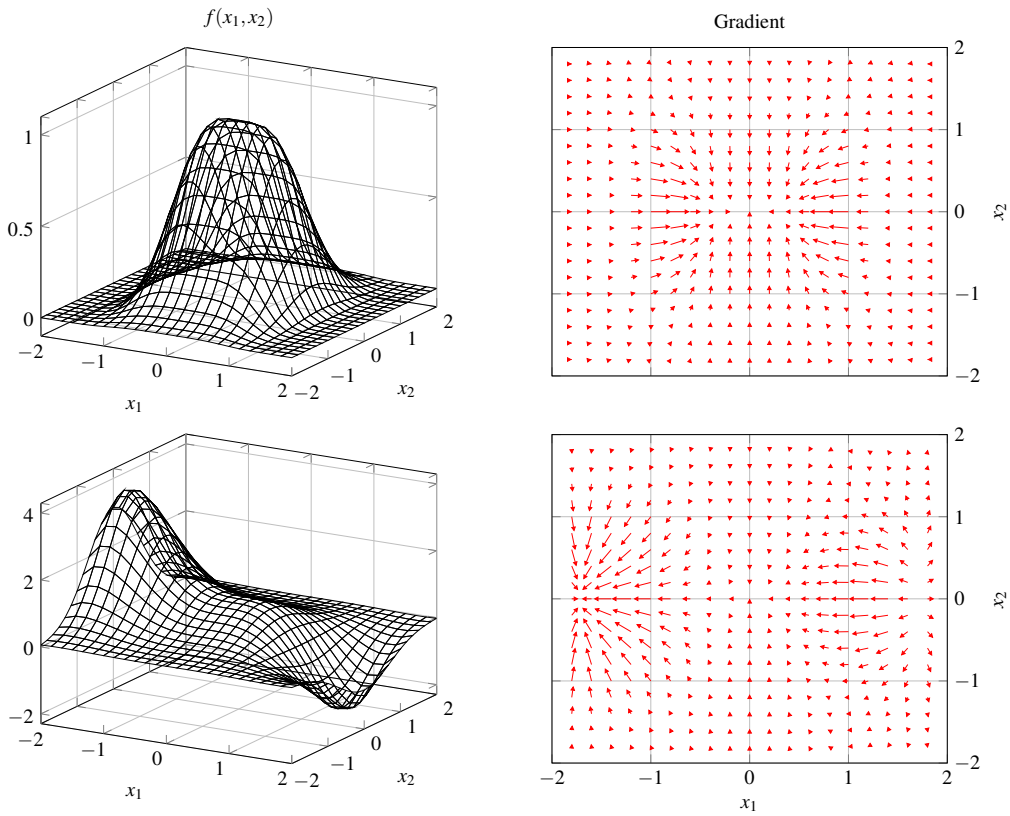


Figure 7.7: On the left we plot two functions $f\colon \mathbb{R}^2 \to \mathbb{R}$ and on the right their gradients at a number of points are represented as arrows. The direction of each arrow indicates the direction of the gradient at that point. The gradient is a vector that points towards the direction of the fastest increase of the function.

Figure 7.6. The gradient is

$$\nabla f(x_1,x_2) = \begin{bmatrix} \partial_1 f(x_1,x_2) \\ \partial_2 f(x_1,x_2) \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x_1} x_1 + \frac{\partial}{\partial x_1} x_2 \\ \frac{\partial}{\partial x_2} x_1 + \frac{\partial}{\partial x_2} x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \tag{7.32}$$

Therefore the fastest increase is obtained by moving diagonally with respect to the two coordinate axes.

7.3.2 *Multivariate Optimisation*

We now turn to local optimisation of a d-variate function $f\colon \mathbb{R}^d \to \mathbb{R}$. The case $d = 1$ was discussed in Section 7.2. We again focus on minimisation as maximisation of f is equivalent to minimisation of $-f$. Let us first present multivariate versions of some definitions and results that are familiar to us from Section 7.2.

Definition 7.15 (Minima in $\mathbb{R}^d$)**.** Let $f\colon \mathbb{R}^d \to \mathbb{R}$ be a function. A point $\mathbf{x}^* \in \mathbb{R}^d$ is a

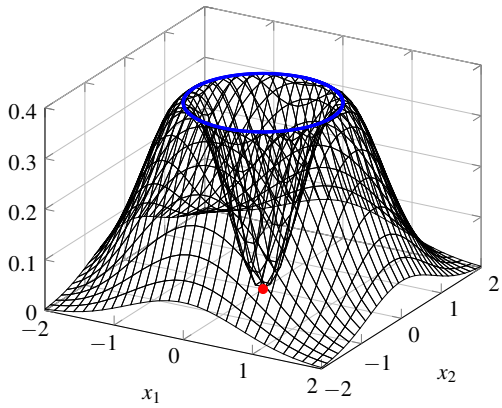


Figure 7.8: A function with an infinite number of critical points. The function $f(\mathbf{x}) = \exp(-\|\mathbf{x}\|^2)\|\mathbf{x}\|^2$ for $d = 2$ has critical points marked in blue and red. The blue critical points, which satisfy $\|\mathbf{x}^*\|^2 = 1$, are local maximum points while the red critical point $\mathbf{x}^* = \mathbf{0}$ is a local minimum point.

- *global minimum point* of f if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^d$;
- *local minimum point* of f if there exists $\varepsilon > 0$ such that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for every $\mathbf{x} \in \mathbb{R}^d$ satisfying $\|\mathbf{x}^* - \mathbf{x}\| \leq \varepsilon$.

The definition of a local minimum requires that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x}$ in *some* non-empty *closed ball* centered at $\mathbf{x}^*$. The closed ball $B(\mathbf{x}^*, \varepsilon)$ of radius $\varepsilon > 0$ centered at $\mathbf{x}^*$ is the set of those $\mathbf{x} \in \mathbb{R}^d$ which are at most ε away from $\mathbf{x}^*$ in the Euclidean distance:

$$B(\mathbf{x}^*, \varepsilon) = \{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}^* - \mathbf{x}\| \leq \varepsilon\}. \tag{7.33}$$

Here the notation reads as "those $\mathbf{x} \in \mathbb{R}^d$ that satisfy the condition to the right of the colon" (i.e., $\|\mathbf{x}^* - \mathbf{x}\| \leq \varepsilon$). Global and local maxima, as well as the concept of extremal points, are defined in the same way as in Section 7.2. In higher dimensions a critical point is a point at which the gradient vanishes [i.e., equals the zero vector $\mathbf{0} = (0, \ldots, 0) \in \mathbb{R}^d$].

Definition 7.16 (Critical points in $\mathbb{R}^d$)**.** A point $\mathbf{x}^* \in \mathbb{R}^d$ is a *critical point* of a differentiable function $f\colon \mathbb{R}^d \to \mathbb{R}$ if $\nabla f(\mathbf{x}^*) = \mathbf{0} = (0, \ldots, 0) \in \mathbb{R}^d$.

Note that each of the d coordinates of the gradient $\nabla f(\mathbf{x}^*) \in \mathbb{R}^d$ has to be zero if $\mathbf{x}^*$ is to be a critical point.

Example 7.17. Consider the function $f\colon \mathbb{R}^d \to \mathbb{R}$ given by

$$f(\mathbf{x}) = \exp(-\|\mathbf{x}\|^2)\|\mathbf{x}\|^2. \tag{7.34}$$

The gradient of this function is

$$\nabla f(\mathbf{x}) = \|\mathbf{x}\|^2 \nabla \exp(-\|\mathbf{x}\|^2) + \exp(-\|\mathbf{x}\|^2)\nabla\|\mathbf{x}\|^2 = 2\exp(-\|\mathbf{x}\|^2)(1 - \|\mathbf{x}\|^2)\mathbf{x}. \tag{7.35}$$

Since the exponential function is positive, the gradient is zero only if $\mathbf{x} = \mathbf{0}$ or $\|\mathbf{x}\|^2 = 1$. Therefore the origin and every point on the unit sphere

$$\{\mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|^2 = 1\} \tag{7.36}$$

is a critical point of f. That is, f has an infinite number of critical points. From Figure 7.8 we see that the critical points on the unit sphere are local maximum points while the origin is a local minimum point.

Fermat's theorem (Theorem 7.4) remains valid in $\mathbb{R}^d$.

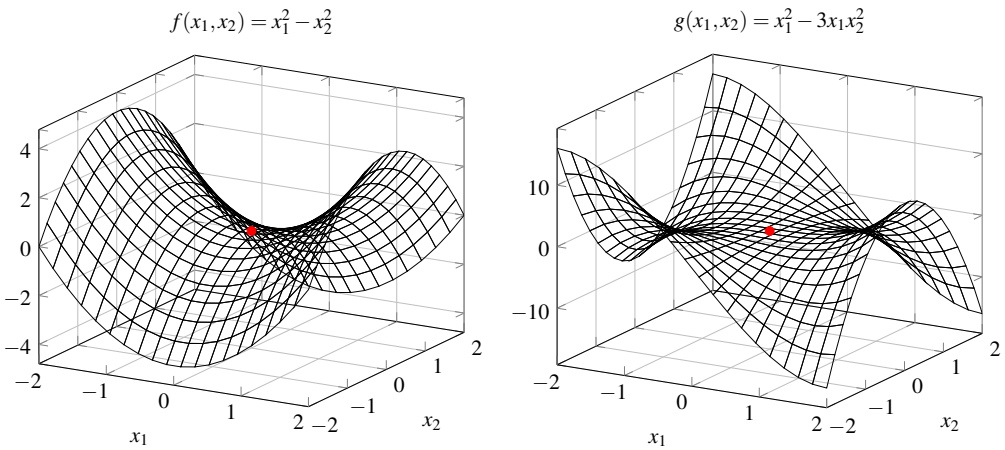


Figure 7.9: The functions $f(x_1,x_2) = x_1^2 - x_2^2$ and $g(x_1,x_2) = x_1^2 - 3x_1x_2^2$ from Example 7.20 both have the saddle point $\mathbf{x}^* = (0,0)$ (marked in red). These figures may explain why saddle points are called saddle points.

Theorem 7.18 (Fermat's theorem in $\mathbb{R}^d$)**.** *Every local extremal point $\mathbf{x}^*$ of a differentiable function $f\colon \mathbb{R}^d \to \mathbb{R}$ is a critical point, in that $\nabla f(\mathbf{x}^*) = \mathbf{0}$.*

Proof. Omitted. □

As in the one-dimensional case, there is no implication to the other direction. That is, from $\nabla f(\mathbf{x}^*) = \mathbf{0}$ one cannot deduce that $\mathbf{x}^*$ is a local extremal point. Non-extremal points at which the gradient vanishes are called saddle points.

Definition 7.19 (Saddle point)**.** A point $\mathbf{x}^*$ such that $\nabla f(\mathbf{x}^*) = \mathbf{0}$ is a *saddle point* of $f\colon \mathbb{R}^d \to \mathbb{R}$ if $\mathbf{x}^*$ is *not* a local extremal point of f.

Example 7.20. The functions

$$f(x_1,x_2) = x_1^2 - x_2^2 \quad \text{and} \quad g(x_1,x_2) = x_1^2 - 3x_1x_2^2. \tag{7.37}$$

have the gradients

$$\nabla f(x_1,x_2) = \begin{bmatrix} 2x_1 \\ -2x_2 \end{bmatrix} \quad \text{and} \quad \nabla g(x_1,x_2) = \begin{bmatrix} 2x_1 - 3x_2^2 \\ -6x_1x_2 \end{bmatrix}. \tag{7.38}$$

Both gradients equal $\mathbf{0} = (0,0)$ only when $\mathbf{x} = (x_1,x_2) = \mathbf{x}^* = (0,0)$. Therefore $\mathbf{x}^* = (0,0)$ is the only critical point for both f and g. It turns out that $\mathbf{x}^*$ is a saddle point; see Figure 7.9.

7.3.3 Gradient Descent

Let us finally introduce a multivariate version of gradient descent whose univariate implementation was introduced in Section 7.2. Gradient descent exploits the fact that the gradient gives the direction and the rate of fastest increase of a function, so that fastest *decrease* is obtained by heading towards the *negative* gradient.

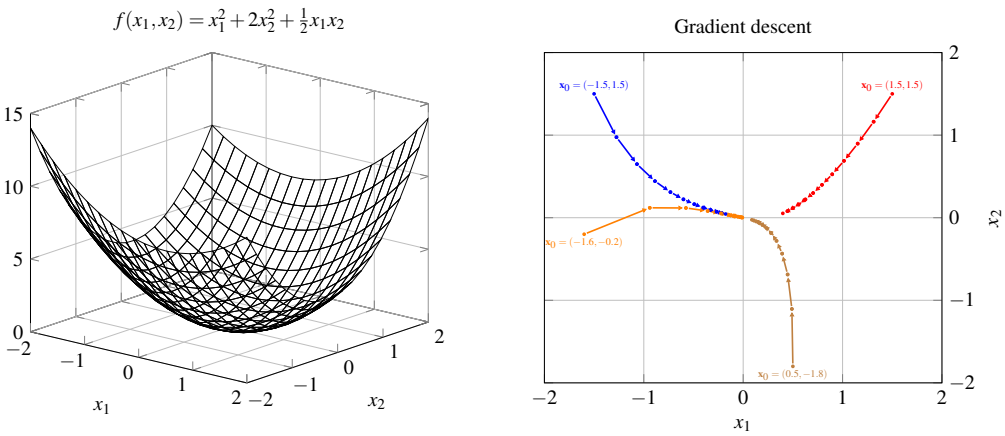


Figure 7.10: Four gradient descent trajectories (right) for the function $f(x_1,x_2) = x_1^2 + 2x_2^2 + \frac{1}{2}x_1x_2$ (left) with different initial points and learning rates. The function attains its minimum at the origin $\mathbf{x}^* = (0,0)$.

Definition 7.21 (Multivariate gradient descent)**.** Let $\eta > 0$ be the *learning rate* and $\mathbf{x}_0 \in \mathbb{R}$ an initial point. The *gradient descent* finds approximations $\mathbf{x}_k$ to a minimum point of a differentiable function $f\colon \mathbb{R}^d \to \mathbb{R}$ with gradient $\nabla f\colon \mathbb{R}^d \to \mathbb{R}^d$ via the iteration

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k). \tag{7.39}$$

The iteration is terminated after sufficiently iterations or when the Euclidean norm $\|\nabla f(\mathbf{x}_k)\|$ is sufficiently small.

That heading towards the negative gradient results in fastest possible decrease is true only *locally*, in the vicinity of the current point $\mathbf{x}_n$. This makes gradient descent (at least in the naive form presented here) a greedy algorithm that at each steps makes a locally optimal choice. However, there is no guarantee that making a sequence of locally optimal choices gives a solution to a minimisation problem that is *globally* good. That is, over multiple steps this may trap gradient descent in a "bad" local minimum point (i.e., one at which f takes a much larger value than at some other local minimum points). Figure 7.10 demonstrates gradient descent for minimising a bivariate (i.e., $d = 2$) function.

7.4 Function Composition and the Chain Rule

In what follows it is necessary to master function composition and the chain rule of differentiation. Complicated functions are often formed by *composing* simple functions.

Definition 7.22 (Function composition)**.** Let $g\colon X \to Y$ and $h\colon Y \to Z$ be functions. Their *composite function* $f = h \circ g$ is a function from X to Z given by

$$f(x) = h(g(x)). \tag{7.40}$$

To compute $f(x) = h(g(x))$ we (i) first map x through g and (ii) after this map the output $g(x)$ through h. We can also compose more than two functions, so that

$$f = r \circ h \circ g \quad \text{is given by} \quad f(x) = (r \circ h \circ g)(x) = r(h(g(x))). \tag{7.41}$$

One can think recursively:

$$f = r \circ h \circ g = r \circ f_1, \quad \text{where} \quad f_1 = h \circ g. \tag{7.42}$$

Example 7.23. Define the functions

$$g(x) = 3\pi, \quad h(x) = \log(1+x), \quad r(x) = 2+x^{-3} \quad \text{and} \quad l(x) = x, \tag{7.43}$$

all from $\mathbb{R}$ to $\mathbb{R}$. Then

$$f(x) = (h \circ g)(x) = h(g(x)) = h(3\pi) = \log(1+3\pi), \tag{7.44a}$$
$$f(x) = (g \circ h)(x) = g(h(x)) = g(\log(1+x)) = 3\pi, \tag{7.44b}$$
$$f(x) = (h \circ r)(x) = h(r(x)) = h(2+x^{-3}) = \log(3+x^{-3}), \tag{7.44c}$$
$$f(x) = (r \circ l)(x) = r(l(x)) = r(x) = 2+x^{-3}, \tag{7.44d}$$
$$f(x) = (r \circ l \circ h)(x) = r(l(h(x))) = r(h(x)) = r(\log(1+x)) = 2+\log(1+x)^{-3}. \tag{7.44e}$$

Note that (7.44a) and (7.44b) are simply the *constant functions* $x \mapsto \log(1+3\pi)$ and $x \mapsto 3\pi$, which map every $x \in \mathbb{R}$ to the constants $\log(1+3\pi)$ and 3π, respectively.

Example 7.24. The *ReLU activation function* is

$$f(\mathbf{x}) = \max\{0, a+\mathbf{w}\cdot\mathbf{x}\} = \max\{0, a+\textstyle\sum_{i=1}^d w_i x_i\} = \begin{cases} 0 & \text{if} \quad a+\mathbf{w}\cdot\mathbf{x} < 0, \\ a+\mathbf{w}\cdot\mathbf{x} & \text{otherwise,} \end{cases}$$

where $a \in \mathbb{R}$ and $\mathbf{w} \in \mathbb{R}^d$ are fixed (i.e., $f\colon \mathbb{R}^d \to \mathbb{R}$). We can write the ReLU activation function as the composition

$$f = h \circ g, \tag{7.45}$$

where

$$g(\mathbf{x}) = \mathbf{w}\cdot\mathbf{x} \quad \text{and} \quad h(x) = \max\{0, a+x\} \tag{7.46}$$

are functions from $\mathbb{R}^d$ to $\mathbb{R}$ and from $\mathbb{R}$ to $\mathbb{R}$, respectively. The composition is *not* unique. We could have alternatively selected $g(\mathbf{x}) = a+\mathbf{w}\cdot\mathbf{x}$ and $h(x) = \max\{0, x\}$.

The *chain rule* is used to differentiate composite functions.

Theorem 7.25 (Chain rule)**.** *Let $f = h \circ g$. The* chain rule *states that*

$$f'(x) = (h \circ g)'(x) = (h' \circ g)(x) g'(x) = h'(g(x)) g'(x). \tag{7.47}$$

The procedure to apply chain rule to compute $f'(x)$ goes as follows:

1. Compute the derivative h' of h.
2. Compute $h'(g(x))$, the value of this derivative at $g(x)$.
3. Multiply by $g'(x)$, the derivative of g at x.

Leibniz's notation provides an easy mnemonic for the chain rule. By treating $\mathrm{d}f$, $\mathrm{d}x$, and other such quantities as numbers, we can formally write

$$f' = \frac{\mathrm{d}f}{\mathrm{d}x} = \frac{\mathrm{d}f}{\mathrm{d}g} \cdot \frac{\mathrm{d}g}{\mathrm{d}x}. \tag{7.48}$$

Here we first differentiate f with respect to g, which means that we treat g as the argument to f. Because $f = h \circ g = h(g)$, this gives us

$$\frac{\mathrm{d}f}{\mathrm{d}g} = \frac{\mathrm{d}h}{\mathrm{d}g}(g) = h'(g) = h' \circ g. \tag{7.49}$$

From (7.48) we then obtain the chain rule in (7.47):

$$f'(x) = \frac{\mathrm{d}f}{\mathrm{d}x}(x) = \frac{\mathrm{d}f}{\mathrm{d}g}(x) \cdot \frac{\mathrm{d}g}{\mathrm{d}x}(x) = (h' \circ g)(x)g'(x) = h'(g(x))g'(x). \tag{7.50}$$

Again, these are purely formal computations (i.e., we treat and manipulate the derivative as if it were the quotient of two numbers, forgetting the presence of a limit in its definition). Although the result is correct, Equation (7.48) is *not* a rigorous mathematical proof of the chain rule.

Example 7.26. Let $h(x) = \log(x)$ and $g(x) = x^n$. By the chain rule,

$$f'(x) = (h \circ g)'(x) = h'(g(x))g'(x) = \frac{1}{g(x)} \cdot g'(x) = \frac{1}{x^n} \cdot nx^{n-1} = \frac{n}{x}. \tag{7.51}$$

Note that we obtain the same derivative directly from $\log(x^n) = n\log(x)$.

Example 7.27. Let $f(x) = \exp(-x^2)$. Set $h(x) = \exp(x)$ and $g(x) = -x^2$, so that $f = h \circ g$. Then the chain rule yields

$$f'(x) = h'(g(x))g'(x) = \exp(g(x)))g'(x) = \exp(-x^2) \cdot (-2x) = -2x\exp(-x^2). \tag{7.52}$$

Example 7.28. Let $f = h(ax)$ for a function h and a constant $a \in \mathbb{R}$. By setting $g(x) = ax$, we obtain from the chain rule and linearity of differentiation that

$$f'(x) = h'(g(x))g'(x) = ah'(ax). \tag{7.53}$$

Note that this differs from computing $h'(ax)$, the derivative of h at point ax.

Example 7.29. The derivative of the logarithm can be derived from the chain rule and $\mathrm{d}e^x/\mathrm{d}x = e^x$. Let $g(x) = e^x$ and $h(x) = \log(x)$. Then $f(x) = h(g(x)) = \log(e^x) = x$, so that $f'(x) = 1$. But we can alternatively apply the chain rule, which gives

$$f'(x) = h'(g(x))g'(x) = \left(\frac{\mathrm{d}h}{\mathrm{d}x}(e^x)\right)e^x = h'(e^x)\,e^x. \tag{7.54}$$

Since we know that (7.54) equals 1, division by e^x gives us

$$h'(e^x) = e^{-x}. \tag{7.55}$$

Note that this equation says the the derivative of the logarithm h' equals e^{-x} when evaluated at e^x. By selecting $x = \log(z)$, so that $e^{\log z} = z$ and $e^{-x} = e^{-\log z} = z^{-1}$, we obtain

$$h'(z) = z^{-1}, \tag{7.56}$$

which is the familiar derivative for the logarithm.

7.5 Automatic Differentiation

To implement gradient descent (as well as Newton's method from Section 5) it is necessary to have access to the derivative of f. While it is essential to have a good grasp of the basic differentiation rules, the derivatives involved in applications quickly become practically impossible (or at least

extremely tedious) to compute by hand. For example, computing the derivative of the loss function in (7.6), which involves a composition of the sigmoid activation function

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

with itself, is quite tedious. In reality there are many more compositions, which makes computing the derivative by hand wholly unrealistic. It would be preferable to have computer take care of differentiation. This is possible in three different ways.

7.5.1 Numerical Differentiation

In *numerical differentiation* the derivative is approximated numerically. The simplest approach is to take the difference quotient definition of the derivative and, instead of going to the limit, use a non-zero h to approximate the derivative:

$$f'(x_0) \approx \frac{f(x_0+h)-f(x_0)}{h}, \quad \text{where} \quad h \approx 0. \tag{7.57}$$

We have seen this in Examples 2.5 and 2.17. This approach is conceptually straightforward and simple to implement (at least in a very naive way). However, numerical differentiation provides only an approximation to the derivative, can introduce round-off errors, and does not scale well to higher-order derivatives and the multivariate setting.

7.5.2 Symbolic Differentiation

Symbolic computation packages can be used to perform *symbolic differentiation*, which automates the process of computing symbolic expressions for derivatives. For example, inputting (here sinh and cosh are the *hyperbolic* sine and cosine functions)

```
differentiate sin(log(x^(sqrt(exp(-cos(x)))) + x^2) + cosh(x))
```

to WolframAlpha yields the lovely expression

$$\cos\left(\log(x^{\sqrt{\exp(-\cos(x))}}+x^2)+\cosh(x)\right) \times \left(\frac{x^{\sqrt{\exp(-\cos(x))}}[x^{-1}\sqrt{\exp(-\cos(x))}+\frac{1}{2}\log(x)\sin(x)\sqrt{\exp(-\cos(x))}\,]+2x}{x^{\sqrt{\exp(-\cos(x))}}+x^2}+\sinh(x)\right)$$

for the derivative of the function

$$f(x) = \sin\left(\log(x^{\sqrt{\exp(-\cos(x))}}+x^2)+\cosh(x)\right). \tag{7.58}$$

This is quite impressive and useful if one needs to study how the derivative behaves as a part of some mathematical problem (or when it is part of an exercise to compute a derivative). However, computing such symbolic expression is computationally expensive and *completely unnecessary*. For in practice it is not a symbolic expression such as this that is needed *but values of a derivative at certain points*: to implement gradient descent we do not need a symbolic expression for the derivative, only its *values* at the iterates x_k. That is, to compute $f'(x_k)$ at a given point x_k (e.g., $x_k = 3$) it is not necessary to have access to a full symbolic expression for the derivative.

7.5.3 Automatic Differentiation

Automatic differentiation utilises the fact that complicated functions reduce to a sequence of compositions and arithmetic operations involving the elementary functions whose derivatives are

available. The chain rule and other differentiation rules then permit algorithmically straightforward computation of derivatives of arbitrarily complicated functions. For example, consider a function f given by

$$f(x) = h(g(h(x)g(x)) + h(x)). \tag{7.59}$$

We may write this function as

$$f(x) = h(r(x)), \tag{7.60}$$

where

$$r(x) = u(x) + h(x), \quad u(x) = g(s(x)), \quad \text{and} \quad s(x) = h(x)g(x). \tag{7.61}$$

By applying the rules of differentiation we can write the derivative of f as

$$f'(x) = h'(r(x))r'(x), \tag{7.62}$$

where

$$r'(x) = u'(x) + h'(x), \quad u'(x) = g'(s(x))s'(x) \quad \text{and} \quad s'(x) = h'(x)g(x) + h(x)g'(x). \tag{7.63}$$

If h and g are elementary functions that we can differentiate, we can easily compute $s'(x_0)$ at any point x_0 we want. Subsequently we can use $u'(x_0) = g'(s(x_0))s'(x_0)$ to obtain the derivative $u'(x_0)$, which then gives us $r'(x_0)$. Finally, we can use $r'(x_0)$ in (7.62) to compute $f'(x_0)$. To obtain $f'(x_1)$ we simply repeat the process with $x = x_1$. At no point do we require access to a symbolic expression for $f'(x)$ or the intermediate derivatives $r'(x)$, $u'(x)$ and $s'(x)$. *It suffices to have have access to expressions for $h'(x)$ and $g'(x)$.*

Implementing very basic automatic differentiation is surprisingly simple. The following Python code suffices to differentiate the function in (7.58). This is the so-called *forward mode* of automatic differentiation.

```
import numpy as np

class fDf:
    def __init__(self, val, deriv):
        self.val = val
        self.deriv = deriv
    def __add__(self, u):
        return fDf(self.val + u.val, self.deriv + u.deriv)
    def __mul__(self, u):
        return fDf(self.val * u.val, self.deriv * u.val + self.val * u.deriv)
    def __rmul__(self, c):
        return fDf(c * self.val, c * self.deriv)

def exp(u):
    return fDf(np.exp(u.val), np.exp(u.val) * u.deriv)
def sin(u):
    return fDf(np.sin(u.val), np.cos(u.val) * u.deriv)
def cos(u):
    return fDf(np.cos(u.val), -np.sin(u.val) * u.deriv)
def log(u):
    return fDf(np.log(u.val), (1.0 / u.val) * u.deriv)
def sqrt(u):
    return fDf(np.sqrt(u.val), (0.5 / np.sqrt(u.val)) * u.deriv)
def cosh(u):
    return fDf(np.cosh(u.val), np.sinh(u.val) * u.deriv)

x = fDf(3.0, 1.0)
fDfx = sin(log(exp(log(x) * sqrt(exp(-1.0*cos(x)))) + x * x) + cosh(x))
Dfx = fDfx.deriv
```

We first define a class `fDf` which holds the value and derivative of a function at a point of interest. By overloading addition (`add`) and multiplication (`mul` and `rmul`) we can perform arithmetic on this class. For example, for two instances `f` and `g` of the class `fDf`, which store $f(x)$ and $g(x)$ as well as $f'(x)$ and $g'(x)$ at some point x, the multiplication `f * g` is an instance of `fDf` that stores the function value $f(x)g(x)$ and the derivative $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$ at x that has been computed using the product rule of differentiation. We then define a number of functions that operate on instances of `fDf`. Each of these functions calls the corresponding `numpy` function to perform function evaluation and implements the chain rule for that particular function in order to evaluate the derivative. For example, `exp` assumes that it is given a function $u(x)$ and its derivative $u'(x)$ at some point x and uses these to compute the function evaluation $e^{u(x)}$ and, via the chain rule, the derivative evaluation $e^{u(x)}u'(x)$. With this code we can differentiate any function that involves sums, products and compositions of exp, sin, cos, log, square root and cosh. At the end we specify that we want to differentiate the function in (7.58) at $x_0 = 3$. The line `x = fDf(3.0, 1.0)` does this by defining that the innermost function in automatic differentiation is $u(x) = x$ and that $u(x_0) = 3$ and $u'(x_0) = 1$ since $u'(x) = 1$. Note that in defining the function being differentiated we use $x^{u(x)} = \exp(u(x)\log(x))$ in order to avoid having to define a differentiation rule for $x^{u(x)}$.

What makes automatic differentiation very powerful is the fact that the function being differentiated does not have to a simple algebraic expression. For example, with the above code we could easily differentiate functions that have been defined using loops and conditional statements. Of course, one should not use home-made automatic differentiation routines. One popular option is `JAX` (https://jax.readthedocs.io/), the usage of which to differentiate (7.58) is illustrated by the following code.

```
import jax.numpy as jnp
from jax import grad

def f(x):
    return jnp.sin(jnp.log(jnp.exp(jnp.log(x) *
        jnp.sqrt(jnp.exp(-1.0*jnp.cos(x)))) + x**2) + jnp.cosh(x))

Df = grad(f)
x = 3.0
Dfx = Df(x)
```

The derivative `Dfx = Df(x)` is now equal to the derivative `Dfx = fDfx.deriv` computed by our earlier code. The crucial JAX function `grad` performs automatic differentiation. Figure 7.11 shows the function in (7.58) and its derivative that has been computed using `JAX`.

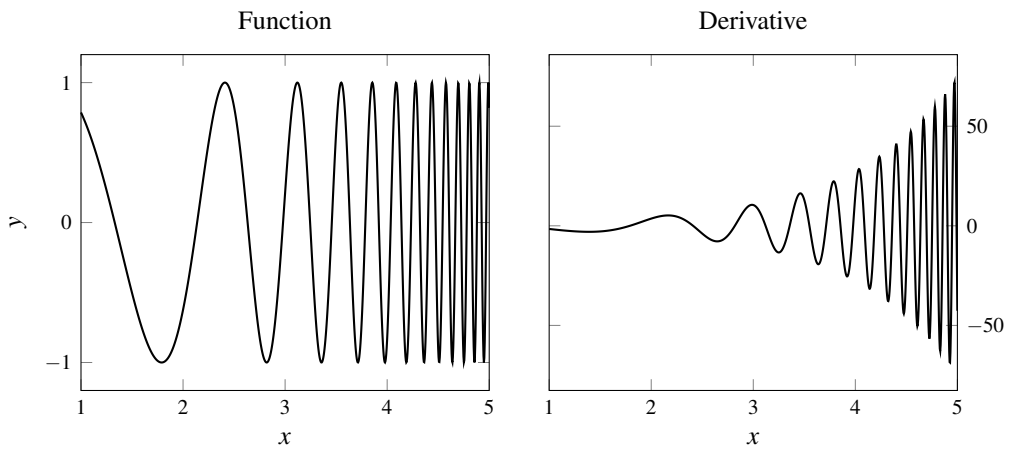


Figure 7.11: The function in (7.58) and its derivative that has been computed using `JAX`.